

Module 1

Introduction to Economics (07 Hours)

Managerial Economics: Meaning, Nature, Scope, & Significance, Uses of Managerial Economics, Role and Responsibilities of Managerial Economist. Theory of the Firm: Firm and Industry, Objectives of the firm, alternate objectives of firm. Managerial theories: Baumol's Model, Marris's Hypothesis, Williamson's Model of growth maximization, Williamson's model of managerial discretion.

Introduction

The word economics come from the ancient Greek word "Oikonomikos" which means skilled in household management. The modern day use of the word refers to the skill in using the country's scarce resources or factors of production to satisfy the desires of its peoples

- Oikos-House
- Nomos-Management

Thus economics means House Management. The head of a family faces the problem of managing unlimited wants of the family members within the limited income of the family. Similarly, considering the whole society as a family, the society also faces the problem of tacking unlimited wants of the members of the society with the limited resources available in that society.

Thus Economics would mean the study of ways in which mankind organizes itself to tackle the basic problem of scarcity of resources

For a long time discipline of economics was not known as a separate discipline. It was related to elements of several other disciplines, which covered non-economic aspects of human knowledge and philosophy as well. The mail reasons for this phenomenon lie in the following facts:

- The primary task of the discipline of economics is to study the working of an economy and the issues by it.
- The nature of economic problems and their solutions is closely linked with the structure, progress and development of the economic system.
- Economic problems of individual and societies are interdependent on other social, cultural and political aspects of human existence.

Micro and Macro Economics

Subject matter of the economics can be classified into two Types

- 1) Micro Economics
- 2) Macro Economics

1) Micro Economics

The **micro economics** is the study of an economic behavior of a particular individual, firm, or household, i.e. it studies a particular unit. Microeconomics is the branch of economics that concentrates on the behaviour and performance of the individual units, i.e. consumers, family, industry, firms. Here, the demand plays a key role in determining the quantity and the price of a product along with the price and quantity of related goods (complementary goods) and substitute products, so as to make a judicious decision regarding the allocation of scarce resources, concerning their alternative uses. **Examples:** Individual Demand, Price of a product, etc.

2) Macro Economics

Macro Economics is the study of the economy as a whole i.e., not a single unit but the combination of all, firms, households, nation, etc. Macroeconomics is the branch of economics that concentrates on the behaviour and performance of aggregate variables and those issues which affect the whole economy. It includes regional, national and international economies and covers the major areas of the economy like unemployment, poverty, general price level, GDP (Gross Domestic Product), imports and exports, economic growth, globalisation, monetary/ fiscal policy, etc. It helps in resolving the various problems of the economy, thereby enabling it to function efficiently. **Examples:** Aggregate Demand, National Income, etc.

• Difference between Micro and Macro Economics

Micro Economics	Macro Economics
1) Microeconomics studies the particular market segment of the economy	1) Macroeconomics studies the whole economy, that covers several market segments
2) Micro economics stresses on individual economic units.	2) Micro economics stresses on individual economic units
3) Microeconomics is applied to operational or internal issues	3) Environmental and external issues are the concern of macro-economics.

Economics for Decision Making (MBA103)

4) Microeconomics deals with an individual product, firm, household, industry, wages, prices, etc.	4) Macroeconomics deals with aggregates like national income, national output, price level, etc.
5) Microeconomics determine the price of a particular commodity along with the prices of complementary and the substitute goods	5) Macroeconomics is helpful in maintaining the general price level.
6) While analyzing any economy, micro economics takes a bottom-up approach	6) Macro-economics takes a top-down approach into consideration.

Modern business has become very vast and Complex. Its rapid expansion took place after the Second World War and particularly after 1950 with the rapid expansion of business all over the world the business executives faced a number of problems due to the changing business environment and consequent variability and unpredictability of their achievements.

Since the modern business world at is full of risks and uncertainties, the business executives, merely with the knowledge of economic theories, could not find proper solutions to their problems in order to achieve or realize the objectives of their business. This is so because the principles of economics on their own do not offer ready-made solutions applicable to a changing business world. These economic principles' could not come to the rescue of the business executives for solving their problems. There was thus a gap between the pure economic theory and the exact procedure to be adopted to arrive at correct decisions in the treatment of their business problems. These problems attracted the attention of the academicians to develop a new branch of discipline known as 'Business Economics'. Of late however, the term 'Managerial Economics' has become more popular and it seems to replace progressively.

Managerial economics is a science which deals with the application of economic theory to business and business management. It is the study of the allocation of resources available to a firm among its activities.

It fills the gap between economic theory and managerial practice. It is concerned with those aspects of economic theory which are directly relevant to the practice of management and decision-making process within the enterprise.

The main function of business executive is decision-making and forward planning '**Decision-Making**', means the process of selecting or choosing a particular course of action

from two or more alternative courses of action whereas '**Forward Planning**' means formulating plans for the future.

In order to perform the functions of decision-making and forward planning under the conditions of risks and uncertainties, the economics theory has to be applied to business management. Economic theory deals with a number of concepts and principles relating to cost, production, pricing profit, demand, supply, competition business cycles, national income etc, These concepts and principles of economics, 'with the aid of other sciences like statistics, accountancy and mathematics, enable the business executives to solve their business problems. The Manner in which economic theory can be used for solving the business problems forms the subject- matter of managerial economics.

Study of Managerial Economics helps in enhancement of analytical skills, assists in rational configuration as well as solution of problems. While microeconomics is the study of decisions made regarding the allocation of resources and prices of goods and services, macroeconomics is the field of economics that studies the behavior of the economy as a whole (i.e. entire industries and economies). Managerial Economics applies micro-economic tools to make business decisions. It deals with a firm.

The use of Managerial Economics is not limited to profit-making firms and organizations. But it can also be used to help in decision-making process of non-profit organizations (hospitals, educational institutions, etc). It enables optimum utilization of scarce resources in such organizations as well as helps in achieving the goals in most efficient manner. Managerial Economics is of great help in price analysis, production analysis, capital budgeting, risk analysis and determination of demand.

Definition

Douglas - "Managerial economics is .the application of economic principles and methodologies to the decision-making process within the firm or organization."

Pappas &Hirschey - "Managerial economics applies economic theory and methods to business and administrative decision-making."

Salvatore - "Managerial economics refers to the application of economic theory and the tools of analysis of decision science to examine how an organisation can achieve its objectives most effectively."

Howard Davies and Pun-Lee Lam - "It is the application of economic analysis to business problems; it has its origin in theoretical microeconomics."

Spencer & Louis Siegelman: The integration of Economic theory with business practice for the purpose of facilitating decision making and forward planning by management

Milton H. Spencer and Louis Siegleman: define managerial economics as the "the integration of economic theory with managerial practice for the purpose of decision-making and forward planning by management".

Colberg: "Managerial economics is primarily concerned with the applicability of economic concepts and analysis to decisions made by business".

The application of economic theory and the tools of decision science to examine how an organization can achieve its aims or objectives most efficiently.

All these definitions reveal the importance of economic theories in decision -making and forward planning. Managerial economics makes use of the analytical tools of economic theory in solving managerial problems. A managerial economist can become a far more help full member of management group by virtue of his studies of economic analysis.

In general, Managerial Economics could be defined as the discipline which deals with the application of economic theory to business management.

Nature of Managerial Economics

Management is the guidance, leadership and control of the efforts of a group of people towards some common objective. It tells about the purpose or function of management. Koontz and O' Donell define management as the creation and maintenance of an internal environment in an enterprise where individuals work together in groups, can perform efficiently and effectively towards the attainment of group goals. Thus, management is coordination, an art of getting things done by other people. On the other hand, economics due to scarcity of resources is primarily engaged in analyzing and providing answers to the various basic economic problems like what to produce? How to produce? And for whom to produce? Science of Economics has developed several concepts and analytical tools to deal with the problem of allocation of scarce resource among competing ends. Close interrelationship between management objectives and economic principles has led to the development of Managerial Economics. Managerial Economics as a link between economic theory and decision science, its purpose is to contribute to sound decision making not only in business but also in government agencies and Non- profit organizations. In particular, managerial economics assists in making decisions about the optimum allocation of scarce resources among competing activities.

Normative and Positive Economics:

Economics is divided into two categories viz, positive economics and normative economics. Positive economics tells the things as they are in the world and hence it is descriptive in nature. But normative economics deals with things as they ought to be or should be and hence it is prescriptive in nature.

Managerial economics is normative rather than positive in character. In other words, it is prescriptive rather than descriptive. It deals with future planning policy-making, decision-making how to make use of economic principles etc. It therefore involves value judgment. It tells what aims and objectives ought to be pursued by a firm and how best they should be achieved.

Managerial economics is pragmatic in nature. It applies economics to. Solve problems faced by the business executives. It is, in fact, a guide to make efficient use of resources for the realization of the objectives of the firm.

Managerial economics is micro-economic in nature: It is interested in the study of an individual business firm and not all the firms in the economy as a whole.

Scope of Managerial Economics

Since managerial economics has emerged as new discipline, there is no unanimity of opinion among the economists regarding its exact scope the following topics have, however, been included in the scope of business or managerial economics.

Well scope is something which tells us how far a particular subject will go. As far as Managerial Economic is concerned it is very wide in scope. It takes into account almost all the problems and areas of manager and the firm. Following aspects are to be taken into account while knowing the scope of ME:

1. Demand Analysis and forecasting: A business firm is an economic organization engaged in transforming productive resources into goods to be sold in the market. A major part of the managerial decisions depends upon an accurate estimate of demand. A forecast of future sales is necessary to serve as a guide to the management for preparing production schedules and employing resources. This will also help the management to maintain or strengthen its market position and enlarge its profits. Demand analysis helps to identify the various factors that influence the demand for the firm's products and provides guidelines to manipulate the demand. Such factors which influence the demand are called demand determinants. Demand analysis and forecasting occupy a strategic place in managerial economics and cover topics such as demand determinants, demand distinctions and demand forecasting.

2. Cost and Production Analysis: Cost analysis is helpful in understanding the cost of a particular product. It takes into account all the costs incurred while producing a particular product. Under cost analysis we will take into account determinants of costs, method of estimating costs, the relationship between cost and output, the forecast of the cost, profit, these terms are very vital to any firm or business.

An accurate cost and production analysis is essential for profit. Cost control and sound pricing policies and practices. Cost estimates are very useful for the management decisions. The factors causing variations in cost estimates should be given due consideration if the management is to arrive at correct cost estimates so essential for planning purposes. Since all factors determining cost are not always known or controllable and element of uncertainty exists in cost calculation. Cost analysis and production analysis are interconnected. However, production analysis frequently proceeds in physical terms while cost analysis proceeds in monetary terms. Cost analysis and production analysis cover such topics as cost concepts, classification, cost output, relationships, economies and diseconomies of scale, production function and cost control.

3. Inventory Management: What do you mean by the term inventory? Well the actual meaning of the term inventory is stock. It refers to stock of raw materials which a firm keeps. Now here the question arises how much of the inventory is ideal stock. Both the high inventory and low inventory is not good for the firm. Managerial economics will use such methods as ABC Analysis, simple simulation exercises, and some mathematical models, to minimize inventory cost. It also helps in inventory controlling.

4. Advertising: Advertising is a promotional activity. In advertising while the copy, illustrations, etc., are the responsibility of those who get it ready for the press, the problem of cost, the methods of determining the total advertisement costs and budget, the measuring of the economic effects of advertising- -- - are the problems of the manager. There's a vast difference between producing a product and marketing it. It is through advertising only that the message about the product should reach the consumer before he thinks to buy it. Advertising forms the integral part of decision making and forward planning.

5. Pricing Decisions, policies and Practices: Here pricing refers to the pricing of a product. Since price is the genesis of the revenue of the firm, it forms the most important area of managerial economics. The success of the business firm depends largely on the correctness of the pricing decision taken by it. Therefore the business manager has to take utmost care in pricing decisions. He has to take into account the price determination theories in order to arrive at correct price decisions. He should know fully the market conditions, degree of

competition prevailing in the market and the steps to be taken for sales promotion. Pricing decisions, policies and practices cover such topics as price determination in different types of markets, pricing methods differential pricing product line pricing and price forecasting.

6. Profit Management: The primary aim of every business firm is to earn as much profit as possible. In fact, the success of the firm is measured in terms of profits that it has made and the rate of dividend that it has declared. But there is an element of uncertainty about profits because of variations in costs and revenues caused by internal and external factors. Therefore, the firms have to work under the conditions of uncertainties. Profit analysis would have been very easy talk, in case the future is certain and known.

7. Capital Management: Capital management implies planning and control of capital expenditure. There are three problems of capital management viz., accurate calculation of the profitability of capital investment, choice of capital investment and optimal allocation of capital. Capital management is the most complex and troublesome problem for the management involving high level decisions. Since the business is full of risks and uncertainties, capital investment requires utmost care. Relatively huge sums are involved and the problems relating to capital investment are so complex that their disposal needs considerable time and labour and top level decisions. Capital management covers such topics as cost of capital, rate of return and selection of projects.

8 Linear Programming and theory of Games: In recent years, there is a trend towards integration of managerial Economics and operations research. Therefore, techniques such as Linear Programming, inventory Control, Theory of Games etc., have also come to be regarded as part of the managerial economics.

9. Resource allocation: Resources are allocated according to the needs only to achieve the level of optimization. As we all know that we have scarce resources, and unlimited needs. We have to make the alternate use of the available resources. For the allocation of the resources various advanced tools such as linear programming are used to arrive at the best course of action.

Managerial economics makes use of the most modern tools of statistics, mathematics and operations research. The principles such as those in accounting, finance, marketing, production, demand forecasting demand analysis, production cost etc, are invariably used in the process of decision making and forward planning

Significance of managerial Economics

Managerial Economics, as a new and developing discipline, has been attaining increasing importance in the modern business world. Managerial economics is very often

called by different names such as Business Economics, economic Analysis for Business Decisions, Theory of the firm, Economics of the firm, Economics of the Enterprise etc, But of them, the names "Business Economics" and "Managerial Economics" are the most fitting ones of these two Managerial Economics is being increasingly used, in place of Business Economics The following points will explain the importance of managerial economics.

1. Tools and Techniques for Managerial Decisions: Managerial economics provides to the management important tools and techniques so essential for decision-making and forward planning. These tools and techniques are most relevant to the firm's decision-making process under the conditions of uncertainties. The managerial economist, with his knowledge of these tools and techniques, research and efficiency, occupies a very predominant place in the modern business world.

2. Useful Fundamental Concepts: Managerial economics provides very useful fundamental concepts or principles to the management of a business firm. These concepts or principles such as opportunity cost principle, incremental principle, time perspective principle, discounting principle and equi-marginal principle are very useful to the management in taking managerial decisions and formulating plans for the future. The knowledge of these concepts or principles is quite essential for demand forecasting, in analysing and solving managerial problems, and in increasing the efficiency and intelligence of the managerial economists.

3. Solution to basic problems: Managerial economics provides appropriate Solution to the following problems of decision making.

- a. What should be the product mix?
- b. What is the least cost production technique and input mix? i.e. what should be the ideal input output relation?
- c. What should be the price output policy?
- d. What should be an ideal investment policy?
- e. What should be the advertising or selling cost?

4. Business Data: Managerial Data are quite essential for solving the complex problems of decision making Managerial economics supplies such data to the management and helps quality of policy making and decisions.

5. Useful in forecasting: managerial economics is very useful in demand forecasting which is essential in planning. Managerial economics helps a business in product improvement product planning sales promotion, inventory control, manpower planning, etc.

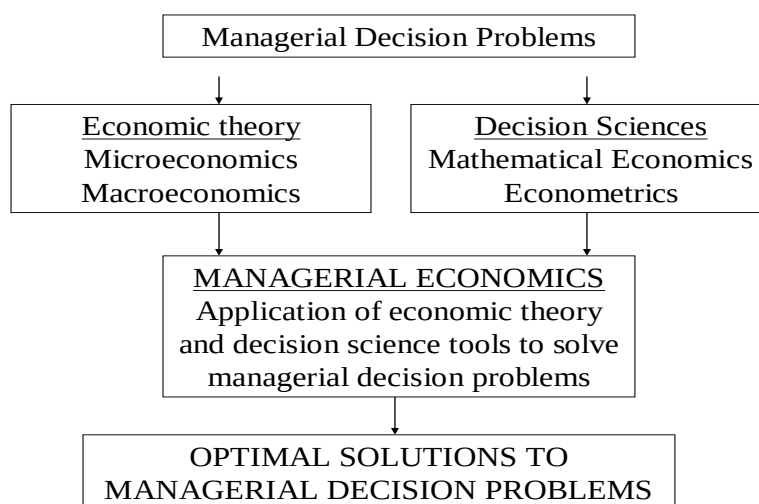
6. Guide to Managerial world: The function of managerial economist is not to take decisions by himself but to analyze conclude and recommend to the management. Thus managerial economics is both" Light giving and fruit bearing.

How is managerial Economics Useful?

A primary emphasis of managerial economics is the application of economic theory and methodology to the practice of business decision making. Because managers of not-for-profit and government agencies must efficiently employ scarce resources, managerial economics is an important tool for them as well. An important secondary emphasis in managerial economics is the study of how managerial decisions are affected by the economic environment. Managerial economics is applied economics; it is the use of economics theory and methodology to solve practical decision problems.

A Evaluating Choice Alternatives: Managerial economics links economic concepts with quantitative methods to develop vital tools for managerial decision making. Managerial economics identifies ways to efficiently achieve goals.

Making the Best Decision: To establish appropriate decision rules, managers must understand the economic environment in which they operate. Once management has set relevant goals, managerial economics can be used to efficiently attain those objectives. Managerial economics can be used to deduce the underlying logic of company, consumer, and government decisions.



Role and Responsibilities of Managerial Economist,

- Demand and supply estimation
- Preparation of sales forecasting
- Analysis of market survey for competition
- Advising management about adequate pricing, investment, budgeting etc
- Briefing management on current domestic and global economic issues
- Discovering new possible field of business, markets, product
- Cost benefit analysis

Objectives of the Firm

Each and every business firm strives to achieve some predetermined objectives. In the conventional economics emphasis was given to the profit maximization objective. In modern society, very few experts will argue that a firm is motivated by the sole objective of **maximization of profit**. In the modern days a firm invariably pursues multiple objectives even though one or some of them may receive priority over others. The objectives of the modern firm could be summarized under the following headings.

Maximizing or Satisficing

1. Profit $P = TR - TC$
2. Market share
3. Revenue growth
4. Return on investment
5. Technology
6. Customer satisfaction
7. Shareholder value

Baumol's Hypothesis of Sales Revenue Maximisation

Prof. William J. Baumol, American economist, does not agree with the traditional view that firms aim at maximizing profit. According to him, the objective of a modern firm is sales maximisation with a profit constraint. Sales maximisation does not mean an attempt to get largest possible physical volume of output. Here the sales means the revenue earned by selling the product.

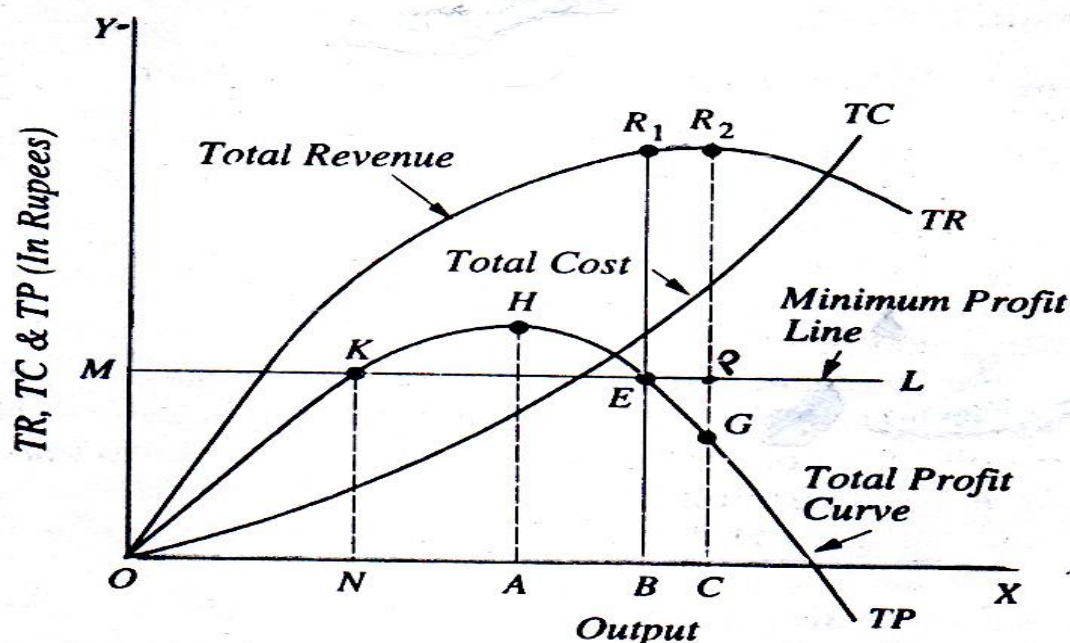
According to Baumol, "maximisation of sales revenue is an alternative to profit maximisation objective". The reason behind this objective is to clearly distinct ownership and management in large business firms. This distinction helps the managers to set their goals other than profit

maximisation goal. Under this situation, managers maximize their own utility function. According to Baumol, the most reasonable factor in managers' utility functions is maximisation of the sales revenue.

The factors, which help in explaining these goals by the managers, are following:

- Salary and other earnings of managers are more closely related to sales revenue than to profits.
- Banks and financial corporations look at sales revenue while financing the corporation.
- Trend in sales revenue is a good indicator of the performance of the business firm. It also helps in handling the personnel problems.
- Increasing sales revenue helps in enhancing the prestige of managers while profits go to the owners.

Managers find profit maximisation a difficult objective to fulfill, consistently over time and at the same level. Profits may fluctuate with changing conditions. salesmaximisation refers to the maximisation of the total revenue that measures the quantity of product sold in Rupee terms. It could be presented with the Support of the figure



In this figure X-axis measures the output and Y-axis measures the Total Revenue (TR), Total Cost (TC) and Total Profit (TP). OM is the minimum profit, which the firm intends to earn. With the increase in the output/sales level the TR goes on increasing up to a certain extent then start falling. Similarly TC goes on increasing with the increase in the sales level.

TP is the difference between the TR and TC; hence TP is the vertical distance between the TR and TC. If the firm intends to get maximum profit it has to produce/ sale OA quantity of output because TP curve is maximum at point H. On the other hand if the firm intends to maximize the sales it has to produce and sell OC amount of the commodity because TR curve maximum at point R₂. At OC level of output profit level (CG) is less than intended level (CP). According to Baumol the firm produce/sell OB amount of output. It maximizes the total revenue subjected to minimum profit shown by ML curve. BE is the profit earned by the firm at OB level of output.

Marris's Hypothesis of Maximisation of Firm's Growth Rate

According to this theory, modern firms are managed by both the manager and the shareholders. A manager aims to maximize the rate of growth of the firm and the shareholders will try to maximize the dividend and the increase the share price

According to Robin Marris, managers maximise firm's growth rate subject to managerial and financial constraints. Marris defines firms' balanced growth rate (G) as follows:

$$G = G_d = G_c$$

where,

G_d = growth rate of demand for firms product.

G_c = growth rate of capital supply to the firm.

In simple words, a firm's growth rate is considered to be balanced when demand for its product and supply of capital to the firm increase at the same rate. The two growth rates according to Marris, are translated into two utility functions such as:

- Manager's utility function
- Owner's utility function

The manager's utility function (U_m) and owner's utility function (U_o) may be specified as follows:

- $U_m = f(\text{salary, power, job security, prestige, status})$ and
- $U_o = f(\text{output, capital, market-share, profit, public esteem})$.

Owner's utility function (U_o) implies growth of demand for firms' products and supply of capital. Therefore, maximisation of U_o means maximisation of demand for a firm's products or growth of supply of capital.

According to Marris, by maximising these variables, managers maximize both their utility

function and that of the owner's. The managers can do so because most of the variables such as salaries, status, job security, power, etc., appearing in their own utility function and those appearing in the utility function of the owners such as profit, capital market, share, etc. are positively and strongly correlated with the size of the firm. These variables depend on the maximisation of the growth rate of the firms. The managers, therefore, seek to maximise a steady growth rate. Marris's theory, though more accurate and sophisticated than Baumol's sales revenue maximisation, has its own weaknesses. It fails to deal satisfactorily with the market condition of oligopolistic interdependence. Another serious shortcoming is that it ignores price determination, which is the main concern of profit maximization hypothesis. In the opinion of many economists, Marris's model too, does not seriously challenge the profit maximisation hypothesis.

Williamson's Hypothesis of Maximisation of Managerial Utility Function

Like Baulmol and Marris, Willamson argues that managers are very careful in pursuing the objectives other than profit maximisation. The managers seek to maximise their own utility function subject to a minimum level of profit. Managers' utility function (U) is expressed below: $V = f(S, M, ID)$

where,

S = additional expenditure on staff

M = Managerial emoluments

ID = Discretionary investments

According to Williamson's hypothesis, managers maximise their utility function subject to a satisfactory profit. A minimum profit is necessary to satisfy the shareholders and also to secure the job of managers. The utility functions which managers seek to maximise, include both quantifiable variables like salary and slack earnings and non-quantitative variable such as prestige power, status, job security, professional excellence, etc. The non-quantifiable variables are expressed in order to make them work effectively in terms of expense preference defined as satisfaction derived out of certain types of expenditures. Like other alternative hypotheses, Williamson's theory too suffers from certain weaknesses. His model fails to deal with the problem of oligopolistic inter dependency, Willinmsoli's theory is said to hold only where rivalry between firms is not strong. In case there is strong rivalry, profit maximisation is claimed to be a more appropriate hypothesis. Thus, Williamson's managerial utility function too does not offer a more satisfactory hypothesis than profit maximisation.

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Questions

3 Marks

- 1) What is Managerial Economics?
- 2) What is Micro Economics?
- 3) What is Macro Economics?
- 4) What is Managerial Economics?
- 5) 7) What is Industry?
- 6) 8) What is Firm?

7 Marks

- 1) Distinguish between Micro and Macro Economics
- 2) Explain importance of Managerial Economics.
- 3) Distinguish between Firm and Industry
- 4) What are the different Forms of Business Ownership?
- 5) Explain Profit maximization theory of the firm
- 6) Explain Baumol's model of the firm
- 7) Explain Williamson's model of the firm.

10 Marks

- 1) Explain Nature of Managerial Economics
- 2) Explain Scope of Managerial Economics
- 3) What are the steps involved in Decision-making process
- 4) Explain Roles and Responsibilities of Managerial Economists
- 5) Explain managerial theories of the firm
- 6) What are the alternative objectives of the firm?

Module 2: Demand analysis (9 Hours)

Law of Demand, Exceptions to the Law of Demand, Elasticity of Demand –Classification of Price, Income & Cross elasticity, Advertising and promotional elasticity of demand. Uses of elasticity of demand for Managerial decision making, Measurement of elasticity of demand. Law of supply, Elasticity of supply, Demand forecasting: Meaning & Significance, Methods of demand forecasting. (No problems)

Demand Analysis

Introduction: The concept of demand studies about the general behavior of the buyer during purchase. The general behavior of the buyers during purchases is such that with the fall in the price of the commodity they buy more and with rise in the price of a commodity they buy less. This behavior has been studied and named in economics as Law of Demand.

The law of demand is an experience of every seller and prudence of every buyer is a famous law in economics developed systematically by Prof. Marshall. Every seller knows that he can sell more when he reduces the price and that the buyers purchase less when price is high. Similarly every buyer purchases less when prices are high and more when prices are low. Maybe a buyer can not purchase when a price is high or when price of one good becomes cheaper. For a common man. For the poor in general, when a commodity becomes costlier buying the same quantity as earlier is either impossible or requires sacrifice of purchase of some other commodity. Simply stated the law establishes an inverse relation between price and demand.

Meaning of Demand : Demand is an important concept in Economics. The theories associated with demand occupy an important place in economic analysis. The modern theories of managerial economics and business management find their basis in demand analysis.

In our day to day life we do not much difference between desires. Wants, needs and demand. Economics attaches a specific meaning to the concept of demand. The word demand has a greater scope than other words. It indicates the willingness of a buyer to possess a good and his readiness to pay the price. It indicates his ability to pay the price. Therefore demand is willingness plus ability. Demand is a claim on a commodity because the buyer is not only willing but also able and ready to pay the price. Demand Refers to the quantity of the commodity which a individual household is willing to purchase in a market at a point of time at a particular price. The demand for a commodity arises from consumers willing or desire.

Mere desire does not constitute demand in economics. Three crucial characteristics of Demand are Wants or

- Desire to acquire it
- Purchasing power or income
- Willingness to spend on that

Demand will not exist for any product in the absence of any of three character mentioned above

Law of Demand: The law of Demand indicates the relationship between the price of a commodity and the quantity demanded in the market. It may be stated as follows “**Other thing being equal, the quantity demanded extends with a fall in price and contracts with rise in price**”. That is to say, the quantity demanded varies inversely in price. In simple language it means that a person will purchase more of a commodity when its price falls and he will purchase less of it when its price rises. The law rests upon from logic. It tells the direction of change in demand with the change in price. ‘Other things remain being equal’ is very important qualifying phrase in law. This means that this law holds good only under static condition, and under dynamic condition, there are other factors which may influence the demand irrespective of change in price. According to Samuelson Law of Demand states that “People will buy more with fall in price and buy less with rise in price”



Demand Schedule

The Law of Demand can be explained with the help of demand schedule. Demand schedule is a table showing the arithmetic relationship between price and corresponding quantities of commodity demanded. In other words Demand Schedule indicates various quantities demanded at different levels of price

An Individual's Demand schedule for product X

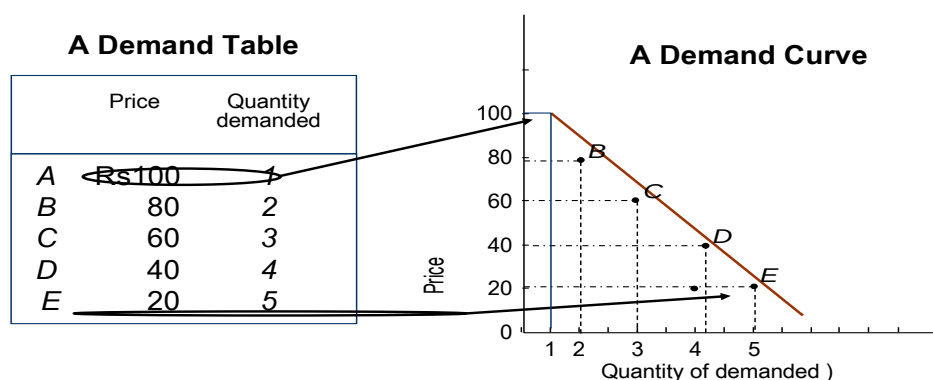
Price In RS	Quantity Demanded in Units
100	1
80	2
60	3
40	4
20	5

It is evident from above statement that when the price becomes the least the quantity demanded is highest and it declines as the price per unit of product increase. This relation ship can be depicted in the form of graph which is called a demand curve

Demand Curve

The demand curve is pictorial representation of the demand schedule which indicates price and quantity demanded at various level following demanded curve constructed based on the scheduled given below

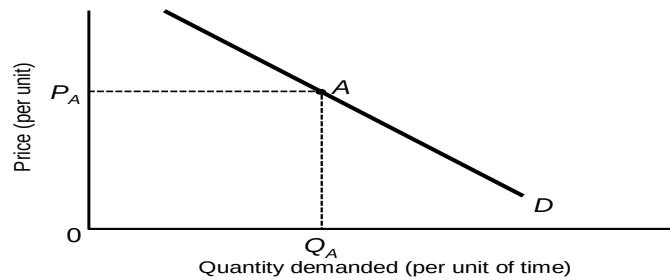
From a Demand Table to a Demand Curve



Independent Variable price which determines the demand is shown on Y axis and dependent variable demand is shown on X axis in terms of Units respectively In the above graph D is the demand curve which has a negative slope from left to right indicating that whenever price rises, demand falls and when price falls demand rise

A Sample Demand Curve

A Sample Demand Curve



Reasons for the Law of Demand

Why does the Law of Demand apply? Why is the Demand Curve negatively-sloped? There are some reasons given for the inverse relationship between price and amount demanded in case of ordinary commodities. These reasons are as follows:

1. Law of Diminishing Marginal Utility. Writers who believe in cardinal utility approach to consumer's demand believe that diminishing marginal utility for the consumer in case of the commodity is the fundamental cause of the Law of Demand. As the price of the commodity falls, consumer purchases more of the commodity so that his marginal utility from the commodity also falls to equal the reduced price. If the price rises, the opposite happens.

2. Substitution Effect. Another group of writers who believe in ordinal utility (Indifference curves) consider the substitution effect of the change in price as the major cause for the application of the Law of Demand. When the price of commodity falls, it becomes cheaper as compared to the other commodities which the consumer is purchasing. As a result the consumer would like to substitute this cheaper commodity for other commodities whose prices remain the same for example, with the fall in price of tea, coffee's price remaining the same, tea will be substituted for coffee. In other words, the demand for tea would go up. This is nothing but the application of the Law of Demand.

3. Income Effect. Another cause behind the Law of Demand is known as Income Effect. As the price of a commodity is reduced, the consumer has to spend less amount of money income for the same amount of the commodity. This may be taken to be a rise in his real income. It is the Income Effect of a fall in price. Part of the increase in his real income can be used to purchase more of the cheaper commodity while the other part may be spent on other goods. Thus, when the price falls, amount demanded rises and vice-versa.

4. New Consumers. When the price of a commodity is reduced then many other' consumers who were not consuming the commodity earlier will start purchasing It now because it is

within their reach now. For example, radio sets have become cheaper and even poor people can easily buy a set the amount demanded of the radio-sets has gone up with a fall in their price. The opposite would happen with a rise in prices.

5. Different uses of the commodity. Commodities have many uses. If their price rises, they are used only for the more important purposes. As a result, their demand will go down. On the contrary, when the price is reduced, the commodity will be put to many other uses where it was not being used earlier. Its demand will go up.

Individual and Market Demand Curves

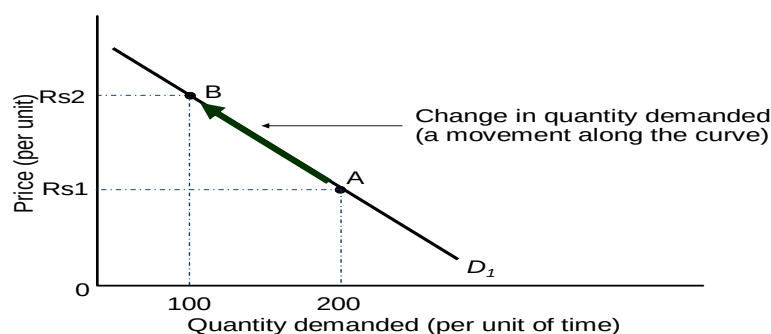
Demand may be expressed as individual demand and market demand. Individual demand is an individual's demand for a product at different prices. The market demand is the sum of the demand of all individuals in a market. Both individual and market demand have an inverse relationship with price.

Individual and market demand.

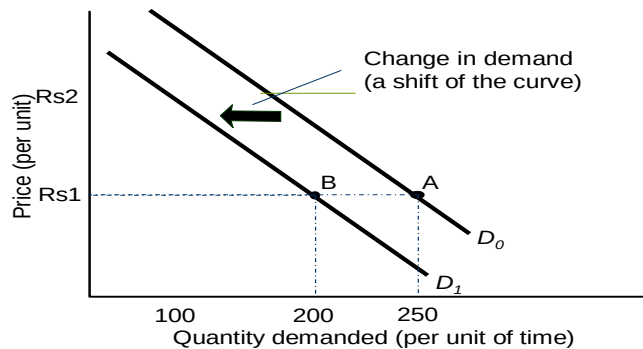
Price Rs	Individual demand					Market Demand
	A	B	C	D	E	
10	5	2	10	4	1	22
8	6	3	12	6	2	29
6	7	4	15	8	3	37
4	8	5	20	9	4	46
2	10	6	25	10	5	56

When prices fall both the individual and market demand increase. When prices increase both of them decline. A movement along a demand curve is the graphical representation of the effect of a change in price on the quantity demanded.

Change in Quantity Demanded



Shift in Demand : A shift in demand is the graphical representation of the effect of anything other than price on demand



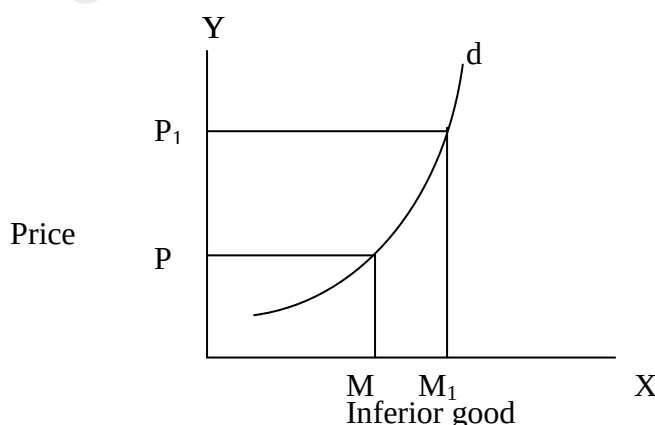
Shift Factors of Demand

Shift factors of demand are factors that cause shifts in the demand curve: Society's income. The prices of other goods., Tastes. Expectations. Taxes on subsidies to consumers. Income: An increase in income will increase demand for normal goods. An increase in income will decrease demand for inferior goods.

Exception to the Law of Demand

The Law of demand state that the price and quantity demanded of a commodity are inversely related. However there are certain peculiar cases in which the law does not hold good. Some time more quantities may be demanded at high price s and less may be demanded at low prices. Such instances are called Exception to the law of Demand.. The following are the important cases where the law does not hold good.

1. Giffen Effect or Giffen Paradox: Sir Robert Giffen found that people buy less commodity at a lower price and more amount of it higher price. People consuming large amount of inferior goods and less amount of superior goods will buy more inferior goods when their price increases by stopping or reducing the purchase of superior goods. In India, such food grains as jowar and bajra are Giffen goods, wheat and rice being the superior goods.



Economics for Decision Making (MBA103)

As shown in the diagram inferior goods have an exceptional demand curve – sloping upwards. It indicates that more is demanded at higher price and only less is demanded at lower price. This is true of all inferior goods. Many times price could be a measure of the quantity and importance of a good. When price declines a consumer may treat it inferior and purchases less when price is low. With changes in income and economic standards of the people new articles are entering the list of inferior goods.

Veblen Effect: Veblen an American Economist in his article “Conspicuous consumption” identified one more Exception to Law of demand. There are products which are purchased mainly for their Snob-appeal or false prestige. The people will buy more product in higher price

Example: when the price of certain product increases rich people may buy more of it for sake of social status or prestige, as they become more attractive when the price increases, if such product become cheaper rich would not like to buy them. Example of veblen goods are Diamonds and products of status symbol

Speculative Effect : In speculative market people expect a price hike to be followed by further rise in prices. Therefore when price rises they buy more thinking that future price will more than present price and give more profit when sales are made. On the other hand when price falls they buy less or wait for further reduction in price. Example in the stock market with the rise in price of share there will be rise its quantity demanded and with fall in price the quantity will also fall. Thus even speculation effect is also an exception to the law of Demand.

Scarcity Effect: In times of scarcity , which may arise due to war or emergency, the general public will demand the commodities with rise in price. Thus scarcity effect also exception to the law of demand . The best example for scarcity effect during gulf war between Iraq and Israel with rise in price of petroleum product demand continued.

Ignorance Effect : Consumer are very much ignorant either for a small fall in price or for a small rise in price their behavior is guided by taste, habits and preference, environment etc.

Brand loyalty : If buyer are attached to their favorite brand they will neglected the rise in price and they continued to buy even with rise in price With all these limitations the law of demand is a guiding principle of various area of economics. Consumer during their day to day purchase adopt this law.

Elasticity of Demand-The Concept of elasticity of demand is an important concept in economics. It is found very useful in the analysis of demand and supply, price determination

or value theory, International trade, taxation and a host of other fields. The concept was developed by Prof. Marshall

The law of demand studies the relationship between demand on one side and price on the other. Change in price brings changes in quantity demanded. The law of demand tells us the **direction of change**. But it does not tell about quantum of change. We do not know how much the quantity changes with the changes in price. This is explained by the elasticity of demand. Elasticity of Demand can be defined as the “**the degree of responsiveness of quantity demanded to a changes in price**” Elasticity of demand is the ratio of relative changes in two variables. It measure the responsiveness of one variable to the changes in another. The concept of elasticity measures the degree or magnitude of changes in demand due to changes in the independent variables like price, income, price of related goods etc. The law demand which states an inverse relationship between price and quantity demanded indicate only the downward movement of demand curve, however it is salient as to say how much demand increase or decrease to a given changes in price, Elasticity of demand describe the amount of change in price and corresponding movement of the demand curve and its shape.

$$\begin{aligned} \text{Ed} &= \frac{\text{Percentage change in Demand}}{\text{Percentage change in price}} \\ &= \frac{\% \Delta D}{\% \Delta P} \end{aligned}$$

Now suppose the price of a commodity is Rs 10 and the present demand is also 10 if the price is increased to Rs 11, the demand immediately responds to it. Consequent upon this increase in price demand contracts. This is the responsiveness or elasticity of demand. Now the percentage change in quantity demanded is 10 and the percentage change in price is also 10

$$\text{Then Ed} = \frac{\% \Delta D}{\% \Delta P} = \frac{10}{10} = 1$$

Therefore Ed = 1

As is dear from the above example a 10% change in price has brought about a 10% change in demand. Or in this particular case demand changes 10% in response to 10% change in price therefore elasticity of demand is 1 this is also called Unitary elasticity of demand when Ed is 1 it indicates that demand changes in the same proportion as price changes.

Elasticity of demand for different goods may be different and at different times it may be different for the same commodity Elasticity is bound to be different for different in come categories. It may vary from person to person, place to place and time to time.

Types of elasticity

1. Price elasticity - (PEd) :-

It refers to responsiveness of demand to a given change in price. If for example price changes 10% and demand changes by 10%

$$\text{then PEd} = \frac{\text{Percentage change in quantity demanded.}}{\text{Percentage change in price 10\%}}$$

$$\text{In our example } \therefore \text{PEd} = \frac{10\%}{10\%}$$

This indicates that D changes in the same proportion as price changes P10%, D10% PEd is greater than 1 when $\Delta D > \Delta P$ P 5%, D 10 %

and similarly PEd is less than 1 when $\Delta D < \Delta P$ D 5% P 10%

PEd is very useful in pricing policy.

1) Unitary elasticity or PEd = 1 :-

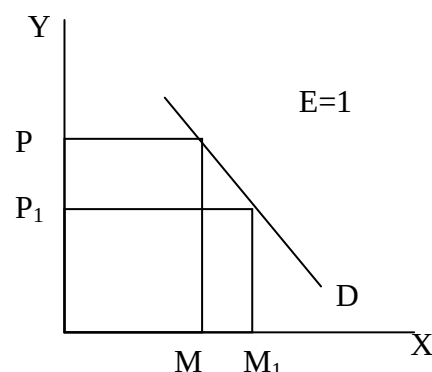
Price elasticity is said to be unitary or equal to 1 when percentage changes of D and P are the same. Suppose quantity demanded changes by 10% in response to 10% change in P then

$$\text{PEd} = \frac{\% \Delta D}{\% \Delta P} = \frac{10\%}{10\%} = 1$$

This means that the demand changes in the same proportion as price changes (Though in opposite direction) this can be explained with the help of the following diagram As shown in the diagram price declines from OP to OP₁, consequently demand increases from OM, to OM₁. Now change in demand is equal to MM₁ and change in price is PP₁ therefore,

$$\text{Ed} = \frac{\% \Delta D}{\% \Delta P} = \frac{MM_1}{PP_1} = 1$$

Since MM₁ is equal to PP₁ PEd = 1



2 Elastic demand :ELASTICITY GREATER THAN 1) It is a situation where a small change in price leads to drastic change in the quantity demanded . The demand changes greater than the proportionate change in price . In this case the value of co-efficient is greater than one ($ED > 1$) and less than infinity as a result of which the demand curve will be flat line. Relatively elastic demand - some times demand for a good may be very sensitive to changes in P. Therefore demand may contract even if the price increases a little. Similarly a small fall in the price may cause a big change in D. In such cases % change in D is greater than % change in P. For instance if a 5% change in P brings about a 10% change in D then

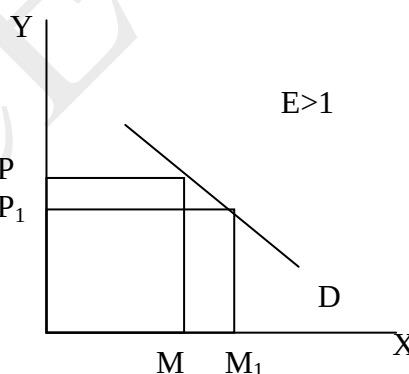
$$Ed = \frac{\% \Delta D}{\% \Delta P} = \frac{10\%}{5\%} = 2$$

It means that a given change in P brings about twice the % change in Demand.

As shown in the diagram change in price is PP_1 and change in demand is MM_1

$$\therefore ED = \frac{MM_1}{PP_1} > 1$$

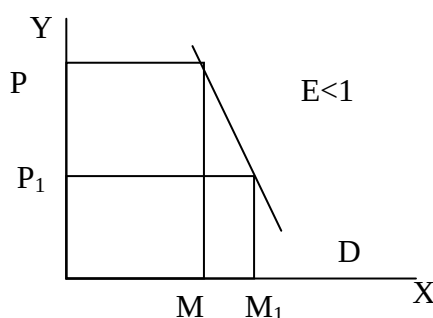
Change in demand MM_1 is greater than change in price. Therefore demand is relatively elastic or $Ed > 1$.



In real market situation luxuries fall in this category. A small reduction in the price of luxuries will lead to a large increase in their demand vice-versa

1) **Elasticity less than 1** or Relatively inelastic demand -Some times demand may be less sensitive to changes in P. It is rigid and sticky. In such cases the % change in D is less than the % change in P. Hence demand is said to be inelastic or elasticity less than unity or 1. This is true of necessities and luxuries. Price does not much influence the quantity demanded. Therefore a 10% change in P may change demand by only 5% then

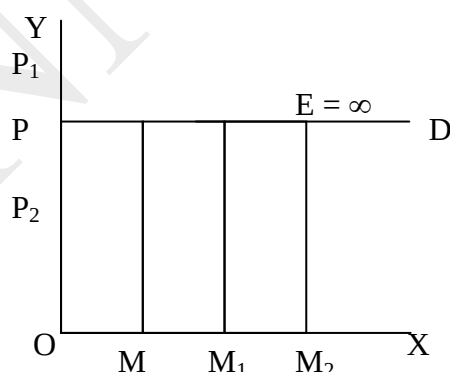
$$Ed = \frac{\% \Delta D}{\% \Delta P} < 1$$



$$\frac{MM_1}{PP_1} < 1$$

In case of food items, only because the price has declined or 'increased the demand neither increases or decreases. This is true of cars, TV sets or refrigerators too. **4) Perfectly Elastic** or infinitely elastic demand. It is a rare and special case, sometimes a small increase or decrease in P may bring about infinite' change in D. In case of monopsony in particular there would be a single buyer. He would be ready to buy any quantity of a commodity at a particular price. Even if the price changes a little he stops buying.

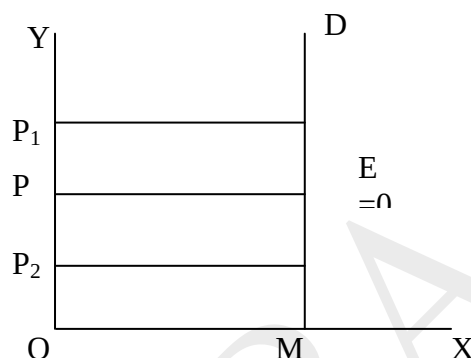
As in the diagram at OP price demand is infinite and the buyer is ready to buy M m₁ m₂ or any quantity. If the price changes to P₁ or P₂ the demand would be zero. Thus demand is highly sensitive to price and the seller would be advised not to make any change in price. In this case demand curve is a horizontal straight line parallel to OX axis



5) Perfectly Inelastic or $E_d = 0$ or zero elasticity. It is also a rare and a special case. In "case of bare necessities, medicines rare commodities, goods of prestige, goods without substitutes have perfectly inelastic demand. This indicates that responsiveness of demand to a change in price is zero. Hence change in D is zero whatever is the change' in p. Suppose there is a 10% change in price and 0 % change in Demand 'then $E_d = 0$. In such a case demand curve

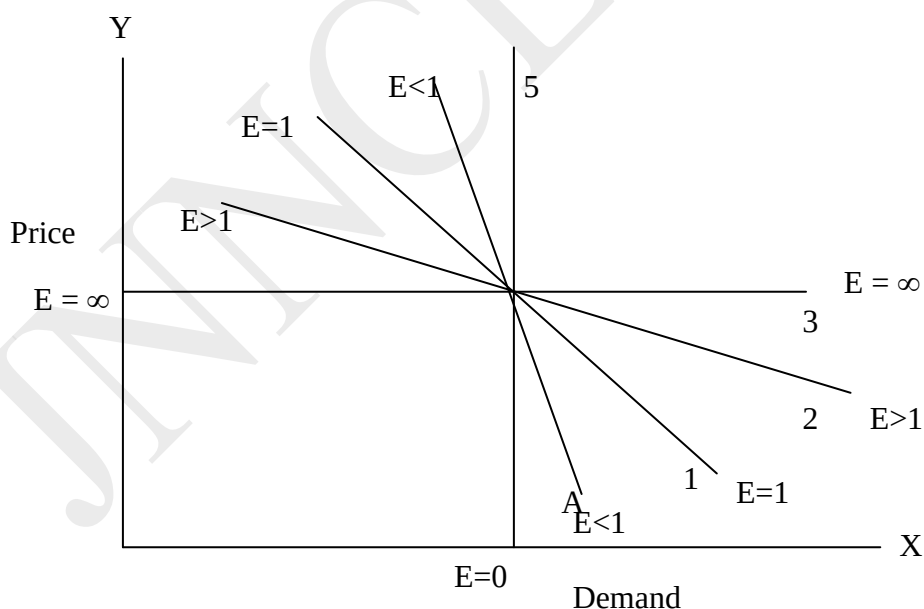
becomes a vertical straight line parallel to OY axis. As shown in the diagram change in price is PP_1 or PP_2 and change in D is zero.

$$\therefore Ed = \frac{M}{PP_1} \text{ or } \frac{M}{PP_2} = 0$$



Thus there are five types of price elasticity of demand. Generally we can make a mention of highly elastic and highly inelastic demand also. When elasticity of demand is more than two and nearing infinity it is called: elastic. Similarly when it is near zero it is called highly inelastic. However these two varieties are not much different from elastic and inelastic demand.

All these five types of elasticities can be summarized and shown in a single diagram as follows.



1. Unitary elasticity: This curve is in the middle of perfectly inelastic (5) and perfectly elastic demand (3) curves.
2. Elastic demand: the curve is in between unitary elasticity(1) and perfect elasticity(3)

3. Perfectly elastic demand - it is horizontal straight line parallel to OX axis (3)
4. Inelastic demand - It is in between perfectly inelastic (5) and unitary elasticity.(1)
5. Perfectly inelastic demand - the curve is a vertical straight line.(5)

Thus on the basis of the slope of a demand curve itself one can be in a position to estimate the elasticity of demand.

Factors that determine Elasticity.

Elasticity of demand varies from commodity to commodity and even for the same commodity from time to time and place to place. The determinants of elasticity are as follows.

1. Nature of the goods and services Necessaries will have to be purchased in a minimum quantity irrespective of prices. Therefore their elasticity is less. The demand for comforts and luxuries can be postponed also. They are purchased more when prices fall. Therefore their elasticity is great.

2. Availability of substitutes: If for any commodity substitutes are available the consumer shifts his demand to the substitutes when the price increases. This makes demand fairly elastic. If the substitute is not available he will have to buy even at a higher price. This makes demand less elastic or inelastic.

3. Extent of use: There are certain goods which satisfy different types of wants. (Electricity) When price is high it is used only for essential purposes and demand is limited. When prices decline demand increases even to meet non essential purposes. Therefore greater the extent of use greater is the elasticity and vice versa.

4. Possibility of postponement of demand: The demand for car, TV etc., can be postponed. The demand for food grains, medicines etc, cannot be postponed. In case of the former demand can be postponed if prices increase. Therefore demand is elastic. In case of the latter demand cannot be postponed even if prices increase. Therefore demand becomes inelastic.

5. Price Range: There are three ranges of prices. Certain goods like car, TV, Bike, etc, belong to high range. Goods like cloth oil, food grains etc are in the middle range. There are many goods in low range of prices. Goods of high range of prices are generally purchased by the rich and the demand is inelastic. Goods of middle range of prices may have elastic demand. Certain cheaper goods are consumed by the poor only. Therefore demand becomes fairly elastic.

6. Proportion of income spent: when proportion of income spent on any commodity is small the consumers don't bother about price changes. Hence demand is inelastic.

7. Time: During a short period of time changing the consumption pattern or finding out a substitute is difficult. Therefore Demand is inelastic During a long period of time a consumer can have both the alternative and therefore demand becomes elastic.

8. Environmental conditions, habits, customs etc, also influence elasticity of demand. If environment habits, and customs make the consumption of a good compulsory then demand becomes inelastic.

Thus there are many factors that govern elasticity of demand for any commodity. Only after considering all these factors we may come to any definite conclusion about elasticity of demand.

Importance of Price Elasticity

1. Price determination: While fixing the price for his commodity a producer or a seller should carefully study the elasticity of demand for his product. Reducing the price while demand is elastic would increase his profit. Similarly when demand is inelastic increasing the prices may be permissible. He will also have to study elasticity of substitutes and complementary goods of his product so that he can maximize sales and profit.

2. It is very useful to the governments while imposing taxes on goods. Taxes increase the price. If demand for it taxed commodity is elastic then people buy less of it and governments revenue may decline. Government revenue may increase only when taxed commodity has inelastic demand.

3. Determination of wages: When the demand for labour is elastic wages are bound to be low. In contrast to it when demand for labour is inelastic wages may be high. Trade unions succeed in getting a wage hike only when demand for labour is inelastic.

4. In international trade: Elasticity concept is useful in measuring gains from international trade. If our demand for foreign goods is elastic and foreigners demand for our goods is inelastic the country gains in terms of better terms of trade.

5. Determination of public utilities: The demand for health, education, transport and communication etc is bound to be inelastic. Then they will have to be well constructed and managed by the governments so that the people's welfare increases.

2. Income elasticity of demand:-

There is a direct relationship between income and demand. An increase in Y leads to an increase in D and a decrease in Y leads to a decrease in D. Income elasticity is responsiveness of demand to a given change in income. This is expressed in percentage terms.

Percentage change in D % Δ D

$$Y_{Ed} = \frac{\text{Percentage change in D}}{\text{Percentage change in Y}} = \frac{\% \Delta D}{\% \Delta Y}$$

If a 10% increase in income leads to a 10% increase in D then

$$Y_{Ed} = \frac{10\%}{10\%}$$

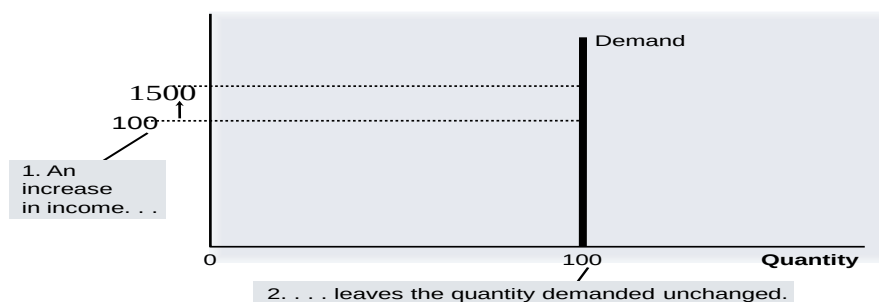
when $Y_{Ed} = 1$ it indicates that demand increases by the same proportion as income increases. Income elasticity of D may be greater or smaller than 1 when $\% \Delta D > \% \Delta Y$ Y_{Ed} is greater than 1 similarly when $\Delta D < \Delta Y$ Y_{Ed} is less than one. The knowledge of Y_{Ed} is very useful in price and out put determination, marketing taxation etc.

Types of Income Elasticity of demand

1. Zero income Elasticity
2. Negative Income Elasticity of demand
3. Positive Income Elasticity of demand

Zero income Elasticity : Zero Income elasticity indicates that any increase or decrease in consumer's income will not reduce the demand for a commodity. In other word demand remains the same regardless of the change in income and the co-efficient is zero. There are certain product like Salt or match box which are purchased without considering the changes in income.

Zero income Elasticity



Negative Income Elasticity of demand : Negative Income Elasticity refers to a situation in which an increase in consumers income leads to decrease in the quantity demanded of a commodity. It happens mainly in case of inferior goods , when income rises people shift demand from inferior to superior goods example shifting from inferior cloth to fine quality cloth.

Positive Income Elasticity of demand : The positive income Elasticity refers to an increase in quantity demanded due to increase in income of the consumer. Demand for superior goods always positive as an increase in income shift the demand from inferior to superior goods

3. Cross Elasticity of demand:-

It measures the relationship of price change of one and demand change of another commodity. There is a direct relationship between P&D when the two goods are substitutes. Similarly when two goods are complementary to each other the would be inverse relationship between price of one commodity & the demand for another commodity

$$\text{Therefore CED} = \frac{\% \text{ change in D for x}}{\% \text{ change in P of y}} = \frac{\% \Delta D_x}{\% \Delta P_y}$$

CED may also be 1, greater than 1 or less than one depending upon change in P & D. Suppose P of Y changes by 10% and D of x changes by 10% then CED = 10/10 =1.

If P of Y changes by 15% & D of x changes by 10 % then CED = 10÷15 = 0.66 etc
This analysis is useful in production and pricing policy

Advertising or promotional elasticity of demand- Modern firm depend heavily on 'advertisements' of their products in order to increase their sales . They face competition in selling their product. The expenditure on Advertisement creates an impact on demand condition .Advertisement is method of sales promotion and competitive advertisement have come to play an important role in business. Advertisement elasticity demand is the ratio of percentage change in quantity demanded of a commodity to percentage change in advertisement outlay or budget on the commodity. It measure impact on a change in advertisement or promotional expenditure on the sales of a product of a company. It indicates the effectiveness of advertisement expenditure in creating market.

It can be calculated with the formula

$$E_a = \frac{\Delta Q}{\Delta AE}$$

Where

Ea=Advertisement or promotional Elasticity of demand

ΔQA =Changes in quantity demanded or sales

ΔPB =Change in the advertisement or promotional expenditure

Advertisement elasticity of demand is always positive as the advertisement are expected to increase sales .

The following are the major features of advertising Elasticity.

If other thing remained constant sales have direct relationship with advertising Increased advertisement will lead to increase in demand upto a certain point at a rate more than proportionate increase, but beyond that point increase in demand will be at a rate less than proportionate and after saturation there will not be any increase in demand to increase in advertisement.

Law of Supply

Introduction to the Law of Supply:

The law of supply reflects the general tendency of the sellers in offering their stock of a commodity for sale in relation to the varying prices.

It describes seller's supply behaviour under given conditions. It has been observed that usually sellers are willing to supply more with a rise in prices.

The law of supply may be written as follows:

“Other things remaining unchanged, the supply of a commodity rises i.e., expands with a rise in its price and falls i.e., contracts with a fall in its price.

In other-words, it can be said that—”Higher the price higher the supply and lower the price lower the supply.”

The law thus suggests that the supply varies directly with the change in price. So, a larger amount is supplied at a higher price than at a lower price in the market.

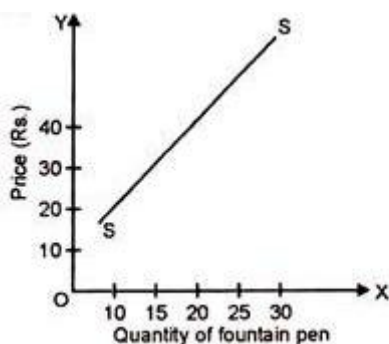
Explanation of the Law:

This law can be explained with the help of a supply schedule as well as by a supply curve based on an imaginary figures and data.

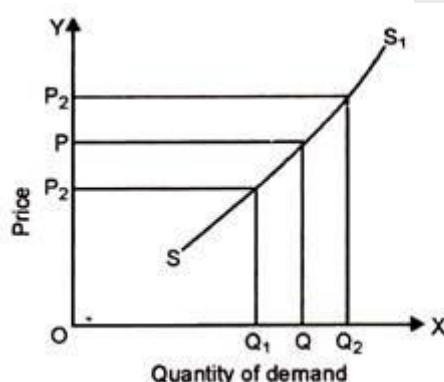
Market Supply Schedule

Price of Fountain Pen Rs.	Quantity Supplied (in '000 per week)
10	10
20	13
30	20
40	25

This can be shown by diagram as follows:



Here, in this diagram the supply curve SS is sloping upward. It suggests with the supply schedule, that the market supply tends to expand with the rise in price and vice-versa. Similarly, the upward slopping curve also depicts a direct co-variation between price and supply.



This law can be shown in this way also.

In the figure above OX axis shows quantity of demand and OY axis shows price. SS_1 line is the line of supply when the price of the commodity is OP then quantity of supply is OQ.

When the price rises from OP to OP_2 and then supply also rises from OQ to OQ_2 . Similarly, if price is reduced from OP to OP_1 , then supply will reduce from OQ to OQ_1 .

By seeing the diagram the conclusion can be drawn that when price rises supply increases and when the price reduces the supply reduces.

Assumptions Underlying the Law of Supply:

Important assumptions of the law of supply are as follows:

1. No change in the income:

There should not be any change in the income of the purchaser or the seller.

2. No change in technique of production:

There should not be any change in the technique of production. This is essential for the cost to remain unchanged. With the improvement in technique if the cost of production is reduced, the seller would supply more even at falling prices.

3. There should be no change in transport cost:

It is assumed that transport facilities and transport costs are unchanged. Otherwise, a reduction in transport cost implies lowering the cost of production, so that more would be supplied even at a lower price.

4. Cost of production be unchanged:

It is assumed that the price of the product changes, but there is no change in the cost of production. If the cost of production increases along with the rise in the price of product, the sellers will not find it worthwhile to produce more and supply more. Therefore, the law of supply will be valid only if the cost of production remains constant. It implies that the factor prices such as wages, interest, rent etc., are also unchanged.

5. There should be fixed scale of production:

During a given period of time, it is assumed that the scale of production is held constant. If there is a changing scale of production the level of supply will change, irrespective of changes in the price of the product.

6. There should not be any speculation:

The law also assumes that the sellers do not speculate about the future changes in the price of the product. If, however, sellers expect prices to rise further in future, they may not expand supply with the present price rise.

7. The prices of other goods should remain constant:

Further, the law assumes that there are no changes in the prices of other products. If the price of some other product rises faster than that of the product in consideration, producers might transfer their resources to the other product—which is more profit yielding due to rising prices. Under this situation and circumstances, more of the product in consideration may not be supplied, despite the rising prices.

8. There should not be any change in the government policies:

Government policy is also important and vital for the law of supply. Government policies like—taxation policy, trade policy etc., should remain constant. For instance, an increase in or totally fresh levy of excise duties would imply an increase in the cost or in case there is fixation of quotas for the raw-materials or imported components of a product, then such a situation will not permit the expansion of supply with a rise in prices.

Exceptions to the Law of Supply or Backward-Slopping Supply Curve:

As we have seen from the study above that supply of a commodity varies directly with its price. But in some exceptional cases where supply may tend to fall with the rise in price or tend to rise with the fall in price.

Such exceptional cases may be described as follows:

1. Exceptions about Future Price:

In this connection if the seller expects a rise in the price in future, he may withhold his stock of the commodity. He will therefore reduce his supply in the market at the present price. Similarly, if he expects a further fall in price in future, he will try to dispose of the commodity and will supply more even at a lower price.

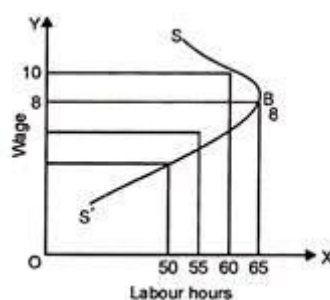
2. Supply of Labour:

Supply of labour after a certain point, when the wage rate rises, its supply will tend to diminish. Why such situation because workers normally prefer leisure to work after receiving a certain amount of wage.

3. Rate of Interest and Savings Position:

When there is rise in the interest rate, more savings are induced. But after a certain point of rise in the rate of interest households may tend to save less than before due to high income from the interest. In that case savings tend to decline even with a rise in the rate of interest.

From the points written above we can observe that the supply tends to fall with a rise in prices at a point. This paradoxical situation of supply behaviour is represented by a backward sloping or regressive supply curve over a part of its length as shown in the figure given below:



In this diagram SS' shows the relationship of supply with price. Backward slopping supply curve BS ' part represents supply curve is bending at B. This curve is also known as an "Exceptional Supply Curve" as such a thing happens only in some exceptional cases like—labour supply or savings.

Further, in this diagram SBS' represents a backward slopping supply curve for labour as a commodity. Here the wage rate has been regarded as the price of labour and the labour supply is determined in terms of Labour-Hours the worker is willing to work at a given wage rate. It has been observed that as wages increase, a worker might work for a lesser number of hours than before.

Demand Forecasting

Modern business produces goods in anticipation of future demand. No producer gets the order first and then starts the process of production. On the basis of cost experiences, sales trends of the past, the fashion of the day and psychology of consumers, markets surveys etc., he makes a forecast of future demand for his product. According to his forecast he will plan the use of resources, production and sales, borrowing and investments etc., therefore demand forecasting is scientific estimation of future demand based on statistical data. Without such a forecast a manufacturer may have to undergo huge losses if overproduction occurs. Demand forecasting is therefore a systematic exercise and not wishful thinking.

Significance:

1. Production planning of a business should be based on estimates of future demand. Otherwise the business may have to face the problem over or under production.
2. Sales forecasting and sales promotion efforts are to be based on the demand forecast.
3. Control of business requires a reliable information about future demand .cost of production, profit etc.,
4. Inventory control: of business inventories, raw materials, semi finished and finished products spare parts etc., requires the accurate knowledge of future demand.
5. Demand forecasting is necessary for determining the growth rate and long term investment programmes of business.
6. Stability: Stability in production and employment over a long period of time needs appropriate steps to counterbalance cyclical fluctuations in business.
7. Economic planning and policy making: for planning the use of resources and national allocation of resources and for deciding upon import and export policies demand forecasting is essential.

Methods of Demand Forecasting

Over the years economists and statisticians have developed various methods of forecasting. One has to select the appropriate method depending upon the purpose, term, product and date of available. Basically there are two methods:

- 1) Survey methods
- 2) Statistical methods.

Survey Methods:

1. Consumers interview methods
2. Collective opinion method
3. Experts opinion method
4. End- use method

Statistical methods:

1. Trend projection method
2. Economic indicators

Survey Methods:

These methods are extensively used in short run and in estimating the demand for new products. Information is gathered from purchase plans of potential buyers through expert opinions or by interviewing the consumers.

1. Consumer interview method: In this method the consumers are directly contacted for information about their future plans to buy. There are different methods adopted to gather information.

A) Survey of buyer's intension or buyers preferences

This is also called as Opinion Survey. To conduct the survey field staff are employed or services of research organizations are hired. Experts in the field of marketing prepare a questionnaire to get information directly or indirectly. The questionnaire contains various questions to estimate buyers intension. The information so collected from consumers is evaluated by experts top executives. Inferences are drawn for taking final decisions.

The success of the method depends upon

1. The nature of the questions
2. The ability of the consumer
3. Nature of the product
4. Characteristic of the market and consumers
5. The techniques of Analysis

The opinion expressed by consumers cannot be taken as [mal. In the light of experiences the business executives will have to arrive at judgments and make regular revision of survey.

(B) Direct Interview Method:-

It is an alternative to survey of buyers intention. As the name suggests the consumers are directly contacted and information is gathered by asking simple budget plan of expenditure, quality, quantity of products and the price they are ready to pay.

Direct interview is done in two ways a) complete enumeration all buyers in a particular region or city are directly contacted and information is collected. This gives us complete picture and is the most reliable. But it is costly and time consuming.

In sample survey only a few potential consumers are selected from the cross section of consumers.

They are asked directly or through a questionnaire about their preferences. The information so gathered is analysed and a quick forecasting is done.

2) Collective opinion method or Opinion survey method:-

This method is also called "sale force polling" "Opinion poll method" under this method sales representatives, experts and market consultants are asked to give their opinions about the volume of trade expected in the future. Since salesmen of sales representatives have close contact with consumers of retailers they can estimate the opinions of customers. Such opinions are systematically analysed to forecast future sales. With this the opinions of experts is also considered. These opinions are revised time to time with reference to selling price, product design, advertisement degrees of competition, pop, destination, of income etc., and demand forecasting is done.

3. Experts opinion or Delphi Method :-

It is a variant of opinion poll and survey method of forecasting .In this outside experts or research organizations are appointed they are supplied with all kinds of information and data. The experts give their expert opinions are more valuable and a firm will prepare its future plan accordingly

4) The end -use Method :-

Or input -output method: This is a method for forecasting the demand for intermediary products metals, chemicals, paper etc., The demand for such products depends upon the demand for final product .For example demand for steel depends upon demand for machinery and so on. The main difficulty with the method is that as end -users increase the analysis becomes more complicated and time consuming .Estimating the demand for such products is really difficult.

Statistical Method:-

Trend projection Method:

- i) The least square Method
- ii) The free hand Method.
- iii) The moving Average Method
- iv) The semi- Averages Method.

2) Economics Indicators:-

Forecasting demand with the help of economic indicators is very recent technique .Economic indicators give us clue or signal for the direction of change. For example if PC income increases the demand for food clothing and such other consumer goods increases .Similarly if distribution of income changes favouring the good class then demand

for food. Clothing etc, would increase. If it is in favour of rich the demand for Luxuries like. Air conditioners TV, car, fridge etc, increases. Similarly indicators like changes in agricultural income salaries GNP, NNP consumer Price index etc, can be used as indicators to estimate demand for various goods. However precise demand forecasting with the help of indicators requires economic expertise.

Model Questions

3. Marks

1. What is Demand?
- 2) Define Law of Demand
- 3) What is demand curve?
- 4) What is Elasticity of Demand?
- 5) What is Price Elasticity of Demand?
- 6) What is Income Elasticity of Demand?
- 7) What is Cross Price Elasticity of Demand?
- 8) What is Advertising elasticity of Demand?
- 9) What is Promotional Elasticity of Demand?
- 10) What is Demand Forecasting?
- 11) What is Supply?
- 12) What is meant by Giffen Paradox?

7. Marks

- 1). Explain five types of Price elasticity of Demand with the help of Diagram
- 2) There is few exception to the law of Demand. What are they? Explain
- 3). State and Explain Law of Supply
- 4). State law of Demand .what are its Exceptions?

10.Marks

- 1) Explain briefly different methods of Demand Forecasting
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Module 3: Cost Analysis & Production analysis (9Hours)

Concepts of Production, production function with one variable input - Law of Variable Proportions. Production function with 2 variable inputs and Laws of returns to scale, Indifference Curves, ISO-Quants & ISO-Cost line, Least cost combination factor, Economies of scale, Diseconomies of scale. Technological progress and production function. Types of cost, Cost curves, Cost – Output Relationship in the short run and in the long run, LAC curve **Break Even Analysis** – Meaning, Assumptions, Determination of BEA, Limitations, Uses of BEA in Managerial decisions (with simple Problems).

Introduction :

A business comes in to being with the main objective of earning profit .To achieve optimum efficiency in production or minimizing cost for a given production is one of the prime concern of the business manager, in fact the survival of a firm in a competitive market depends on their ability to produce at a competitive cost. There fore manager of business firm endeavor to minimize the production cost or what is same things maximize out put from a given quantity of inputs.

Production: “Production” means a process by which resources men material, time etc) transformed into different and more useful commodity or services.In general production means transforming inputs (labour, machines, raw materials, time etc) into an output. Various inputs are combined in different quantities to produce various levels of output.”Inputs and output: An inputs is a good or service that goes into the process of production.

According to Boumol “ An inputs is a simply anything which the firm buys for use in its production or other process” The term inputs needs some more explanation Production process require wide variety of input depending on the nature of product but Economist have classified inputs as 1)labour 2)Capital 3)Land 4) Raw materials and Time. All these variable are flow variable since they are measured per unit of time

Output: An output is any good or service that comes out of production process.

Fixed and Variable inputs: A fixed inputs can be defined as a fixed factor is one that remains fixed (constant) for certain level of output.A variable input is defined as one whose supply in the short run is inelastic i.elabour and raw material etc the user of such factor can employ a large quantity in the short run . Technically a variable inputs is one that change with the change in output in the long run all inputs are variable

Short Run and long Run

Short run: refer to a period of time in which the supply of certain inputs (e.g plant, building , machinery etc) is fixed or inelastic .in short run therefore production of a commodity can be increased by increasing the use of only variable inputs like labour and raw material

Long run: refer to period of time in which the supply of all the inputs is elastic but not enough to permit a change in technology that is, in the long run all inputs are variable . Therefore, in long run production of commodity can be increased by employing more of both variable and fixed inputs

Production Function: It is a tool to analysis used to explain the input-output relationship. A production function describes the technological relationship between inputs and output in physical terms. In other word production Function showing the relationship between the quantities of inputs used and the output produced. There is a production function for every good that shows the maximum output you can get from any quantities of inputs. The production function is the description of the current best technology for making a good.

Production function indicates the highest output that a firm can produce for every specified combination of inputs given the state of technology. The firm's production function for a particular good (q) shows the maximum amount of the good that can be produced using alternative combinations of capital (k) and labor (l) .The production function for two inputs:

$$Q = F(K,L)$$

Q = Output, K = Capital, L = Labor

For a given technology

Characteristics of a Production Function:

- Production refers to the transformation of inputs into output;
- A production function is a flow concept;
- A production function is a physical, technological, or engineering concept, and yet it has an economic dimension to it; and
- In production analysis, factors of production can be separated into fixed and variable factors.

Short-Run Production with One Variable Input (Labour) or Short Run Production Function:

In the short run, some inputs (land, capital) are fixed in quantity. The output depends on how much of other variable inputs are used. For example if we change the variable input namely (labour) the production function shows how much output changes when more labour is used. In the short run producers are faced with the problem that some input

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factors are fixed. The firms can make the workers work for longer hours and also can buy more raw materials. In that case, labour and raw material are considered as variable input factors. But the number of machines and the size of the building are fixed. Therefore it has its own constraints in producing more goods.

In the long run all input factors are variable. The producer can appoint more workers, purchase more machines and use more raw materials. Initially output per worker will increase up to an extent. This is known as the **Law of Diminishing Returns** or the **Law of Variable Proportion**. To understand the law of diminishing returns it is essential to know the basic concepts of production

Amount of Labour(L)	Total Capital(K)	Output(q)
0	10	0
1	10	10
2	10	30
3	10	60
4	10	80
5	10	95
6	10	108
7	10	112
8	10	112
9	10	108
10	10	100

Observation: with additional workers, output (Q) increases, reaches a maximum, and then decreases.

Measures of Productivity

Total production (TP): the maximum level of output that can be produced with a given amount of input.

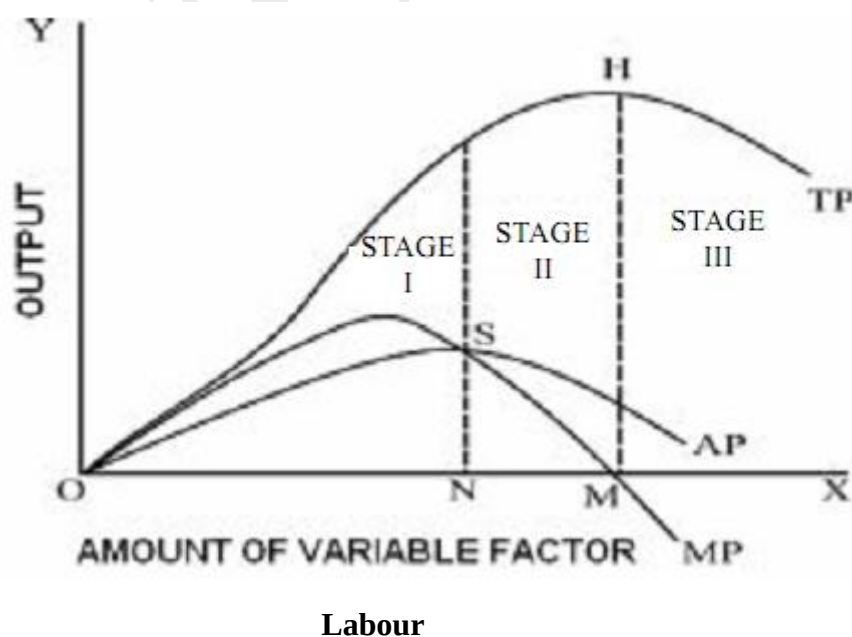
Average Production (AP): output produced per unit of input $AP = Q/L$

Marginal Production (MP): the change in total output produced by the last unit of an input

Short-Run Production with One Variable Input (Labor)

Amount of labour (L)	Amount of Capital(K)	Total output (Q)	Average Product	Marginal Product
0	10	0	0	0
1	10	10	10	10
2	10	30	15	20
3	10	60	20	30
4	10	80	20	20
5	10	95	19	15
6	10	108	18	13
7	10	112	16	4
8	10	112	14	0
9	10	108	12	-4
10	10	100	10	-8

From The above table it is possible to understand the Law of Return. The Total returns are increasing continuously with the increase in the factor labour. The marginal return remain constant during fifth and sixth units of labour and it is falling with the additional unit of labour. The law of diminishing return is operating from fourth unit of labour. The marginal physical product clearly show the operation of the law of diminishing returns. The law of return are shown with the help of following diagram.



Relationships Among Total, Average, and Marginal Product Curves

From the above Diagram ox axis measures number of workers and the OY axis measures total returns. TPC is the total production curve, which is the outcome of increase in number of workers and total returns. In the first stage of the TPC there is a continuous and steep increase in the total production. In the second stage the total production curve is not that steep and comparison to first stage. The marginal return in this stage are almost constant. In the third stage TPC will have a negative slope.

The Law of Diminishing Marginal Returns: As the use of an input increases in equal increments, a point will be reached at which the resulting additions to output decreases (i.e. MP declines). The slope of the MP curve in Figure illustrates an important principle, *the law of diminishing marginal returns*. As the number of units of the variable input increases, the other inputs held constant (fixed), there exists a point beyond which the MP of the variable input declines. Table illustrates this law. Observe that MP was increasing up to the addition of 4th worker (input); beyond this the MP decreases. What this law says is that MP may rise or stay constant for some time, but as we keep increasing the units of variable input, MP should start falling. It may keep falling and turn negative, or may stay positive all the time. Consider another example for clarity. Single application of fertilizers may increase the output by 50%, a second application by another 30% and the third by 20% and so on. However, if you were to apply fertilizer five to six times in a year, the output may drop to zero.

Three stages of production

Increasing Returns: When employment of variable inputs increased, a combination of fixed factor and variable factor tend to be near the optimum. Thus in the short run production function is adjusted to optimization, resulting output tends to be greater proportion to be increase in the variable factor units.

Diminishing Returns: The reason for diminishing returns is not for seek to increase output by employing more and more units of variable factors, there by trying to substitute fixed factor by variable factor. But due to imperfect substitutability of factors, when the fixed factor is over utilized there emerge internal diseconomies and the diminishing returns follow.

Negative Returns: Stage III is the stage of negative returns, when the inputs of variable factor is much excessive in relation to the fixed components in the production function. For instance excessive use of chemical fertilizer on a farm may eventually spoil the farm output.

Total Physical Product	Marginal Physical Product	Average Physical Product
Stage I Increasing at an increasing rate	Increases, reaches its maximum and then declines till $MR = AP$	Increases and reaches its maximum
Stage II Increases at diminishing rate till it reaches maximum	Is diminishing and becomes equal to zero	Starts diminishing
Stage III Starts declining	Becomes negative	Continues to decline

Production Functions with Two Variable Inputs

(Isoquants & LR Production Functions)

We have discussed in the preceding section, the technological relationship between inputs and output assuming labour to be the only variable inputs capital remain constant. This is a short run phenomenon. In this section we will discuss the relationship between inputs and output under condition that both the inputs capital and labour are variable factor this is long run phenomenon. In this long run, supply of both the inputs is supposed to be elastic and firms can hire larger quantities of both labour and capital with larger employment of capital and labour of the scale of production increases the technological relationship between changing scale of inputs and output is explained under the law of return to scale. Can be explained through the production function and isoquant curve technique, The most common and simple tool of analysis is isoquant curve.

Isoquant curve: A production function with two variable can be represented by a family of isoquant. The word isoquant simply means equal quantities;

Isoquant curve is defined is a curve representing all various input combinations which produces the same output (various capital-labor combinations that produces the same output). Or All those combination of two variable inputs which yield a given quantity of product Or an isoquant curve is locus of point representing various combination of two inputs capital and labour yielding same output

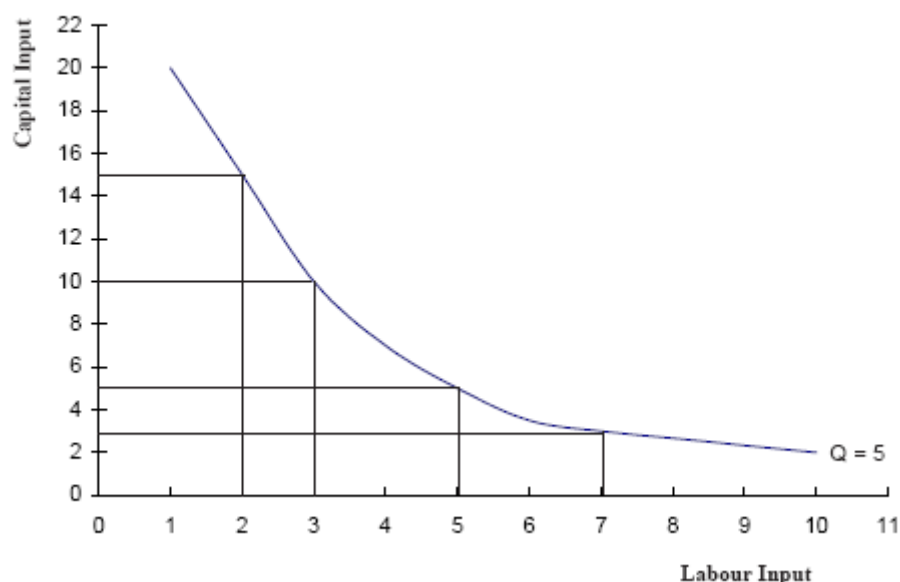
Capital, K	Labour, L					
	1	2	3	4	5	6
1	10	14	17	20	22	24
2	14	20	24	28	32	35
3	17	24	30	35	39	42
4	20	28	35	40	45	49
5	22	32	39	45	50	55
6	24	35	42	49	55	60

The different combination of labour and capital can be used to produce a given level of output. Thus if one labour is used and 6 capital need be employed to produce 24 units of output alternatively 24 units of output can be produce with the help of 1 units of capital and 6 units of labour in between there are other alternative. Alternative combination of labour and capital that firm can employs in order to produce 24 units of output

Isoquants analyze and compare the different combinations of K & L and output.

An Isoquant is a curve representing all various input combinations which produces the same output (various capital-labor combinations that produces the same output). Or Isoquants show combinations of two inputs that can produce the same level of output.

Figure 7.2: Production Isoquant: This isoquant shows various combinations of capital and labour inputs that can produce 5 units of output.



Properties of Isoquants

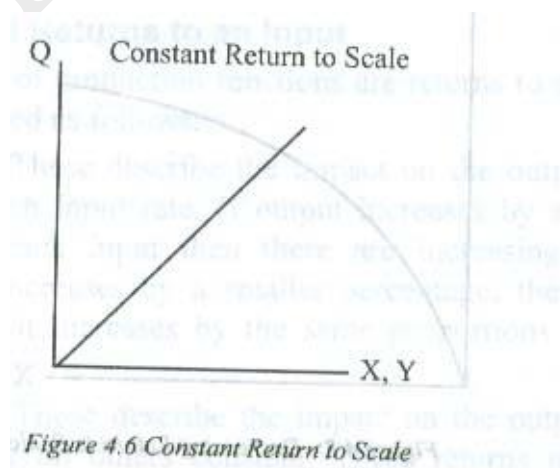
All the above-mentioned isoquants are featured with some common properties, which are as follows:

- An isoquant is downward sloping to the right, i.e., negatively inclined. This implies that for the same level of output, the quantity of one variable will have to be reduced in order to increase the quantity of other variable.
- A higher isoquant represents larger output. that is, with the same quantity, of one input and larger quantity of the other input, larger output will be produced.
- No two isoquants intersect or touch each other. If two isoquant intersect or touch each other, this would mean that there will be a common point the Two curves; and this would imply that the 'same amount of two inputs could produce two different levels of output (i.e., 400 and 500 units), which is absurd.

Production Functions with all variable inputs

A closely related question in production .economics is how a proportionate increase in all the input factors will affect total production. This is the question of returns to scale, which brings to mind three possible situations:

- If the proportional increase in all inputs is equal to the proportional increase in output, returns to scale are constant. For instance, if a simultaneous doubling of all inputs results in a doubling of production then returns to scale are constant. The following figure 4.6 shows a constant rate to scale.



- If the proportional increase in output is larger than that of the inputs, then we have increasing returns to scale. The following Figure 4.7 shows increasing returns to scale.

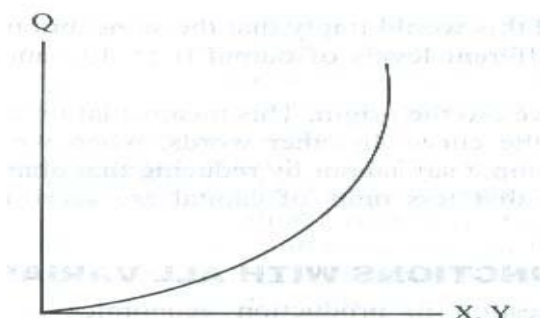


Figure 4.7: Increasing Returns to Scale

- If output increases less than proportionally with input increase, we have decreasing returns to scale. The following Figure 4.8 shows decreasing returns to scale.

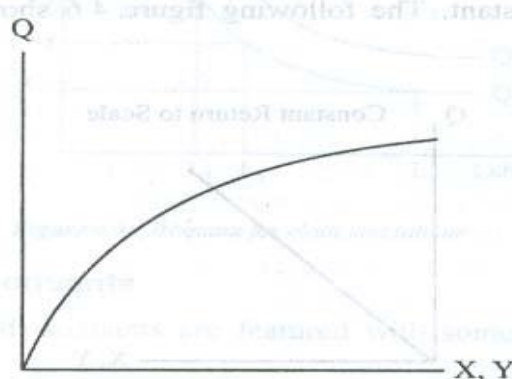


Figure 4.8: Decreasing Return to Scale

The most typical situation is for a production function to have first increasing then decreasing returns to scale is shown in Figure 4.9.

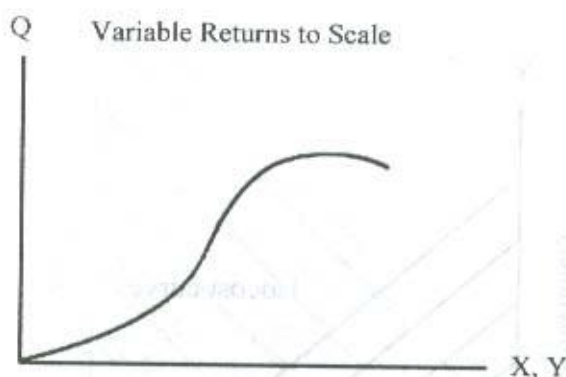


Figure 4.9: Variable rate of returns

The increasing returns to scale attribute to specialisation. As output increases, specialized labour can be used and efficient, large-scale machinery can be employed in the production process. However beyond some scale of operations further gains from

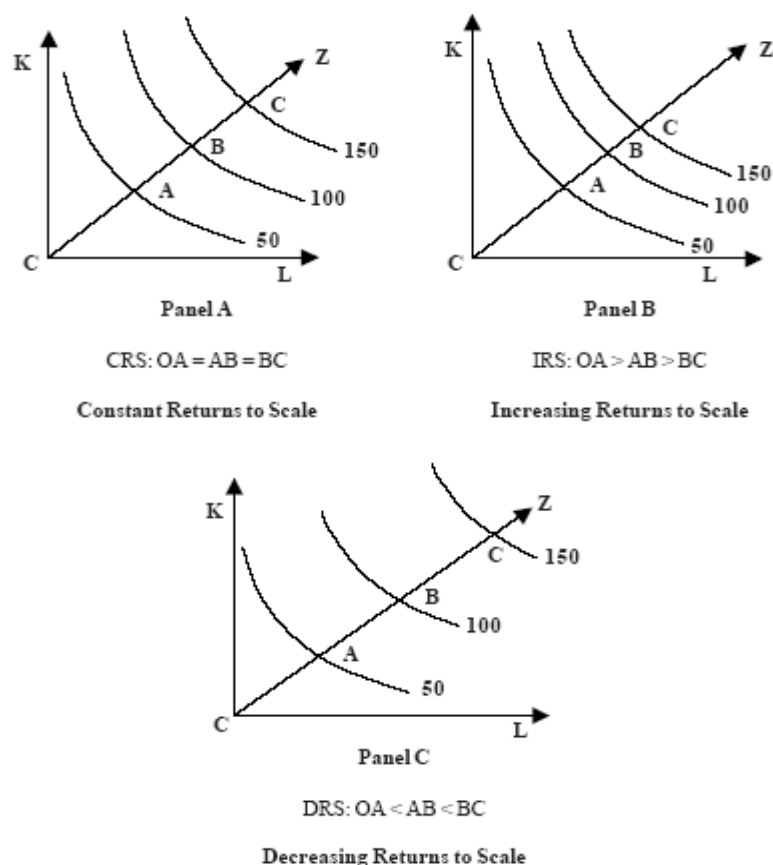
specialisation are limited, and co-ordination problems may begin to increase costs substantially. When co-ordination price is more than offset additional benefits of specialisation, decreasing returns to scale begin.

Returns to Scale and Returns to an Input

Two important features of production functions are returns to scale and returns to input, which are explained as follows:

Returns to scale: Another important attribute of production function is how output responds in the long run to changes in the scale of the firm i.e. when all inputs are increased in the same proportion (by say 10%), how does output change. Clearly, there are 3 possibilities. If output increases by more than an increase in inputs (i.e. by more than 10%), then the situation is one of **increasing returns to scale (IRS)**. If output increases by less than the increase in inputs, then it is a case of **decreasing returns to scale (DRS)**. Lastly, output may increase by exactly the same proportion as inputs. For example a doubling of inputs may lead to a doubling of output. This is a case of **constant returns to scale (CRS)**.

Figure 7.10: Isoquants Showing Returns to Scale



Returns to input: These describe the impact on the output when only one input is

varied, holding all others constant. These returns may be increasing, 'diminishing, or constant.

Optimal Input Combinations

From the overall discussion so far it is obvious that production function, has a pure 'physical or technological' character. However, it does not tell which input combinations are optimal. For that purpose, one has to take into account the input prices. The following Figure 4.10 shows the isocost curves.

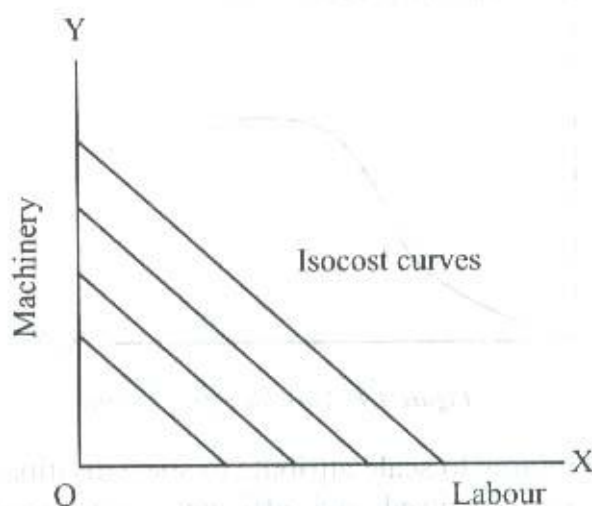


Figure 4.10: Isocost curves

Isocost Curves

In this connection, one has to consider yet another but important diagram consisting of isocost curves. Here also, the axes represent quantities of the inputs X and Y. Suppose that the prices of the inputs are given, and there are no quantity discounts for the firm to get larger quantities at lower prices. The next step will be to plot the various quantities of X and Y which may be obtained from the given monetary outlays. Figure 4.10 shows the resulting isocost curves, which are straight lines under the assumption made here. One isocost showing the quantities of X and Y that can be purchased for Rs. 1,000 and another isocost curve showing the quantities of X and Y which can be purchased for an expenditure of Rs. 2,000 and so on.

Now we can easily superimpose the isocost diagram on the isoquant diagram (as the axes in both the cases represent the same variables). With the help of Figure 4.11, it can be ascertained that the maximum output for a given outlay, is say Rs. 2,000. The isoquant tangent represents this maximum output, which is possible with this outlay, to the isocost curve. The optimum combination of inputs is represented by point E, the point of tangency. At this point, the marginal rate of substitution (MRS, sometimes known as the rate of technical substitution), between the inputs is equal to the ratio between the prices of the inputs.

Likewise, in order to minimize the cost for a given output, one may again refer to the isoquant and isocost curves in Figure 4.11. In this case one moves along the isoquant representing the desired output. It should be clear that the minimum cost for this input is represented by isocost line tangent to the isoquant.

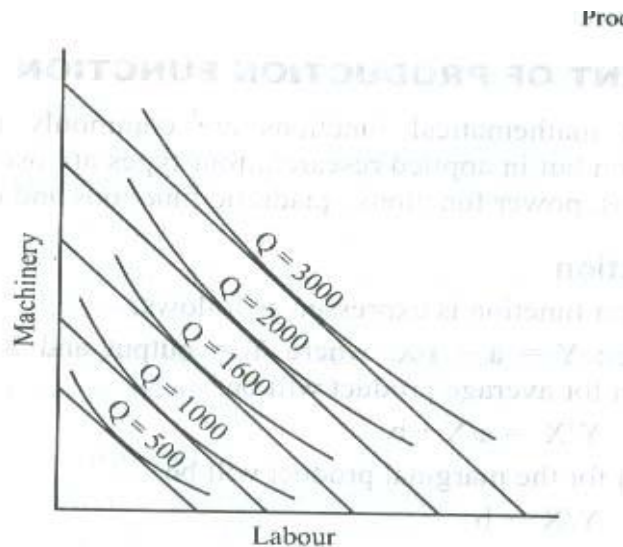


Figure 4.11: Isocost curves of labour and machinery

Firm's Expansion Path

A firm's expansion path is defined by the cost-minimising combination of several inputs for each output level. Thus the line representing least cost combination for different levels of output is called firm's expansion path or the scale line shown by

Economies of Scale: There are Economies of scale in production when the long run average cost decreases as output increases. Economies of scale (increasing returns to scale) are cost savings associated with larger scale of production. Diseconomies of scale or decreasing returns to scale refer to the increasing long run average costs as output increases. Diseconomies occur for a number of reasons as the firm increases its size. Coordination of a large firm is more difficult. Information costs and communication costs increase as firm increases.

Economies of Scale: The study of economies of scale is of greater importance for all business manager because it serves as basis for whole process of decision making. The term economies of scale is a technical term developed by economist to describes the benefits enjoyed by those firms which produce large scale and participate in long run it will enjoy certain special benefits which are called economies of scale or

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Economies of scale refers to the phenomena of decreased per unit cost as the number of units of production increase Economies of scale means a reduction in the per unit costs of a product as a firm's production increases

	Capital	Land	Labour	Output	TC	AC
Scale A	5	3	4	100	57	0.57
Scale B	10	6	8	300	164	0.54

Doubling the scale of production (a rise of 100%) has led to an increase in output of 200% - therefore cost of production PER UNIT has fallen

Economies of scale have been broadly classified under two group's

A) Internal Economies B) External Economies

Internal Economies

Internal economies are also known as operational economies which are obtained during the day to day process of production and hence they are even called operational economies. These economies are within the purview of the firm and the firm itself is responsible for these economies. These important internal economies are

Labour Economies: Large firms normally employ factor labour in different quantities, and qualities. Large firms practice the large process of Division of labour which is nothing but dividing a particular work into many smaller parts. If a small work is performed repetitively many fold benefits are obtained to the large business firm. They are like increase in efficiency of work, increase in productivity, economy of time, improvement in skills, avoiding wastage of resources etc.

As a result of all these benefits the task performed by worker will be superior quality. Thus all labour economies will be enjoyed only by large firm and not by small firm

Technological Economies: Large firms normally adopt capital intensive technology by making huge investment on plant and machinery. Large firms divide the entire total fixed cost on all the units of the output. Costly machinery and equipment brings with them both quantity and quality in production. Large firms will enjoy the benefit of superior technology and also benefit of indivisibility of factor. Latest technological improvement and other benefits will be enjoyed by large firms.

Financial Economies: Financial Economies also favor large firms but not small firms.

Normally large firms enjoy the support of stock market, Financial institutions and also support of general public. Large firm obtain fund at cheaper rate of interest with sufficient time of repayment. Small firm will not enjoy any financial economies.

Managerial Economies : Managerial economies are also called functional economies. Large firms employ skilled, highly qualified and specialized personal for both buying and selling. The functional benefits will be inform of team work, decentralization of power and adoption of modern management concept like PERT, (Progress evaluation review Technique) TQM, etc. The large firm during purchase price will be comparatively low with free transportation sufficient times for payment are benefits enjoyed during purchase and in sales too. Large firm also adopt modern technique to market the product through channels of distribution

Risk Spreading and Survival Economies : Risk spreading is easy for large firm and hence they survive with whatever be the magnitude of risk and uncertainties. Large firms will diversify their risk and future they insure against risk. The majority of the activities in the large firms will be carried out by appropriate massive insurance policy. Large firms meet accidents, unexpected stricks or events causing damage, risk of transportation etc, effectively . Small firms will perish due to risk and uncertainties.

External Economies of Scale These are economies made outside the firm as a result of its location, and occur when: A local skilled labour force is available. Specialist, and local back-up firms can supply parts or services. An area has a good transportation network. An area has an excellent reputation for producing a particular good

Diseconomies of Scale

Economies of increasing size do not continue indefinitely. After a certain point, any further expansion of the size leads to diseconomies of scale. For example, after the division of labour has reached its most efficient point, further increase in the number of workers will lead to a duplication of workers. There will be too many workers per machine for really efficient production. Moreover, the problem of co-ordination of different processes may become difficult. There may be divergence of views concerning policy problems among specialists in management

and reconciliation may be difficult to arrive. Decision-making process becomes slow resulting in missed opportunities. There may be too much of formality, too many individuals between the managers and workers, and supervision may' become difficult. The management problems thus get out of hand with consequent adverse effects on managerial efficiency.

The limit of scale economics is also often explained in terms of the possible loss of control

and consequent inefficiency. With the growth in the size of the firm, the control by those at the top becomes weaker. Adding one more hierarchical level removes the superior further away from the subordinates. Again, as the firm expands, the incidence of wrong judgments increases and errors in judgement become costly.

Last but not the least, is the limitation where the larger the plant, the larger is the attendant risks of loss from technological changes as technologies are changing fast in modern times.

Concept of Cost

Introduction

Business decisions are generally taken on the basis of money values of the inputs and outputs. The cost production expressed in monetary terms. It is an important factor in almost all business decisions, especially those pertaining to (a) locating the weak points in production management; (b), minimising the cost; (c) finding out the optimum level of output; and (d) estimating or projecting the cost of business operations. Besides, the term 'cost' has different meanings under different settings and is subject to varying interpretations. It is therefore essential that only relevant concept of costs is used in the business decisions.

Concept of Cost

In managerial economics, cost is normally considered from the producer's point of view. In producing a commodity, a firm has to employ an aggregate of various factors of production such as **land, labour, capital and entrepreneurship**. The compensation for these factors is the cost. Thus the cost of production of a commodity is the aggregate of price paid for the factors of production used in producing that commodity. A firm's costs depend on the rate of output and we will show how these costs are likely to change over time. The characteristics of the firm's production technology can affect costs in the long run and short run.

The Importance of Cost Analysis

Managers seek to produce the highest quality products at the lowest possible cost. Cost analysis is helpful in the task of finding lower cost methods to produce goods and services.

Basic Definitions

Profit: The money that business makes: Revenue minus Cost

Cost: the expense that must be incurred in order to produce goods for sale

Revenue : the money that comes into the firm from the sale of their goods

Types of Costs: Short-Run, Short-run total costs (TC), Fixed costs (FC), Short-run variable costs (VC)

Long-Run. All costs are Variable 'No fixed costs

Fixed and Variable Costs

Fixed costs are those, which are fixed in volume for a given output. Fixed cost does not vary with variation in the output between zero and any certain level of output. The costs that do not vary for a certain level of output are known as fixed cost. The fixed costs include cost of managerial and administrative staff, depreciation of machinery, building and other fixed assets and maintenance of land, etc.

Variable costs are those, which vary with the variation in the total output. They are a function of output. Variable costs include cost of raw materials, running cost on fixed capital, such as fuel, repairs, routine maintenance expenditure, direct labour charges associated with the level of output and the costs of all other inputs that vary with the output.

Total, Average and Marginal Costs

Total cost represents the value of the total resource requirement for the production of goods and services. It refers to the total outlays of money expenditure, both explicit and implicit, on the resources used to produce a given level of output. It includes both fixed and variable costs. The total cost for a given output is given by the cost function.

The Average Cost (AC) of a firm is of statistical nature and is not the actual cost. It is obtained by dividing the total cost (TC) by the total output (Q), i.e.,

$$AC = \frac{TC}{Q} = \text{average cost}$$

Marginal cost is the addition to the total cost on account of producing an additional unit of the product. Or marginal cost is the cost of marginal unit produced. Given the cost function, it may be

defined as

$$AC = \frac{aTC}{aQ}$$

These cost concepts are discussed in further detail in the following section. Total, average and marginal cost concepts are used in economic analysis of firm's production

activities.

Short-run and Long-run Costs

Short-run and long-run cost concepts are related to variable and fixed costs, respectively, and often appear in economic analysis interchangeably. Short-run costs are those costs, which change with the variation in output, the size of the firm remaining the same. In other words, short-run costs are the same as variable costs. Long-run costs, on the other hand, are the costs, which are incurred on the fixed assets like plant, building, machinery, etc. Such costs have long-run implication in the sense that these are not used up in the single batch of production.

Long-run costs are, by implication, same as fixed costs. In the long-run, however, even the fixed costs become variable costs as the size of the firm or scale of production increases. Broadly speaking, the short-run costs are those associated with variables in the utilisation of fixed plant or other facilities whereas long-run costs are associated with the changes in the size and type of plant.

Cost-Output Relations

Cost-output relations play an important role in business decisions relating to cost minimisation, profit maximisation and optimisation of output. Cost-output relations are specified through a cost function expressed as

$$T(C) = f(Q) \quad (1)$$

where,

TC = total cost

Q = quantity produced

Cost functions depend on production function and market-supply function of inputs. Production function specifies the technical relationship between the input, and the output. Production function of a firm combined with the supply function of inputs or prices of inputs determines the cost function of the firm. Precisely, cost function is a function derived from the production function and the market supply function. 'Depending on whether short or long-run is considered for the production, there are two kinds of cost functions: such as short-run cost-function and long-run cost function. Cost-output relations in relation to the changing level of output will be discussed here under both kinds of cost-functions.

Short-run Cost Output Relations

The basic analytical cost concepts used in the analysis of cost behaviour are total average and

marginal costs. The total cost (TC) is defined as the actual cost that must be incurred to produce a given quantity of output. The short-run TC is composed of two major elements: total fixed cost (TFC) and total variable cost (TVC). That is, in the short-run,

$$TC = TFC + TVC \quad (2)$$

As mentioned earlier, TFC (i.e. the cost of plant, building, equipment, etc.) remains fixed in the short-run, whereas TVC varies with the variation in the output.

For a given quantity of output (Q), the average total cost, (AC), average fixed cost (AFC) and, average variable cost (AVC) can be defined as follows:

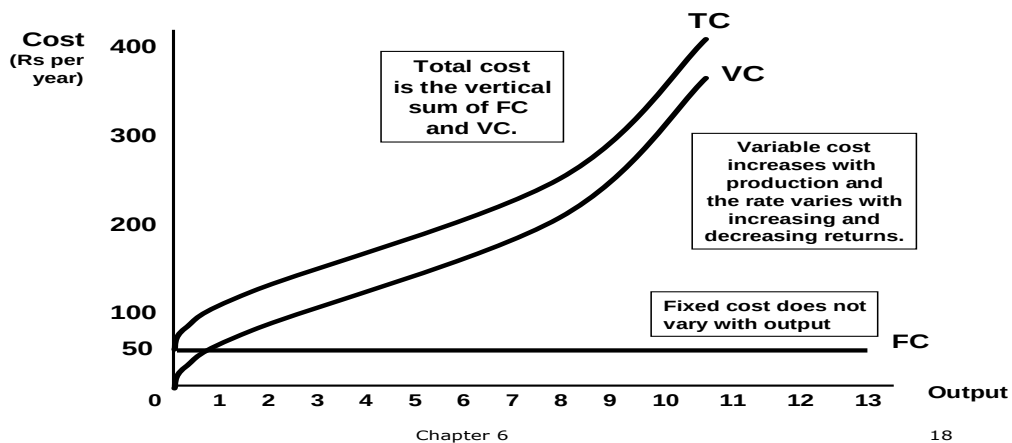
$$AC = \frac{TC}{Q} = \frac{TFC + TVC}{Q}$$

$$AFC = \frac{TFC}{Q}$$

$$AVC = \frac{TVC}{Q}$$

and

$$AC = AFC + AVC \quad (3)$$



Marginal cost (MC) is defined as the change in the total cost divided by the change in the total output, i.e.,

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$$MC = \frac{\Delta TC}{\Delta Q} \quad \text{or} \quad \frac{aTC}{aQ}$$

(4)

Since $\Delta TC = \Delta TFC + \Delta TVC$ and, in the short-run, $\Delta TFC = 0$, therefore, $\Delta TC = \Delta TVC$

Furthermore, under marginality concept, where $\Delta Q = 1$, $MC = \Delta TVC$.

Cost Function and Cost-output Relations

The concepts AC, AFC and AVC give only a static relationship between cost and output in the sense that they are related to a given output. These cost concepts do not tell us anything about cost behaviour, i.e., how AC, AVC and AFC behave when output changes. This can be understood better with a help of numerical example .

Numerical Example

Output	TVC	TFC	TC	MC*	ATC	AVC	AFC
0	0	8500	8500				
100	2500	8500	11000	25	110	25	85
200	3800	8500	12300	13	62	19	43
300	4800	8500	13300	10	44	16	28
400	6000	8500	14500	12	36	15	21
500	7500	8500	16000	15	32	15	17
600	9500	8500	18000	20	30	16	14
700	12500	8500	21000	30	30	18	12
800	17000	8500	25500	45	32	21	10.6
900	22500	8500	31000	55	34	25	9.4
1000	32500	8500	41000	100	41	32.5	8.5

*MC is per 100

From this schedule, the following inferences can be drawn Total cost is rising continuously in all units of output. The rate of fall in marginal cost is at a faster rate and even it is rising at an increasing rate. Average cost is falling continuously but the rate of fall is minimum and even the rising rate is minimum. When compared to the rate of fall and rise, marginal

cost and average cost, marginal cost falls and rises sharply.

(b) Average Cost, Marginal Cost

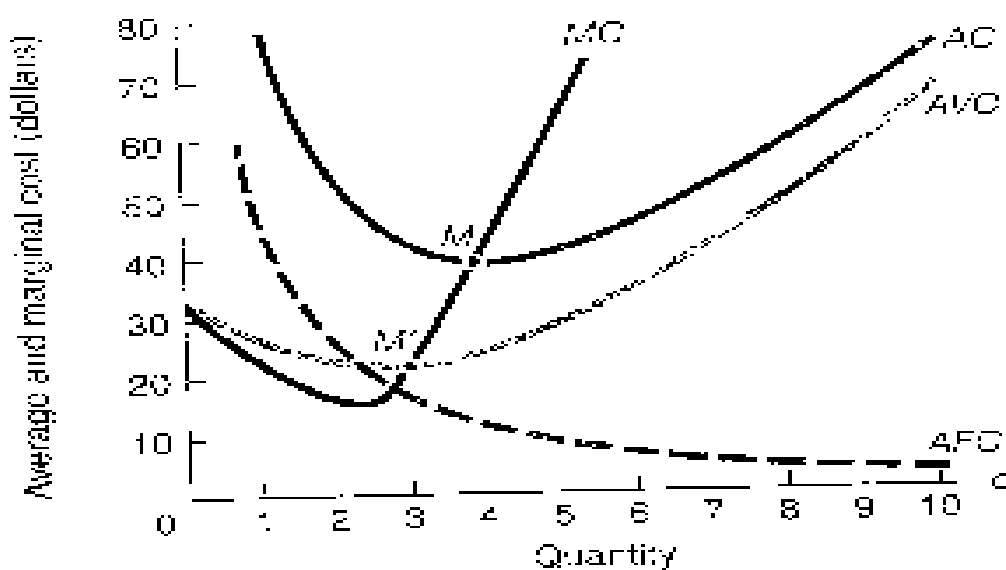


Figure 3.2

Cost Curves and the Laws of Diminishing Returns

We now return to the laws of variable proportions and explain it through the cost curves. The short-term laws of production, i.e., the laws of diminishing returns. Let us recall the law: it states that when more and more units of a variable input are applied to those inputs which are held constant, the returns from the marginal units of the variable input may initially increase but will eventually decrease. The same law can also be interpreted in terms of decreasing and increasing costs. The law can then be stated as, if more and more units of a variable inputs are applied to the given amount of a fixed input, the marginal cost initially decreases, but eventually increases. Both interpretations of the law yield the same information: one in terms of marginal productivity of the variable input, and the other, in terms of the marginal cost. The former is expressed through production function and the latter through a cost function.

Figure 3.2 represents the short-run laws of returns in terms of cost of production. As the figure shows, in the initial stage of production, both AFC and AVC are declining because of internal economies. Since $AC = AFC + AVC$, AC is also declining, this shows the operation of the law of increasing returns. But beyond a certain level of output, while AFC continues to fall, AVC starts increasing because of a faster increase in the TVC. Consequently,

the rate of fall in AC decreases. The AC reaches its minimum when output increases to 10 units. Beyond this level of output, AC starts increasing which shows that the law of diminishing returns comes in operation. The MC, curve represents the pattern of change in both the TVC and TC curves due to change in output. A downward trend in the MC shows increasing marginal productivity of the variable input mainly due to internal economy resulting from increase in production. Similarly, an upward trend in the MC shows increase in TVC, on the one hand, and decreasing marginal productivity of the variable input, on the other.

Some Important Cost Relationships

Some important relationships between costs used in analysing the short-run cost behaviour may now be summed up as follows:

- As long as AFC and AVC fall, AC also falls because $AC = AFC + AVC$.
- When AFC falls but A VC increases, change in AC depends on the rate of change in AFC and AVC then any of the following happens:
 - If there is decrease in AFC and increase in A VC, AC falls,
 - if the decrease on AFC is equal to increase in Ave, AC remains constant, and
 - if the decrease in AFC is less than increase in A VC, AC increases.
- The relationship between AC and MC is of varied nature. It may be described as follows:

When MC falls, AC follows, over a certain range of initial output. When MC is falling, the rate of fall in MC is greater than that of AC. This is because in case of MC the decreasing marginal cost is attributed to a single marginal unit while; in case of AC, the decreasing marginal cost is distributed over the entire output. Therefore, AC decreases at a lower rate than MC.

Similarly, when MC increases, AC also increases but at a lower rate, the reason given in the above point. There is however a range of output over which this relationship does not exist. When MC starts increasing, it increases at a relatively lower rate, which is sufficient only to reduce the rate of decrease in AC, i.e., not sufficient to push the AC up. That is why AC continues to fall over some range of output even, if MC falls. MC intersects AC at its minimum point. This is simply a mathematical relationship between MC and AC curves when both of them are obtained from the same TC function. In simple words, when AC is at its minimum, then it is neither increasing nor decreasing it is constant. When AC is constant, $AC = MC$.

Optimum Output in Short-run

An optimum level of output is the one, which can be produced at a minimum or least average cost, given the required technology is available. Here, the least 'cost' combination of inputs can be understood with the help of isoquants and isocosts. The least-cost combination of inputs also indicates the optimum level of output at given investment and factor prices. The AC and MC cost Curves can also be used to find the optimum level of output, given the size of the plant in the short-run. The point of intersection between AC and MC curves determine the minimum level of AC. At this level of output $AC = MC$. Production below or beyond this level will be in optimal. If production is less than 10 units (Figure 3.2) it will leave some scope for reducing AC by producing more, because $MC < AC$. Similarly, if production is greater than 10 units, reducing output can reduce AC. Thus, the cost curves can be useful in finding the optimum level of output. It may be noted here that optimum level of output is not necessarily the maximum profit output. Profits cannot be known unless the revenue curves of firms are known.

Long-run Cost-output Relations

By definition, in the long-run, all the inputs become variable. The variability of inputs is based on the assumption that, in the long run, supply of all the inputs, including those held constant in the short-run, becomes elastic. The firms are, therefore, in a position to expand the scale of their production by hiring a larger quantity of all the inputs. The long-run cost-output relations, therefore, imply the relationship between the changing scale of the firm and the total output; conversely in the short-run this relationship is essentially one between the total output and, the variable cost (labour). To understand the long-run cost-output relations (In order to derive long-run cost curves it will be helpful to imagine that a long run is composed of a series of short-run production decisions. As a corollary of this, long-run cost curves are composed of a series of short-run cost curves. We may now derive the long-run cost curves and study their relationship with output.

Long-run Total Cost Curve (LTC)

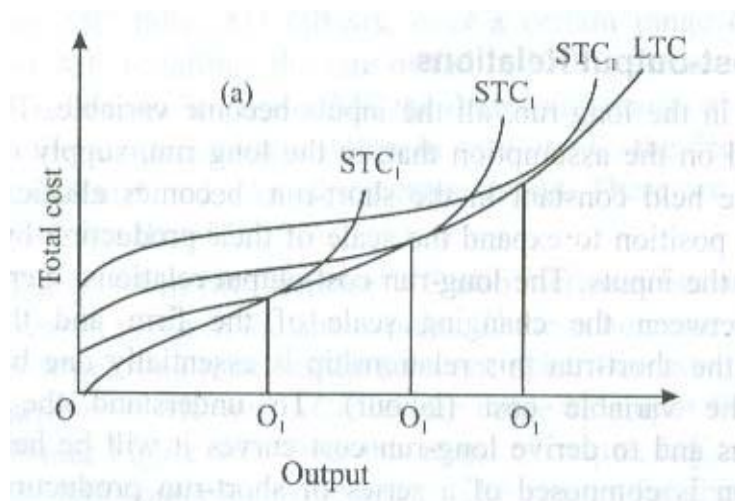
In order to draw the long-run total cost curve, let us begin with a short-run situation. Suppose that a firm having only one plant has its short-run total cost curve as given by STC_1 in panel (a) of Figure 3.3. In this example if the firm decides to add two more plants to its size over time, one after the other then in accordance two more short-run total cost curves are added to STC_1 in the manner shown by STC_2 and STC_3 in Figure 3.3 (a). The LTC can now be drawn

through the minimum points of STC_1 , STC_2 and STC_3 as shown by the LTC curve corresponding to each STC.

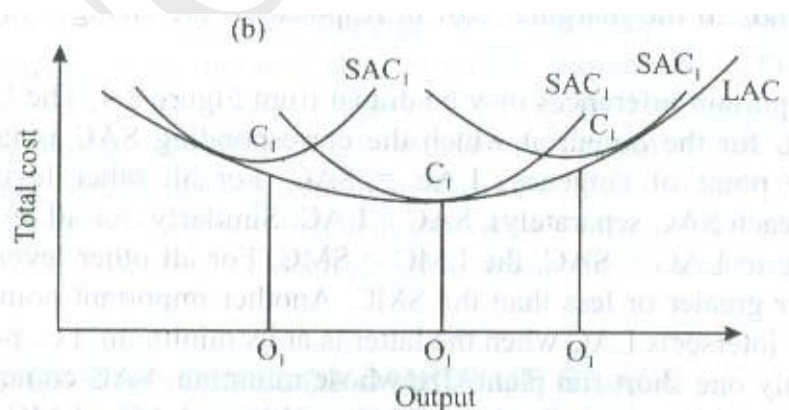
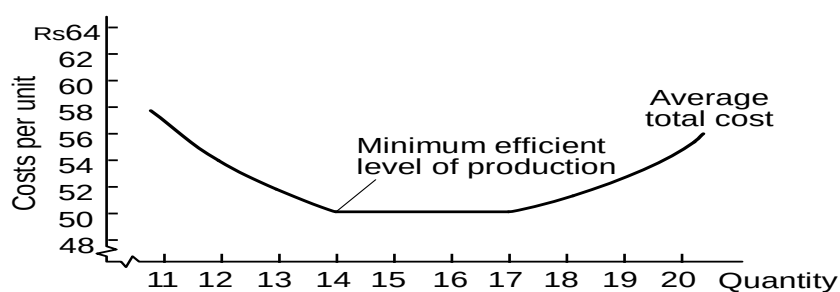
Long-run Average Cost Curve (LAC)

Combining the short-run average cost curves (SACs) derives the long-run average cost curve (LAC). Note that there is one SAC associated with each STC. Given the STC_1 , STC_2 , and STC_3 curves in panel (a) of Figure 3.3, there are three corresponding SAC curves as given by SAC_1 , SAC_2 and SAC_3 curves in panel (b) of Figure 3.3. Thus, the firm has a series of SAC curves, each having a bottom point showing the minimum SAC. For instance, C_1Q_1 is the minimum AC when the firm has only one plant. The AC decreases to C_2Q_2 when the second plant is added and then rises to C_3Q_3 after the inclusion of the third plant. The LAC can be drawn through the bottom of SAC_1 , SAC_2 and SAC_3 as shown in Figure 3.3 (b). The LAC curve is also known as 'Envelope Curve' or 'Planning Curve' as it serves as a guide to the entrepreneur in his planning to expand production.

Quantity	Total Costs of Labor	Total Cost of Machines	Total Costs = $TC_L + TC_M$	Average Total Costs = TC/Q
11	381	254	Rs635	58
12	390	260	650	54
13	402	268	670	52
14	420	280	700	50
15	450	300	750	50
16	480	320	800	50
17	510	340	850	50
18	549	366	915	51
19	600	400	1,000	53
20	666	444	1,110	56



A Typical Long-Run Average Total Cost Curve



Figures 3.3 (a), (b): Long-run Cost Curves

The SAC curves can be derived from the data given in the STC schedule, from STC function or straightaway from the LTC-curve. Similarly, LAC can be derived from LTC-schedule, LTC function or from LTC-curve. The relationship between LTC and output, and between LAC and output can now be easily derived. It is obvious. from the LTC that the long-run cost-output relationship is similar to the short-run cost-output relationship. With the

subsequent increase in the output, LTC first increases at a decreasing rate, and then at an increasing rate. As a result, LAC initially decreases until the optimum utilisation of the second plant and then it begins to increase. From these relations are drawn the 'laws of returns to scale'. When the scale of the firm expands, unit cost of production initially decreases, but it ultimately increases as shown in Figure 3.3 (b).

Long-run Marginal Cost Curve

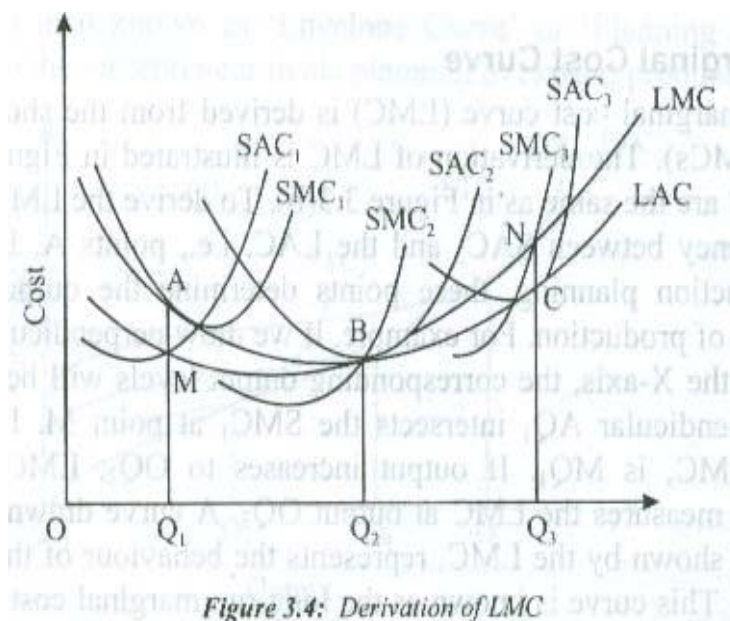
The long-run marginal, cost curve (LMC) is derived from the short-run marginal cost curves (SMCs). The derivation of LMC is illustrated in Figure 3.4 in which SAC_3 and LAC are the same as in Figure 3.3(b). To derive the LMC_3 , consider the points of tangency between SAC_3 and the LAC, i.e., points A, B and C. In the long-run production planning, these points determine the output levels at the different levels of production. For example, if we draw perpendiculars from points A, B and C to the X-axis, the corresponding output levels will be OQ_1 , OQ_2 and OQ_3 . The perpendicular AQ_1 intersects the SMC_1 at point M. It means that at output OQ_1 , LMC, is MQ_1 . If output increases to OQ_2 , LMC rises to BQ_2 . Similarly, CQ_3 measures the LMC at output OQ_3 . A curve drawn through points M_1 , B and N, as shown by the LMC, represents the behaviour of the marginal cost in the long run. This curve is known as the long-run marginal cost curve, LMC. It shows the trends in the marginal cost in response to the change in the scale of production.

Some important inferences may be drawn from Figure 3.4. The LMC must be equal to SMC for the output at which the corresponding SAC is tangent to the LAC. At the point of tangency, $LAC = SAC$. For all other levels of output (considering each SAC separately), $SAC > LAC$. Similarly, for all levels of output corresponding to $LAC = SAC$, the $LMC = SMC$. For all other levels output, the LMC is either greater or less than the SMC. Another important point to notice is that the LMC intersects LAC when the latter is at its minimum, i.e., point B. There, is one and only one short-run plant size whose minimum SAC coincides with the minimum LAC. This point is B where, $SAC_2 = SMC_2 = LAC = LMC$.

Optimum Plant Size and Long-run Cost Curves

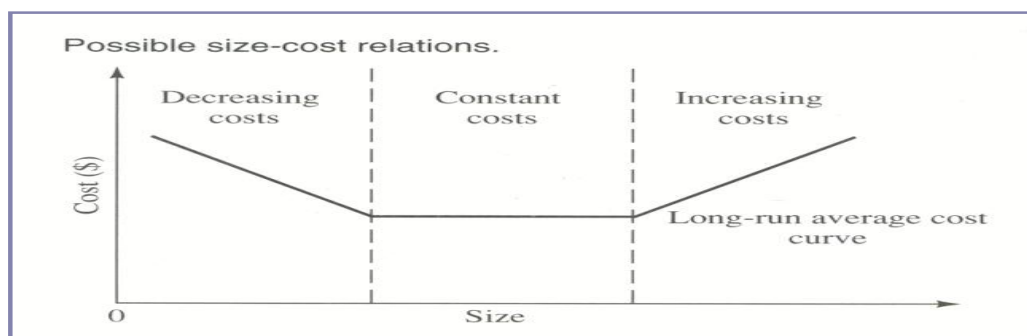
The short-run cost curves are helpful in showing how a firm can decide on the optimum utilisation of the plant-which is the fixed factor; or how it can determine the least-cost output level. Long-run cost curves, on the other hand, can be used to show how the management can decide on the optimum size of the firm. An Optimum size of a firm is the one, which ensures the most efficient utilisation of resources. Given the state of technology overtime, there is

technically a unique size of the firm and level of output associated with the least cost Concept. This unique size of the firm can be obtained with the help of LAC and LMC. In Figure 3.4 the optimum size consists of two plants, which produce OQ_2 units of a product, at minimum long-run average cost (LAC) of BQ_2 .



The downtrend in the LAC indicates that until output reaches the level of OQ_2 , the firm is of non-optimal size. Similarly, expansion of the firm beyond production capacity OQ_2 causes a rise in SMC as well as LAC. It follows that given the technology, a firm trying to minimise its average cost over time must choose a plant which gives minimum LAC where $SAC = SMC = LAC = LMC$. This size of plant assures most efficient utilisation of the resource. Any change in output level, i.e., increase or decrease, will make the firm enter the area of in optimality.

Possible Size-Cost Relations

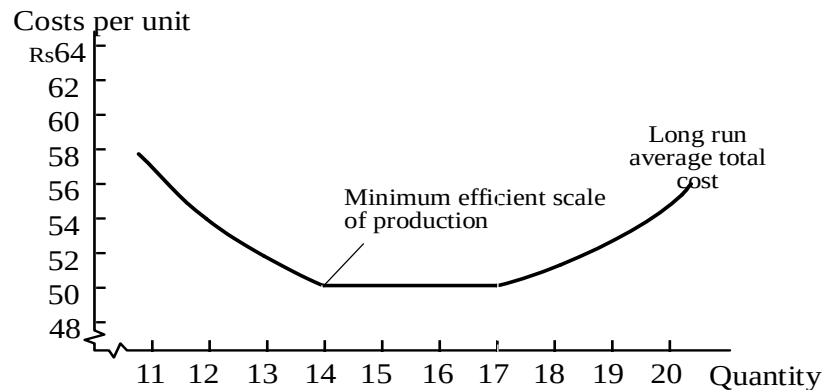


Determinants of the Shape of the Long-Run Cost Curve

The law of diminishing marginal productivity does not hold in the long run since all inputs are variable.

The shape of the long-run cost curve results from the existence of economies and diseconomies of scale

A Typical Long-Run Average Total Cost Curve,



Break Even Analysis

Introduction : An important part of running a company is the determination of how the company should be financed and how the prices of the products the company sells should be set.

Break-even analysis is a study of Costs, Revenue and sales of a firm and finding out the volume of sales where the firm's costs and revenues will be equal. The break even point is that level of sales where the net income is equal to zero

The Break even point is the zone of no profit and no loss, as the cost equal revenues

Breaking even is when a firm is just covering its costs, making neither a profit nor a loss.

Variable cost – which vary with the level of activity taking place. These would include the costs of the raw materials used, the fuel needed for vans or equipment & the wage costs of people working on the machines

Fixed cost – which remain constant over a given range of activity. These would include the rent of the premises, insurance on the firm's vehicles & rates

Break-Even Analysis

$$\text{Total Costs} = \text{Fixed Costs} + \text{Variable Costs}$$

$$\text{Total Variable Costs} = \text{Variable costs per unit} \times \text{number of units produced}$$

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Total Revenue is the money received by a firm from the sale of its goods or services

Total Revenue = Selling price x Quantity sold

Contribution

This is a measure of the amount of money that each unit sold contributes towards, covering the fixed cost of a business. Once FC are covered all further contribution is profit.

Break-Even Analysis

Contribution = Total Revenue – Variable cost

Contribution per unit: Selling price per unit – Variable costs per unit

Break-Even : This is the level of output & sales at which a firm generates just enough income to cover fixed & variable costs, earning neither a profit nor a loss

If the selling price of a product exceeds its variable cost, each unit sold will earn a contribution towards fixed costs. If total contribution covers fixed costs, the firm breaks even

Break-Even Analysis: Break even point is the level of sales at which the firm breaks even.

$$\text{Break-Even} = \frac{\text{Fixed Cost}}{\text{Contribution Per Unit}}$$

Margin of Safety

This is the difference between sales volume & break even point, & is the amount by which sales can fall before a firm incurs a loss

Break-Even Charts

Out put in units	Total Revenue (price Rs 4 per Units)	Total fixed cost	Total variable cost	Total cost
0	0	300	0	300
100	400	300	300	600
200	800	300	600	900
300	1200	300	900	1200
400	1600	300	1200	1500
500	2000	300	1500	1800
600	2400	300	1800	2100

Break-even analysis is very commonly presented by means of break even charts. Break-even charts are also known as profit-graphs. A break-even chart prepared on the basis of example 1 above is given in Figure 5.2. In this figure, units of product are shown on the horizontal axis OX while revenues and costs are shown on the vertical axis OY. The fixed costs of Rs. 10,000 are shown by a straight line parallel to the horizontal axis. Variable costs are then plotted over and above the fixed costs. The resultant line is the total cost line, combining both variable and fixed costs. There is no variable cost line in the graph. The vertical distance between the fixed cost and the total cost lines represents variable costs. The total cost at any point is the sum of Rs. 10,000 plus Rs. 2.00 per unit of variable cost multiplied by the number of units sold at that point. Total revenue at any point is the unit price of Rs. 4.00 multiplied by the number of units sold. The break-even point corresponds to the point of intersection of the total revenue and the total cost lines. A perpendicular from the BEP to the horizontal axis shows the break-even point in units of the product. Dropping a perpendicular from BEP to the vertical axis shows the break-even sales value in rupees. The firm would suffer a loss at any point below the BEP. Total costs are more than total revenue. Above the BEP, total revenue exceeds total costs and the firm makes profits. Since profit or loss occurs between costs and revenue lines, the space between them is known as the profit zone, which is to the right of the BEP, and the loss zone, which is to the left of the BEP. The following Figure 5.2 shows Break-even Chart.

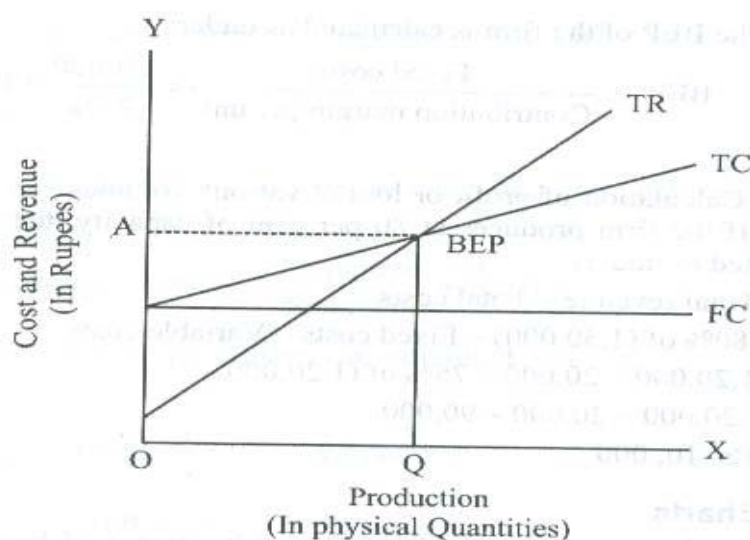
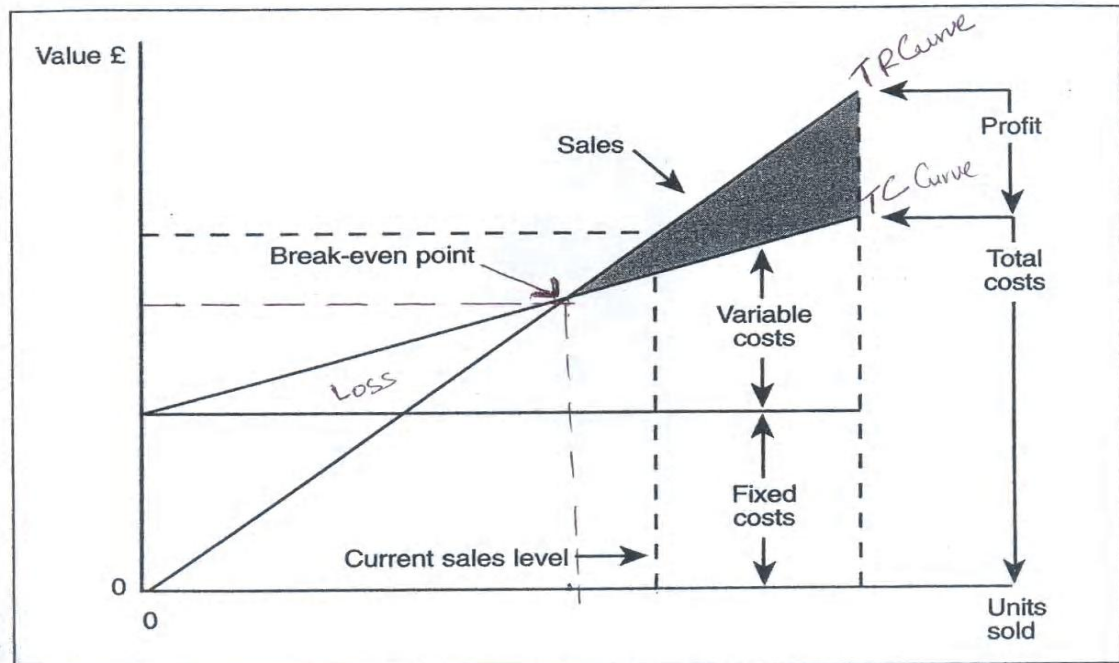


Figure 5.2: Break-even chart



Source: *Finance for the Non-Financial Manager*, J. Harrison/Lucas Productions, HarperCollins, London, 1989.

The break-even chart remains where the BEP is measured in terms of sales value rather than in physical units. The only difference is that the volume on the X-axis is measured in terms of sales value. In that case, a perpendicular from the point BEP to either axis would show the break-even rupee sales value. The same type of chart could be used to depict the BEP in relation to full capacity. In this case the horizontal axis would represent the percentage of full capacity, instead of physical units or the sale value.

Assumptions

1. All costs are either variable or fixed over the entire range of the volume of production. But in practice, this assumption may not hold well over the entire range of production.
2. All revenue is variable in nature. This assumption may not be valid in all cases such as the case where lower prices are charged to large customers.
3. The volume of sales and the volume of production are equal. The total products, produced by the firm, are sold and there is no change in the closing inventory. In practice, sales and production volumes may differ significantly. However, these assumptions are not so unrealistic so as to weaken the validity of the break-even analysis.
4. In the case of multi-product firms, the product-mix should be stable. For multi-product firm, the BEP is determined by dividing total fixed costs by an average ratio of variable profit, also called contribution to sales. If each product has the same contribution ratio, the BEP is not

affected by changes in the product-mix.

However, if different products have different contribution ratios, shift in the product-mix may cause a shift in the break-even point. In real life, the assumption of stable product-mix is somewhat unrealistic

Managerial Uses of Break-even Analysis

To the management, the utility of break-even analysis lies in the fact that it presents a picture of the profit structure of a business firm. Break-even analysis not only highlights the areas of economic strength and weaknesses in the firm but also sharpens the focus on certain in leverages which can be operated upon to enhance its profitability. Through break-even analysis, it is possible for the management to examine the profit structure of a business firm to the possible changes in business conditions. For example, sales prospects, changes in Cost structure, etc. Through break-even analysis, it is possible to use managerial actions to maintain and enhance profitability of the firm. The break-even analysis can be used for the following purposes:

- Safety margin
- Volume needed to attain get profit
- Change in price Change in price
- Expansion of capacity
- Effect of alternative prices
- Drop or add decision
- Make or buy decision
- Choosing promotion-mix
- Equipment selection
- Improving profit performance
- Production planning

Safety Margin

The break-even chart helps the management to know the profits generated at the various levels of sales. But while deciding the volume at which the firm would operate, apart from the demand, the management should consider the safety margin associated with the proposed volume. The safety margin refers to the extent to which the firm can afford a decline in sales before it starts occurring losses. The formula to determine the safety margin is:

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$$\text{Safety Margin} = \frac{(\text{Sales} - \text{BEP}) \times 100}{\text{Sales}}$$

Example 5: Assume that our sales in Example 1 are 8,000 units.

$$\text{Safety Margin} = \frac{(8,000 - 5,000) \times 100}{8,000} = 37.5\%$$

Before incurring a loss, a business firm can afford to loose sales up to 37.5 per cent of the present level. A decreasing safety margin indicates that the firm's resistance capacity to avoid losses has become poorer. A margin of safety can also be negative. A negative safety margin is the percentage increase in sales necessary to reach the BEP in order to avoid losses. Thus, it reveals the minimum extent of effort in terms of sales expected by the management. Suppose in the same example sales are as low as 4,000 units. The safety margin would be:

$$\text{Safety Margin} = \frac{(4,000 - 5,000) \times 100}{4,000} = 25\%$$

In other words, the management must strive to increase sales at least by 25 per cent to avoid losses.

Volume Needed to Attain Target Profit

Break-even analysis is also utilised for determining the volume of sales, necessary to achieve a target profit. The formula for target sales volume is:

$$\text{Target Sales Volume} = \frac{\text{Fixed costs} + \text{Target profit}}{\text{Contribution margin per unit}}$$

Example 6: Continuing with the same example, if the desired profit is Rs. 6,000, the target sales volume would be calculated as follows:

$$\frac{10,000 + 6,000}{2} = 8000 \text{ units}$$

Change in Price

The management is also faced with a problem whether to reduce the prices or not. The

management will have to consider a number of points before taking a decision related to the change in the prices. A reduction in price results in a reduction in the contribution margin as well. This means that the volume of sales will have to be increased to maintain the previous level of profit. The higher the reduction in the contribution margin, the higher will be the increase in sales needed to maintain the previous level of profit. However, reduction in prices may not always lead to an equal increase in the sales volume, which is affected by the elasticity of demand. But the information about elasticity of demand may not be easily available. Breakeven analysis helps the management to know the required sales volume to maintain the previous level of profit. On the basis of this knowledge and experience, it becomes much easier for the management to judge whether the required increase in sales will be feasible or not. The formula to determine the new sales volume to maintain the same level of profit, given a reduction in price, would be as under:

$$Q_n = \frac{FC + P}{SP_n - VC}$$

where Q_n = New volume of sales

FC = Fixed cost

P = Profit

SP_n = New selling price

VC = Variable cost per unit (n denotes new)

Example 6(a): Continuing with the same example 6, if we propose a reduction of 10 per cent in price from Rs. 4.00 to Rs. 3.60, the new sales volume needed to maintain the previous profit of Rs. 6,000 will be:

$$\frac{10,000 + 6,000}{3.60 - 2.00} = \frac{16,000}{1.60} = 10,000 \text{ units}$$

This shows that there is an increase of 2,000 units or 25 per cent in sales. The management can also easily decide whether this increase in sales volume is profitable for the business firm or not.

If a firm proposes the price increase, the question to be considered is by how much the sales volume should decline before profitable effect of the price increase gets eliminated.

Example 6(b): If the firm in example 6 considers an increase in price by 12% per cent to Rs. 4.50, the new volume to maintain the old profit would be:

$$Q_2 = \frac{10,000 + 6,000}{4.50 - 2.00} = \frac{16,000}{2.50} = 6,400 \text{ units}$$

In other words, if the fall in sales, due to an increase in price, were less than 1,600 units or 20 per cent, it would be profitable for the firm to increase the price. But if the decline were more than 1,600 units, the proposed price increase would reduce the profit.

Limitation of Break Even analysis

It is static in character: In Break even analysis, we keep every thing constant. The selling price is assumed to be constant and cost function is linear in practice it will not be so.

Projection of future with the past is not correct

The profits are a function of not only output relationship is linear is true

Model Questions

3 Marks

- 1) What is Cost?
- 2) What is Total Fixed Cost?
- 3) What is Total Variable Cost?
- 4) What Marginal Cost?
- 5) What is Average Cost?
- 23) What is Average Revenue?
- 24) What is Marginal Revenue?
- 25) What is Production?
- 26) What is Production Function?
- 27) What is Total Product (TP)?
- 28) What is Average Product (AP)?
- 29) What is Marginal Product (MP)?
- 30) What is law of variable proportions?
- 31) What is law of returns to scale?
- 33) What are Isoquants?
- 35) What is Isocost line?

- 36) What is MRTS?
- 38) What is Economies of Scale?
- 39) What is diseconomies of scale?
- 40) What is Profit?
- 41) What do you mean by break even analysis?
- 42) What is Break-even Point?
- 43) What is Contribution?
- 44) What is Profit-volume ratio?
- 45) What is Margin of safety?
- 46) What is Contribution Margin Ratio?

7 Marks

- 1) Explain different types of short-run costs.
- 2) Write a Short note on Long-run average cost curve
- 3) Why Long-run average cost curve is “U” shaped?
- 5) Explain different concepts of revenue with example
- 7) Explain relationship between Average Product and Marginal Product
- 9) Explain ISOQUANT Curve
- 10) Explain properties of Isoquant Curve.
- 11) Explain internal economies of scale.
- 12) Explain external economies of scale.
- 13) Explain internal diseconomies of scale.
- 14) Explain external diseconomies of scale.
- 15) What is the Use of Break-even analysis in decision making?
- 16) What are the limitations of Break-even Analysis?

10 Marks

- 1) What are the Different types costs?
- 4) Explain short-run Production Function?
- 5) What are the three stages of law of variable proportions? Explain with example
- 6) Explain Producers equilibrium (Isoquant analysis).
- 7) What are the three stages of law of returns to scale? Explain with example
- 8) Explain types of economies of scale
- 9) Explain diseconomies of scale.
- 12) What is Break even Chart? Draw a breakeven chart and explain its elements.

Economics for Decision Making (MBA103)

1. Explain the Law of Variable Proportions with the help of diagrams, indicating the Increasing, diminishing and negative returns
2. Construct a hypothetical cost schedule and bring out the cost-output relationship in the short run between output and TC, TFC, TVC, AC, AFC, AVC
3. Explain Economies and Diseconomies of Scale.
4. What are the uses or application of break-even analysis and limitations
5. Explain the short-run cost-output relationship with statistical examples

Module 4: Market structure and pricing practices (9 Hours)

Perfect Competition, Features, Determination of price under perfect competition, Monopoly: Features, Pricing under monopoly, Price Discrimination. Monopolistic Competition: Features, Pricing Under monopolistic competition, Product differentiation. Oligopoly: Features, Kinked demand Curve, Cartels, Price leadership. **Descriptive Pricing Approaches:** Full cost pricing, Product line pricing, Product life cycle pricing, Pricing Strategies: Price Skimming, Penetration Pricing, Loss leader pricing, Peak Load pricing.

Introduction

Market: Market has several Meaning and interpretations. Ordinarily the “market” refers to a particular place where goods are purchased and sold. But in economics, market is used in wide perspective in economics the term “market” does not mean a particular place but the whole area where the buyers and seller of a product are spread .

Economist: It is commodity Which is being bought and sold. The market consists of buyers and sellers who can communicate with one another.

Market Structure

Every firm has to decide how much to produce and sell and what price to charge.

This decision depends on market-structure considerations. Economists classify market structures into four broad categories:

Perfect competition	Monopoly
Monopolistic competition	Oligopoly

Perfect competition

As the name itself suggests, in perfect competition market, competition will be complete and sharp. Perfect Competition is an hypothetical market situation in which there are large number of buyer and sellers, each of them are too small to influence anything in the market , though they are dealing with homogeneous product and fully aware of the market condition

Characteristics or Feature

1. Existence of Large Number of Buyers and Sellers:

There will be a large number of buyers and sellers in the perfect competition market. Sellers compete among each other intensely with an intention to earn large profits by selling goods at a high price. On the other hand, buyers compete among each other to purchase commodities at a low price. Hence any single buyer or seller in the market will not be capable of influencing the price and output behavior This is because each seller is selling a very small

quantity of products out of the total quantity offered for sale, and each buyer is buying a very small quantity out of the total product. Thus it is rightly compared that the role of an individual is like a drop of water in a vast ocean. Price will be determined as a result of collective decision and-action of all the producers (sellers) and consumers (buyers). Every producer and consumer accepts the price so determined.

2. Homogeneity of Commodities: Any commodity which is offered for sale will be homogeneous in the whole market. In other words, any commodity will be a perfect substitute from consumers point of view. Commodity similar in all respects like quality, quantity, taste, package, size, colour with any seller. Therefore there cannot be price differences.

3 Single Price:

As a result of homogeneity of the product, only one price rules. That is, all the sellers in the whole market will be selling any commodity at a single price. Because, if any one seller raises the price, consumers will be distracted from him. If he reduces price, he has to incur loss and will not be in a position to face threat of competition.

4 Entry and Exit of Firms:

Firms enjoy complete freedom to enter the industry or to leave it. At any time a firm can join the industry or it can take an exit. Firms leaving the industry when the latter is incurring losses and the firms joining industry when it is earning profits will be a natural process.

5. Perfect Mobility of Factors of Production: Perfect competition provides scope for free movement of any factor of production from one industry to the other. It is a common feature that a factor of production moves to a profitable enterprise. For example, if there are differences in wage rates in different industries, labourers move away from one industry to another industry in which wages are high.

6. Perfect Knowledge of Market Conditions

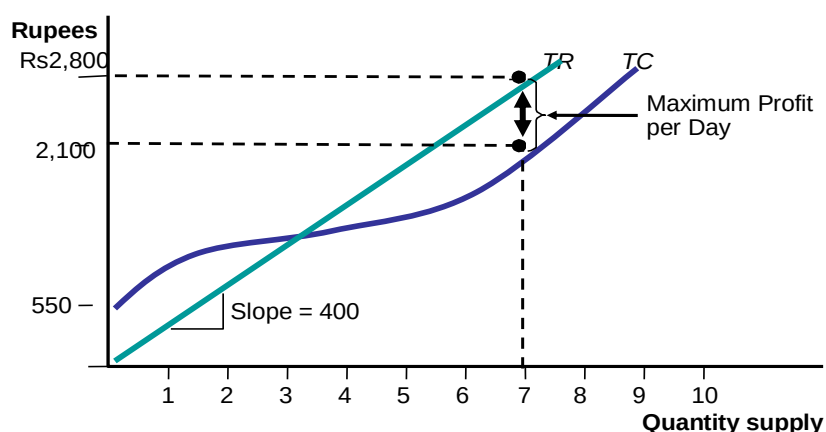
There will be close contact between buyers and sellers in perfect competition. Buyers and sellers have complete knowledge about the price in which goods are sold and bought.

7 Absence of transportation cost

Price Determination Under Perfect Competition

In perfect competition individual participants have no control over the price. Price is determined by market supply and demand → the perfectly competitive firm is a price taker → it must “take” or accept, the market price. Once the market establishes the price, each firm is free to produce whatever quantity maximizes profit.

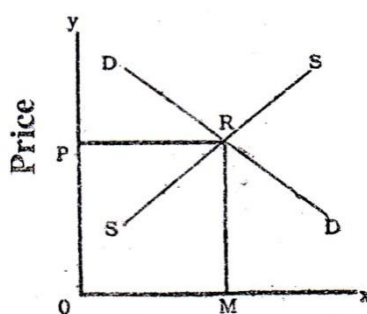
Profit Maximization: find greatest TR - TC



Under the conditions of perfect competition, price is determined by the interaction of the forces of demand and supply. Demand represents marginal utility, whereas supply represents cost of production. Demand will be more when marginal utility of the commodity to the consumer is high. Similarly, supply of commodities will be more at a lesser cost of production.

In the market, buyers try to buy the commodity at the lowest price, whereas sellers make efforts to sell the commodity at the highest price possible. An increase in demand results in a rise in price, whereas a rise in supply brings down the price level. The forces of demand and supply move in the opposite direction, and determine the price at a certain stage. The price so determined is called equilibrium price.

In the figure 5.1, supply curve SS and demand curve DD intersect at point R. OP price determined by the mutual forces of demand and supply is the equilibrium price. OM is equilibrium quantity of demand and supply. In other words, at OP price, OM quantity of goods will be in supply and in demand.

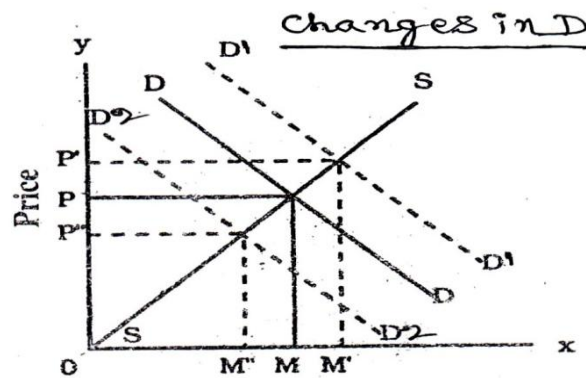


Demand and Supply Fig 5.1

Price determination under perfect competition

Effect of a Change in Demand and Supply on Equilibrium Price ,

Equilibrium price will vary in accordance with a change in demand or supply. Price goes up with on increase in demand and it falls with a decline in demand. As against this, price falls with an increase in supply and it rises with a fall in supply. Separate diagrams are required to facilitate this explanation.



Quantity Fig 5.2

In figure 5.2, SS is the supply curve, and DD is the original demand curve, Thus OP is the equilibrium price. When demand has increased to D^1 to D^1 , price has risen to OP^1 Instead, curve D^2D^2 represents a fall in demand. Price has come down to OP^1 with a fall in demand. Thus, it is evident from this explanation that change in demand is the essential cause for a change in equilibrium price.

We shall now understand the effect of a change in supply on the price.

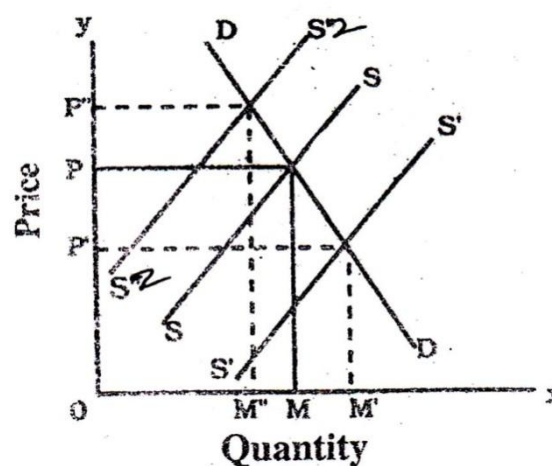


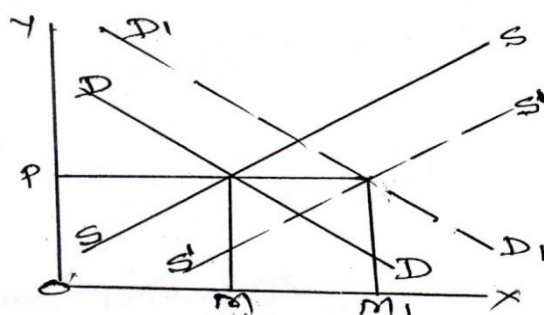
Fig 5.3

In figure 5.3 demand is shown through the curve DD Demand has remained constant without witnessing a change. SS is the original supply curve. OP is the equilibrium price

supply has come down from SS to S^1S^1 , In contrast to this, when Supply has declined to S^2S^2 price has risen to OP_{11} . Thus, the diagram clearly elucidates that price falls with an increase in supply and it rises with a fall in supply. Thus there is a direct relation between D and P and an inverse relation between S and P .

Changes both in Demand and Supply

Some times both demand and supply may change and change in the same direction and in the same proportion. Then the effect of one may be neutralized by the other.

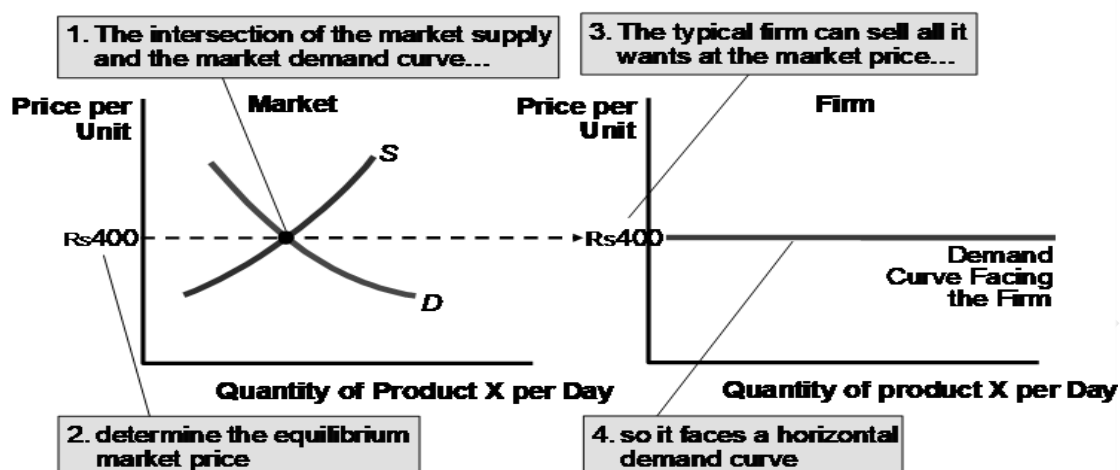


As shown in the diagram demand increases to D^1D^1 and supply increases to S^1S^1 . The equilibrium price OP remains unchanged. Only equilibrium supply and equilibrium demand increase.

In the similar fashion when both D and S decline in the same proportion price remains unchanged. Generally an increase in D brings about an increase in price and a decrease in D brings about a decrease in price. Increase in supply reduced P and decreases in supply increases P . When both demand and supply change the price follows the direction and proportion of the stronger force. Demand is more powerful in short period and supply is more powerful in long period.

Thus price in perfect competition is influenced by both demand and supply. They are like two blades of a scissors.

The Competitive Industry and Firm



Short-Run Profit Maximization

How does the perfectly competitive firm maximize profit? The perfectly competitive firm has no control over price, however, what the firm does control is the amount produced – the rate of output → the question facing the Producer (wheat farmer) is How much should I produce to earn the most profit?

Total Revenue Minus Total Cost The firm maximizes economic profit by finding the rate of output at which total revenue exceeds total cost by the greatest amount Total revenue is simply output times the price per unit

Revenue (Sales) and Costs of a Perfectly Competitive Firm

OUTPUT	PRICE	TOTAL REVENUE	TOTAL COST	PROFIT
1	20	20	28	-8
2	20	40	30	10
3	20	60	34	26
4	20	80	40	40
5	20	100	50	50
6	20	120	70	50
7	20	140	100	40
8	20	160	162	-2

Marginal Revenue Equals Marginal Cost Approach :Another way to find the profit-maximizing rate of output is to focus on marginal revenue and marginal cost

Marginal revenue, MR, is the change in total revenue from selling another unit of output Since the firm in perfect competition is a price taker, marginal revenue from selling one more unit is the market price $\rightarrow MR = P$ Marginal cost is the change in total cost resulting from producing another unit of output

Marginal Revenue and Marginal Costs

Marginal Revenue Equals Marginal Cost Approach

Marginal Revenue and Marginal Costs

OUTPUT	MR	MC
1	Rs 20	4
2	Rs 20	2
3	Rs 20	4
4	Rs 20	6
5	Rs 20	10
6	Rs 20	20
7	Rs 20	30
8	Rs 20	62

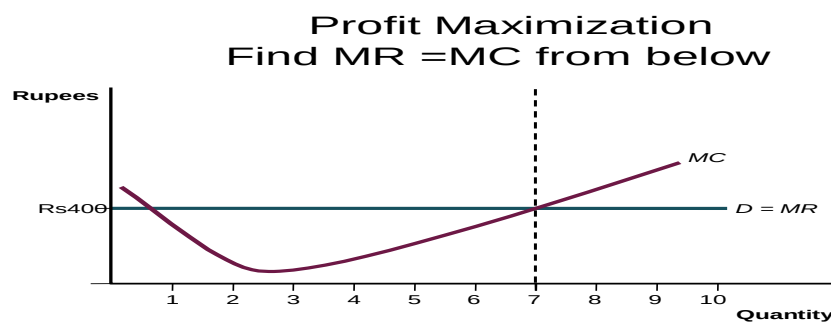
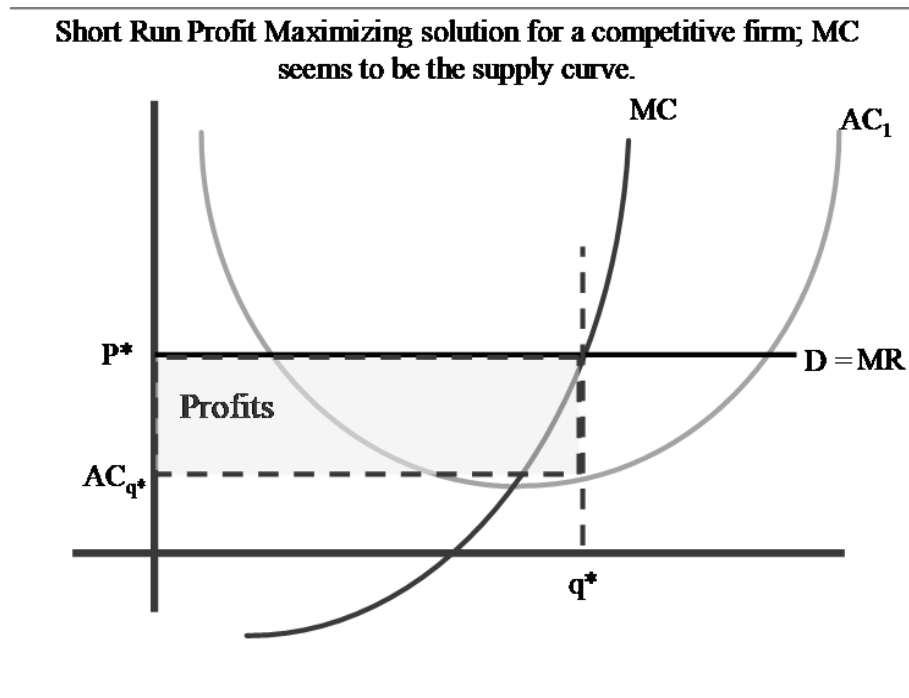
Golden rule of profit maximization: Generally, a firm will expand output as long as marginal revenue exceeds marginal cost and will stop expanding output before marginal cost exceeds marginal revenue

Firm and Industry Short-Run Supply Curves :If the price exceeds average variable cost, the firm will produce the quantity where marginal revenue equals marginal cost Further, the firm will vary output as the market price changes

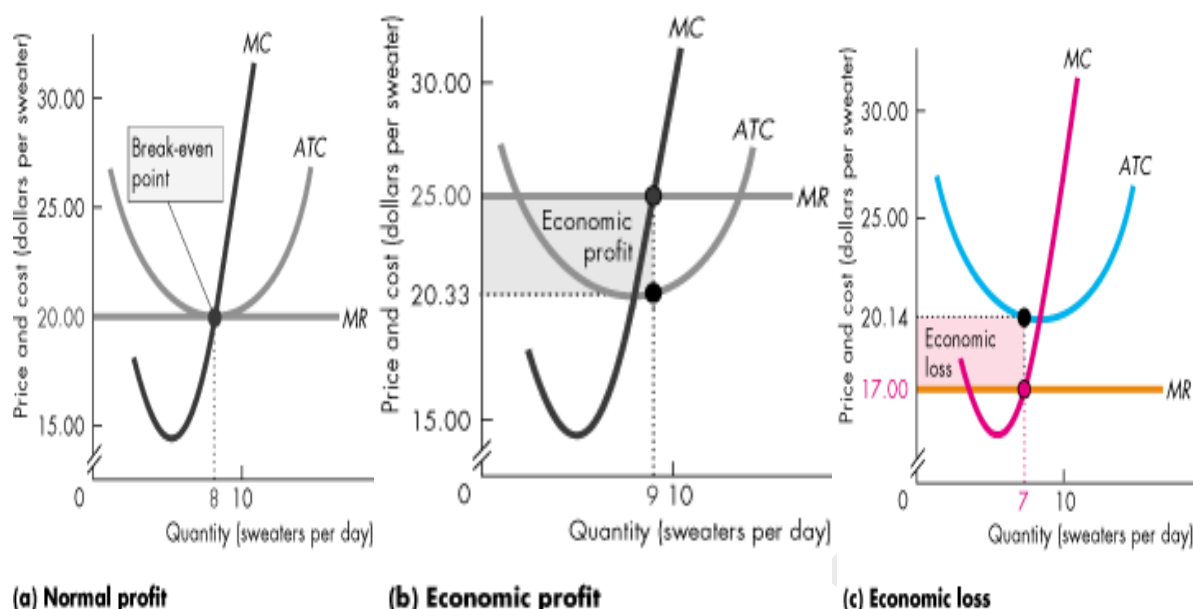
Short-Run Firm Supply Curve

As long as the price covers average variable cost, the firm will supply the quantity resulting from the intersection of its upward-sloping marginal cost curve and its marginal revenue, or demand curve Thus, that portion of the firm's marginal cost curve that intersects and rises

above the lowest point on its average variable cost curve becomes the *short-run firm supply curve*



Profit Max. in Perfect Comp.



Summary

A perfectly competitive firm selects the short-run output rate that maximizes profit or minimizes loss. When confronting a loss, a firm either produces an output that minimizes that loss or shuts down temporarily.

Given the conditions for perfect competition, the market will converge toward the equilibrium price and quantity.

Perfect Competition in Long Run: In the short run, the quantity of variable resources can change, but other resources, which generally determine firm size, are fixed. However, in the long run, firms have time to enter and exit and to adjust their size → adjust the scale of their operations → there is no distinction between fixed and variable cost because all resources under the firm's control are variable.

Short-run economic profit will in the long run encourage new firms to enter the market and may prompt existing firms to expand the scale of their operations.

Economic profit will attract resources from industries where firms earn only normal profit or suffer losses. The expansion in the number and size of firms will shift the industry supply curve rightward in the long run, driving down the price.

New firms will continue to enter a profitable industry and existing firms will continue to increase in size as long as economic profit is greater than zero. Conversely, a short-run loss will, in the long run, force some firms to leave the industry or to reduce the scale of operation. In the long run, departures and reductions in scale shift market supply to the left → market price increases until the remaining firms just earn a normal profit.

In the long-run, how much will a competitive firm produce?

Free-entry implies that economic profits are driven down to zero in the long run.

Price has to be equal to average costs if profits are zero.

In fact, since the firm can produce any output it wants at the market price, price has to be equal to the lowest value of its long-run average cost curve.

In addition, since the firm maximizes profits, price has to equal its marginal cost as well.

Monopoly

Meaning of Monopoly

The term 'monopoly' is derived from two Greek words, 'mono' meaning 'single' and 'polie' meaning 'seller'. Therefore, monopoly refers to a market situation in which there is a single seller of the product for which there are no close substitutes. There are some substitutes available in the market but they are not close substitutes. Since he is the single producer or seller of the product, he has complete control over the supply of his product. Control on both price and supply is, however, not possible. The monopolist can either fix the supply for his product and leave the price to be determined by the demand in the market. Alternatively, he can fix a particular price and leave the quantity of output to be determined by the demand in the market at that price. Further, since the monopolist is the only seller of the product in the market, the distinction between the industry and the firm disappears under monopoly. The monopoly firm is not only the firm but it is also the industry as well. According to Chamberlin “Monopoly refers to control over supply”

Features of Monopoly

The following are the important characteristic features of monopoly:

1. **Only one producer or seller** :Monopoly is a market situation where there is only one producer or seller of the product. Therefore, the monopolist firm represents entire industry. Therefore, the monopoly is called a single firm industry. The buyers of his products are however in a large numbers.
2. **The product sold by the monopolistic has no close substitutes.** Therefore there are no other firms producing similar products under monopoly. Therefore, the cross elasticity of demand between the product of the monopolist and its substitutes is very little or even zero. There are, however, distant substitutes available in the market the consumers there have to purchase either his products or go without them.
3. The monopolist, being the only producer has complete control over the supply of his product.

4. The monopolist can **fix any price** he likes. But if he fixes a very high price, he will have to be content with a lower output. On the other hand if he decides to produce more, he will have to charge a lower price. But he cannot fix both the price and the quantity of output to be produced

5 Since the monopolist is the only seller in the whole industry, he can sell more only at a lower price and less at a higher price. Therefore, the demand curve or the average revenue curve of the monopolist slopes down wards from left to right in the same manner as a demand curve of the industry descends.

6. There are **strong barriers to the entry** of rivals into the monopoly industry. Such barriers may be economic, legal artificial or nature. Therefore the monopolist has theoretically unlimited powers over the price as well as the output of his product without bothering about the reaction of his rivals

7. In monopoly, since Average Revenue (Price) falls as more units are to be sold. Marginal Revenue is always less than Average Revenue. In other words the MR curve under monopoly lies below the AR. curve.

8. The monopolist- may charge a single uniform price from all customers for his product. Such a monopoly is called 'Simple Monopoly' "But sometimes, the monopolist may charge different prices from different customers in which case it is called 'Discriminating Monopoly'.

Cases of monopolies are found in the actual world. Railways, Life Insurance, Post and Telegraph, Electricity Supply etc. are the best examples of a monopoly.

Price and Output Decisions under Monopoly

An unregulated monopolist will choose the price and output that maximizes profits.

Unlike a perfectly competitive firm, a monopolist is aware that the market price declines if the quantity sold in the market increases. Under monopoly, the marginal revenue perceived by the monopolist is lower than the market price for each level of output.

The monopolist will maximize profits if it sets output at the level at which marginal revenue equals marginal cost.

At this level, the market price charged is higher than the marginal cost and the firm earns positive economic profits.

Price and output decisions for a monopolist

OUTPUT	PRICE	TR	TC	MC	MR	PROFIT
1	700	700	640			
3	350	1050	790	150	350	260
4	342	1370	960	170	320	410
5	331	1655	1150	190	285	505
6	311	1866	1361	211	211	505
7	278	1946	1590	229	80	356
8	250	2000	1840	250	54	160

Monopoly profits are maximized where $TR - TC$ is maximized

Marginal costs, marginal revenue and profits

3	150	350	260
4	170	320	410
5	190	285	505
6	211	211	505
7	229	80	356
8	250	54	160

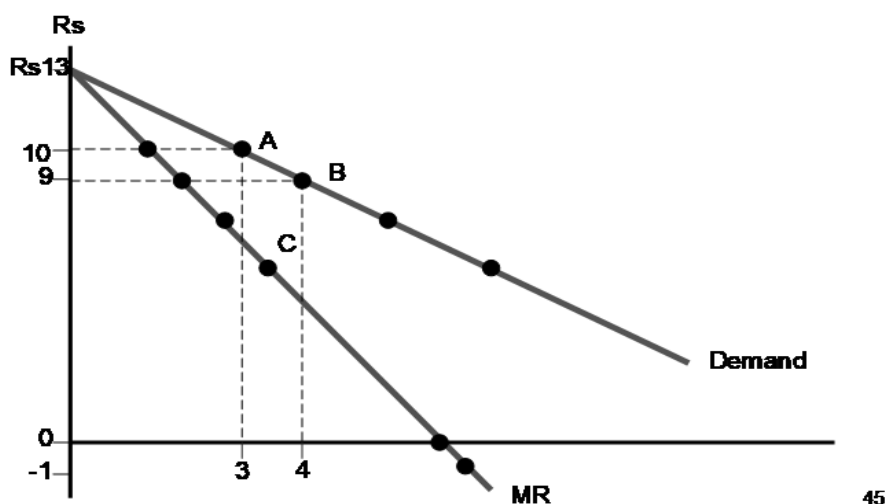
Marginal Revenue & Demand

Increasing output will, on one hand, increase revenue (as the additional output is sold).

Increasing output, on the other hand, will decrease revenue, as the price must be cut to sell the extra output.

Demand and Marginal Revenue

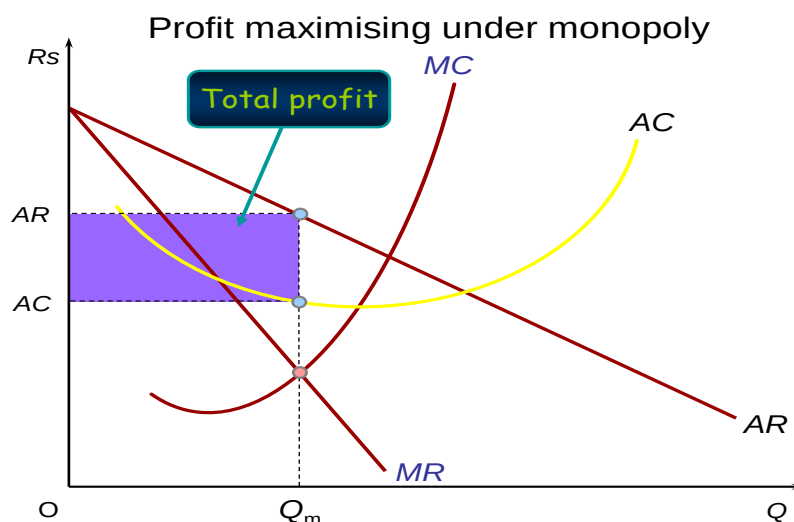
Demand and Marginal Revenue



The Profit-Maximizing Output Level

To maximize profit Produce the quantity where $MC = MR$

MC curve crosses the MR curve from below Price and output are not independent decisions, but different ways of expressing the same decision



Profit maximising under monopoly Profit and Loss

$$\text{Profit} = TR - TC$$

Equilibrium in Monopoly Markets

Monopoly market – equilibrium

–The monopoly is maximizing profit

–Produce where $MR=MC$ Profit if $P>ATC$ Loss if $P<ATC$

If $P>AVC$ – produce If $P<AVC$ – shut down

Long run Monopolies may earn economic profit

–Monopoly suffering an economic loss - should always exit the industry

If regulated - the government may decide to subsidize it If privately owned – exit the industry.

Monopoly Equilibrium

The ultimate objective of every business firm is to make the maximum profit. A monopoly firm, unlike a firm under perfect competition, is always after earning super normal profit. Because of its control over market a monopoly firm can earn abnormal profit both in the short period and long period.

1) Short period Equilibrium

A monopoly firm is in equilibrium when 1) $MR = MC$ and 2) MC curve is cutting the MR curve from below or when MC is rising. However the second condition is not compulsory. Equilibrium of monopoly firm is explained with the help of the following diagram. The firm is in Equilibrium where $MR = MC$. Therefore it produces OM output. The AR curve is demand curve also. Therefore at point P Demand = Supply and the price will be MP . When output is OM the average Cost (CAC) is MT and the AR is MP . Therefore it earns a supernormal profit represented by the shaded area. During short period the firm cannot and does not usually expand output even if there is demand. There is no possibility of other firms entering into the business. Therefore profit is usually abnormal.

2) Long period equilibrium

For a monopoly firm long period does not make much difference from short period. There is no possibility of entry of new firms because of natural monopoly or barriers to entry. Therefore a firm may not produce as much is demanded. Therefore a monopoly firm earns abnormal profit in the long period also. But it may expand its output a little more so that its total profit is larger. The long run equilibrium is explained with the help of the following diagram. At point E , $MR = MC$ and therefore the firm is in equilibrium it produces OM Output. When output is OM the average cost is MT and the Price or AR is MP . Therefore the average profit is TP and the total profit is equal to the shaded area. Thus it earns super normal profit in the long period also.

As compared to perfect competition a monopoly firm can earn abnormal profit both in short and long period. Long period price under perfect competition is equal to AC and the profit is only normal. But under monopoly during the long period also $AR > AC$ and the price is higher and profit is larger as compared to perfect competition.

Price discrimination under monopoly

Price discrimination means selling the same or slightly differentiated product to different section of consumers at different price

Consumer are discriminated on the basis of their income, purchasing power, geographical location, age, sex, colour, quantity of purchased etc. when consumer are discriminated on the basis of these factors in regard to price charged from them it is called price discrimination

Price discrimination by degree,

First-degree or perfect discrimination is a situation in which each unit of product is sold at each customer at the highest price the customer is willing to pay for it.

Second degree: or block pricing method. Monopolist divides the potential buyers into blocks first highest price to rich then at a lower price to the middle class and so on

Third degree price discrimination :

Third degree: Monopolist sets different prices in different market having demand curve with different elasticity's

Monopolistic Competition

Meaning of Monopolistic Competition

Neither perfect competition nor pure monopoly is to be found in the real world market. The market in the real world lies somewhere between the two extremes of perfect competition and pure monopoly. In real life, the market exhibits the characteristics of monopoly as well as competition. In some cases, the monopoly element predominates while in some others, the competitive element holds the sway. Such a market situation where neither pure monopoly nor perfect competition prevails but where a combination of both monopoly and competition prevails is referred to by the economists as the imperfectly competitive market. In such a market, there are many producers producing similar products which are close substitutes of each other. They compete among themselves but the competition is imperfect. The imperfect competition is mainly of two varieties- Monopolistic and Oligopoly.

Monopolistic competition is a market situation where there are many producers producing goods which are very close substitutes of each other. Competition exists between these producers but the competition is imperfect. However, though the producers compete among themselves, each one of them enjoys some monopoly power over his products. Thus, in monopolistic competition, there are many 'monopolists' competing with each other. They produce goods which are not identical but which are similar.

Every producer has a monopoly control over his product but at the same time competition prevails between these varieties. There is thus a combination of monopoly and competition in such a market. Such a market is called 'monopolistic competition' or 'group equilibrium'. The concept of monopolistic competition was introduced into the economic theory by Prof. Chamberlin of the U.S.A. in his famous work, "The Theory of Monopolistic Competition" published in 1933. At the same time, Mrs. Joan Robinson of Great Britain published her famous work "Economics of Imperfect competition" in which she uses the term imperfect competition for the real world market. There are many examples: Grocery stores, Hairdressers, Petrol stations, Video rental stores, Restaurants, Real estate agencies.

Features or characteristics

An important feature of the monopolistic competition is that there, is product variation or differentiation. Different varieties of the same commodity are treated by the consumers as different products under monopolistic competition. For example, cigarette is one product, but there are different brand cigarettes. These different brand of cigarettes are treated as different products by consumers. Such product differentiation is brought by the producers in a variety of ways. Product variation is the term used by Prof. Chamberlin for quality competition.

In the first place, the producers may bring about product differentiation through differences in the quality of - the raw materials used, workmanship, durability, size, colour, shape, style, brand, trade mark, trade name and such other devices.

Secondly, the producers may. bring about differences in their products by offering special services to their customers before and after the sale of their products e.g. credit; home delivery; collection of orders at home, guarantee of repairs and servicing, acceptance of unwanted goods etc.

Thirdly, the producers may bring about, product differentiation through sales promotion, such as advertisement, publicity and propaganda.

Fourthly, differences are also brought about by the location and distances of the selling depots.

Product differentiation is very common in the case of manufactured goods. But it may also be found in the case of many food grains like wheat, rice, flour, tea, coffee, etc. when they are sold in bags or packets or boxes, with labels or trademarks.

Existence of a Large- Number of Firms: The number of firms in the market is large, but it is not so large as under perfect competition. Since the number of firms is large, each firm acts independently without bothering about the reactions of its rivals. In other words, each monopolistic firm follows an independent price-output policy.

No Possibility of Combination: There is no possibility of combination of all the competing monopolists in order to eliminate competition" and raise the price and restrict the output.

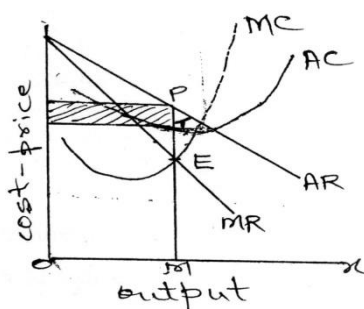
Free Entry or Free Exit of Firms: There are no restriction either on the entry of new producers into the industry or on the exit of the old ones from the industry

Price-Output Determination in the Short Period

Economics for Decision Making (MBA103)

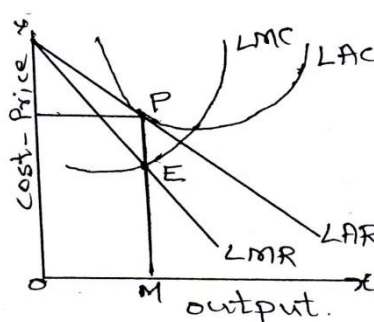
Under monopolistic competition every firm has a different AR or demand curve and MR curve because of product differentiation and selling costs. There AC and MC curves are common to all. Every firm produced that output which brings max profit to it as explained by the diagram. At point E the firm attains equilibrium because of equality between MR and MC when output is OM the price (AR) is MP and the AC is MT. This brings an extra profit shown by shaded area.

But it should be remembered that during a short period there is neither the possibility of entry of new firms nor exit of firms. Therefore there are all the three possibilities of super normal, normal and sub normal profits. Different firms may be making different profit normal supernormal and sub normal.



Price Output determination in the long period

In the long term as there is freedom of entry and exit there is generally only normal profit. If a few firms are under losses they quit the industry. Production and therefore total supply to the market declines and prices increase. This brings some firms to normalcy. If there were abnormal profits in the short period it would encourage new firms to enter the industry and compete with the old firms. This reduces prices and wipes out supernormal profit. Thus all firms earn only normal profit when output is OM here $MR=MC$ and $AR=AC$.



Oligopoly

Meaning And Characteristics of Oligopoly

The term 'oligopoly' is derived from two Greek words 'oligos' meaning 'a few' and 'pollen' meaning 'to sell'. Therefore, 'oligopoly' is that form of imperfect competition where there are a few firms in a market producing either homogeneous products or differentiated products which are close but not perfect substitutes of each other. "Thus, oligopoly or competition among a few sellers occurs when an industry consists of a few firms either producing an identical product or producing products which are close but not perfect substitutes.

Oligopoly is sometimes called limited competition, incomplete monopoly or multiple monopoly. Such oligopoly markets are generally found in modern capitalist countries. The prices and outputs of several important products like steel, gasoline, aluminum, automobiles, tires, heavy electrical equipments etc. are determined by a few firms that produce them. In fact, oligopoly appears to be a characteristic of industries where modern methods of production are applicable. Examples of Oligopolies Petrol Retailing, National Food Retailers, Hotel Industry, Leading Commercial Banks, Telecommunications Industry, Pharmaceutical companies, Soft drinks manufacturers

Characteristics of Oligopoly

The important characteristics of oligopoly are as follows:

- 1. Monopoly Power:** The most important characteristic of oligopoly is that there is an element of monopoly power in such a market. Though there are only a few firms in such a market, each of them producing a differentiated product enjoys a monopoly position in the market.
- 2. Inter-dependence:** In an oligopoly market, there are only a few firms each firm producing a homogenous or a slightly differentiated product. Since the number of firms in the market is small, each firm enjoys a sufficiently large share of the market. It is also in a position to influence significantly the price and output policy of the industry.
- 3. Indeterminateness of Demand Curve:** Mutual interdependence of firms in an oligopoly market creates an element of uncertainty for all the firms. No firm under oligopoly can correctly visualize the consequences of its price-output policy. When the quantity to be sold at a particular price is itself uncertain, the demand curve would also be indeterminate, indefinite and uncertain.
- 4. Conflicting Attitudes of Firms:** In an oligopoly market, we observe the existence of two conflicting attitudes on the part of the firms. At times, the firms realising the disadvantages of competition, may desire to come together to maximize their joint profits. Such a tendency among the oligopoly firms towards collusion is to protect their common interests. But at other times, each of the oligopoly firms aims at earning maximum profit and therefore, may initiate

conflicting and antagonistic attitude towards the other firms. Thus two conflicting trends, one of co-operation and combined action and the other of conflict and antagonism, appear to be operating in the oligopoly market.

5. Price Rigidity:

Under oligopoly prices tend to be rigid because of price war. If a firm reduced prices to sell more others also reduce price. Therefore its sales do not increase. If a firm increases prices to earn more profit. Then other firms do not increase price. It reduces its own sales. Therefore all firms maintain prices rigid.

Oligopoly = market dominated by a few sellers, at least several of which are large enough relative to the total market that they can influence the market price

Oligopoly $\Rightarrow$ more intense competition than pure competition

6. Importance of Advertising and selling Costs: A direct effect of interdependence of oligopolists is that various firms have employ various aggressive and defensive marketing weapon to gain a greater share in the market or to prevent a fall in their market share For this various firms have to incur a good deal of cost on advertising and on the measure of sales promotion

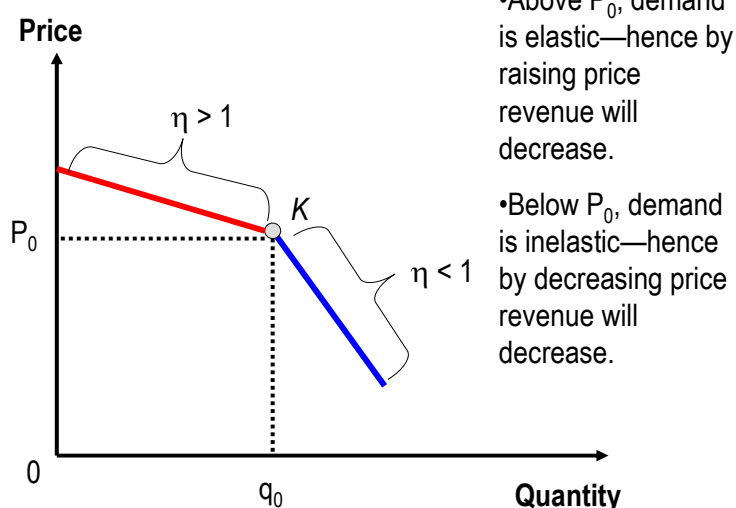
7. Group Behavior: Another important feature of oligopoly is that for the proper solution to the problem of determination of price and output under it analysis of group behavior is important.

Price Rigidity (Kinked Demand

Oligopolistic firms are interdependent – when one firm changes, others will have to consider whether action is required on their part. When firm cuts price, Q will increase – if other firms also cut price, increase in Q will be minimal (inelastic)

When firm raises price, Q will decrease. If other firms do not raise their price, increase in Q will be more substantial (elastic). Difference in relative elasticity will cause kink in demand curve at current price.

Incentive to Price At the Kink



Price leadership: A pricing strategy in which a dominant firm sets the price for an industry and the other firms follow. Price Leadership is an important form of collusive oligopoly. Under it one firm sets the price, others follow it. Price leadership also comes into existence either through tacit or formal agreement.

Types of Price Leadership:

Price leadership by Low cost firm: In order to maximize profits the low cost firm sets a lower price than the profit maximizing price of the high cost firms.

Price Leadership by Dominant firm

Barometric Price leadership: Under which old Experienced, largest or most respected firm assumes the role of a custodian who protects the interest of all.

What is a cartel?

A group of firms formally agreeing to control the price and output of a product.

Collusive Oligopoly : Cartel As a Cooperative Model

In order to avoid uncertainty arising out of interdependence and to avoid price war and cut-throat competition, firms working under oligopolistic conditions often enter into agreements regarding a uniform price-output policy pursued by them.

Summary : Oligopoly is a market structure characterized by a small number of firms and interdependence.

There is no single model of oligopoly. Since the firms recognize their interdependence, in some cases they collude (explicitly or implicitly) and act as a cartel.

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Collusive arrangements are hard to maintain since each firm has an incentive to increase the quantity sold and profits. Another model of oligopolistic behavior is price leadership (or a dominant firm).

Model Questions

3 Marks

- 1) What is an Industry?
- 2) What is a Market?
- 3) What is Perfect Competition?
- 4) What is Equilibrium?
- 5) What is Monopoly?
- 6) What is Price Discrimination?
- 7) What is Monopolistic Competition?
- 8) What is Oligopoly?
- 9) What is a Cartel?
- 10) What is Price Stickiness?
- 11) What is Price Rigidity?
- 12) What is pricing?
- 13) What is Cost-Plus Pricing?
- 14) What is Markup Pricing?
- 15) 20) What is Skimming Pricing?
- 16) 21) What is Penetration Pricing?
- 17) 22) What is Peak-load pricing?
- 18) 24) What is Price Stickiness?
- 19) 25) What is Transfer pricing?

7 Marks

- 1) Explain Equilibrium of the industry under Perfect Competition
- 2) Explain Long Run Price output determination in Perfect Competition
- 3) "Firm under perfect competition is a Price Taker" Explain
- 4) Explain different Characteristics of Monopoly Market
- 5) Explain long-Run Equilibrium of the firm under monopoly
- 6) What are the Types of Discriminating pricing

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- 7) Explain Degrees of Price Discrimination
- 8) Explain Features of Monopolistic Competition
- 9) Explain long-Run Equilibrium of the firm under monopolistic competition
- 10) What are the Advantages and disadvantages of monopolistic competition?
- 11) Explain different Characteristics of Oligopoly Market
- 12) Write a short note on Price Stickiness
- 13) What are the different pricing objectives?

10 Mark

10 Marks

- 1) What are the characteristics of Perfect Competition?
 - 2) Explain Price output determination/Equilibrium of the firm Under Perfect Competition
 - 3) Explain Short Run Price output determination in Perfect Competition
 - 5) What are the different types competitions under the imperfect competition?
 - 6) Explain Equilibrium/Price output determination in the monopoly
 - 7) Explain Short-Run Equilibrium of the firm under monopoly.
 - 8) Explain Equilibrium under price discrimination
 - 9) Explain Equilibrium/Price-output determination under monopolistic competition
 - 10) Explain Short-Run Equilibrium of the firm under monopolistic competition
 - 11) Explain the Sweezy Model of Kinked Demand Curve
 - 12) What are the factors Affecting Pricing?
 - 13) Explain Different types of pricing strategies.
 - 14) What is Cost based pricing? Explain its types of cost-based pricing.
 - 15) What is Skimming Pricing? Explain conditions that are favoring Skimming Pricing
 - 16) What is Penetration Pricing? Explain conditions that are favoring Penetration Pricing
-
1. How is Price-Output determined in the short-run in monopoly market? Discuss
 2. How is Price-Output determined in the short-run in monopolistic market? Discuss
 3. How is Price-Output determined in the perfectly competitive market
 4. Briefly explain different types of pricing
 5. Clarify the pricing strategies of Price skimming and penetration pricing what is the major difference between these two

Module -5
Indian Business Environment

Indian Business Environment

Nature, Scope, Structure of Indian Business Environment, Internal and External Environment. Political and Legal Environment, Economic Environment, Socio– Cultural Environment, Global Environment. Private Sector, Growth, Problems and Prospects, SMEs, Significance in Indian economy, challenges and prospects. Fiscal policy and Monetary Policy: Meaning of Fiscal policy, three main types of fiscal policy – neutral policy, expansionary, and contractionary. Monetary policy: Meaning, Objectives of monetary policies: Controlling inflation, Managing employment levels, and Maintaining long-term interest rates. (Theory only)

Introduction to Business Environment

Business is an important institution in society. Be it for the supply of goods or services, creation of employment opportunities, offer of better quality life, or contribution to the economic growth of a country, the role of business is crucial

Business Environment:

To understand any business the critical step is to explore all the factors related to business and properly judging its impact on the business. There are many factors and forces which have considerable impact on any business. All these forces come under one word called environment. Hence understanding the business means understanding its environment. Environment refers to all external forces which have a bearing on the functioning of business

Business Environment: It refers to all external forces which have a bearing on the functioning of the business. According to Barry M. Richman and Melvyn Copen “Environment consists of factors that are largely if not totally, external and beyond the control of individual industrial enterprise and their managements. These are essentially the ‘givers’ within which firms and their management must operate in a specific country and they vary, often greatly, from country to country”.

William F. Gluck defines business environment “as the process by which strategists monitor the economic, governmental, market, supplier, technological, geographic, and social settings to determine opportunities and threats to their firms.

Economics for Decision Making (MBA103)

From the above definitions we can extract that business environment consists of factors that are internal and external which poses threats to a firm or these provide opportunities for exploitation.

In business all the activities are being organized and also carried out by the people to satisfy the needs of the consumers. So, it is an activity carried out by the people for the people which means people occupy a central place around which all the activities revolve. It means business is people and a human is always a dynamic entity who believes in change and it may be right to say that the only certainty today is change. It poses a huge challenge for today's and especially tomorrow's businessmen and managers to be aware of specific changes so as to keep themselves abreast with the latest happenings in the field of business to maintain their survival and sustainability in the market. Therefore, the study of business environment is of at most importance for the managers and practitioners. Business environment comprises Internal Environment and External Environment .



Internal Environment

The prominent internal factors which have implications on the tactics and decisions of a business organization are given below:

1. Vision, mission and objectives

Concepts like vision, mission and objectives of a company plays a key role in deciding business province, preferences, course of development, business philosophy, business policy etc. The mission of Tata Consultancy Services (TCS), that is to help customers achieve their business objectives by providing innovative, best-in-class consulting, IT solutions and services, enabled it to be one of the triumphant companies of the world.

2. Values.

Factors like mission and objectives of the organization, business policies and practices are moulded based on the values of the founders and pioneers of the organisation. Endorsing and dissemination of strong values by the whole organization leads to its success while weak value base ends up in its failure.

3. Management structure and nature

Factors like organizational structure, the composition of the board of directors, level of professionalization of management etc have a sway upon business decisions and strategies. Quick decision making is easily enabled by some management structures while some others cause delay in it.

4. In-house authority liaison.

It intends that certain features like support from different levels of employees to the top management, shareholders and board of directors etc are swaying decisions and strategies.

5. Human Resources

The attributes of the employees like soft and hard skills, eminence, self-esteem, dedication, attitude and aptitude etc could add to the strength and weaknesses of an organization.

6. Owners

Owners are people who have invested their fund in the company and have property rights and claims on the organization. Owner can be an individual or a group of persons who created the company; or who purchased the shares from the share market. They have the right to change the company policies at any time. Hence it can be said that they are a decisive component of the internal environment.

7. Board of Directors

The board of directors is the governing body of the company who are elected by stockholders, and they are given the responsibility of administering a firm's top managers and other officials. They have crucial control over the internal environment.

8. Company image and brand equity.

The image of the company have a say in doing operations like mobilization of finance, materialization of joint ventures or other alliances, entering purchase or sales contracts, marshalling raw materials, opening new product lines etc.

9. Research and development

Good research activities undertaken by an organization is a strong internal component that can contribute to its prosperity as such a venture will enhance its reputation and can exploit first mover advantage.

10. Technological factors

Factors like introduction of new technology, its effective exploitation etc have a sway on the affluence of the business. There are a number of other internal factors like physical assets and facilities like production capacity, marketing capabilities like marketing personnel and distribution networks etc. The top management of the firm can renovate the internal environment with more strength and less weaknesses.

External Environment

The external business environment of a firm can be classified in to micro and macro environment.

Micro Environment

The micro environment or task environment encompasses those forces in the close surrounding area of an organization that influence it's functioning. Even if it is external to an organization, micro factors need not affect all the firms in a particular industry in an equivalent manner. Some of the micro factors may be unique to a firm. It embraces the following factors:

1. Suppliers

Suppliers of raw materials, components and semi finished goods are very prominent for a firm. They operate as an important force within the micro environment of the firm.

2. Marketing intermediaries

It includes the firms that assist the company in promoting, selling and distributing its goods to final buyers. They are operating in the micro environment.

3. Customers

As far as any business firm is concerned, creation and maintenance of customers are of utmost importance. Triumph of a business principally depends on realising the needs, desire and tastes of customers.

4. Competitors

Every organization has a competitive environment. Activities of a business should be adjusted according to the actions and reactions of competitors. An enterprise will be facing direct and indirect competition from many rivalries. A firm should monitor the activities of the competitors in its micro environment and should counteract accordingly.

5. Public

Public refers to any cluster that has actual or potential interest in the business activities. Such clusters can exert influence on the business. *e.g.*, growth of consumer groups may affect the working of newly developed businesses.

6. Financiers

The term financiers include commercial banks, money lending institutions, private persons etc who have lent money for business operations. In addition to the financing capabilities, their policies and strategies, attitudes, ability to provide non-financial assistance etc are vital.

7. Regulators

Regulators are units in the task environment that have the authority to control, regulate or influence an organization's policies and practices. Government agencies are the main player of this environment.

8. Strategic Partners

They are the organizations and individuals with whom the firm is in an agreement or understanding for the benefit of the organization. Such strategic partners may influence the organizations activities in various ways.

Macro Environment

A company along with its micro environment situate in a bigger macro environment. This micro environment provides opportunities and poses threats to a firm. The macro forces are generally more uncontrollable and the success of a company depends on its adaptability to the environment. A firm cannot exercise effective control on the factors of macro environment and only the degree of adaptability it has to that particular environment can direct it to success. There are two more factors which are not included in definition and which exercise considerable influence on business. They are physical or natural environment and global environment. Therefore, we will study the following environmental factors one by one.

- **Political – Legal Environment**
- **Economic Environment**
- **Socio-Economic Environment**
- **Technological Environment**
- **Global Environment**

Business environment is becoming very complex day by day as some environmental issues such as deforestation, global warming, depletion of the ozone layer, pollution of land, air and water are no longer strictly the issues related to books and conferences. The leading politicians and managers around the world have picked up the environmental banner. The green marketing movement has been gaining momentum around the world. The businesses are challenged today to develop creative ways to make profits without unduly harming the existing environment. Considering the variety of these sources of change in the environment, global managers are challenged to keep themselves abreast and adjust as necessary. Some companies like Daewoo, Hyundai, Maruti, Tata and Hero-Honda in India, with their pollution prevention programmes are leading the way. Indeed, cleaning up the environment promises to generate whole new classes of jobs in the future.

Political Environment

The influence of political environment of business is enormous. The political system prevailing in a country decides, promotes, fosters, encourages, shelters, directs and controls the business activities of that countries. A political system which is stable, honest, efficient and dynamic and which ensures political participation of the people, and assures personal security to the citizens, is primary factor for growth of any business.

Two basic political philosophies are in existence all over the world, viz., democracy and totalitarianism. In its pure sense, democracy refers to a political arrangement in which supreme power is vested in the people. Democracy may manifest itself in any of two fundamental manners. If each individual is given the right to rule and vote on every matter, the result is pure democracy which is not, however, workable in a complex society with a large constituency. Hence, the republican forms of organization follows whereby the public, in a democratic manner, elect their representatives who do ruling.

In totalitarianism, also called authoritarianism, individual freedom is completely subordinated to the power of authority of the state and concentrated in the hands of one person or in a small group which is not constitutionally accountable to the people. Societies ruled by a pressure clique - political, economy or military - or by a dictator plus most oligarchies and monarchies belong to this category. The doctrine of fascism and erstwhile Russian Communism Russian Communism are example of totalitarianism.

India is a democratic country. Our political system comprises three vital institutions :-

1. Legislature

2. Executive or government
3. Judiciary

1. **Legislature**

Out of three, legislature is most powerful political institution vested with such powers as policy making, law-makings, budget approving, executive control and acting as mirror of public opinion. The influence of legislature on business is considerable. It decides such vital aspects as the type of business activities, the country should have, who should own them, what should be their size of operation, what should happen to their earnings and other related factors.

2. **Government as Executive**

Also called the 'state' the term government refers to "the centre of political authority having the power to govern those it serves". For business consideration, we should know what are government's responsibilities to business.

Specifically, government's responsibilities towards business are as follows :

- a) Establishment and enforcement of law
- b) Maintenance of order
- c) Money and credit
- d) Orderly growth
- e) Infrastructure
- f) Information
- g) Assistance to small industries
- h) Transfer of technology
- i) Tariffs and Quotas

3. **Judiciary** : The third political institution is judiciary. Judiciary determines the manner in which the work of executives has been fulfilled. It settles the relationship between private citizens, on one hand, and between citizens and the government upon the other.

The power of the judiciary are of dual type :

- i) The authority of the courts to settle legal disputes.
- ii) Judicial review - the authority of the courts to rule on the constitutionality of legislation.

Economic Environment

Economic environment refers to all those economic factors which have a bearing on the functioning of a business unit. Business depends on the economic environment for all the needed inputs. It also depends on the economic environment to sell the finished goods.

Naturally, the dependence of business on the economic environment is total and it is not surprising because, as it is rightly said, business is one unit of the total economy.

It is difficult to be precise about the factors which constitute the economic environment of a country. But still there are some factors which have considerable influence. These factors are :-

- (a) Growth strategy
- (b) Economic system
- (c) Economic planning
- (d) Industry
- (e) Agriculture
- (f) Infrastructure
- (g) Financial and fiscal sector
- (h) Removal of regional imbalances
- (i) Price and distribution control
- (j) Economic Reforms

Out of the above said factors, two are of prime importance:-

1. Economic system
2. Industry

1. **Economic System** : The scope of a private business and the extent of government regulation of economic activities depend to a very large extent on the nature of economic system, which is an important part of business environment. Broadly the economic system is divided into three groups.

- (a) Capitalism
- (b) Socialism
- (c) Communism

(a) **Capitalism** :The system of capitalism stresses the philosophy of individualism believing in private ownership of all agents of production, in private sharing of distribution processes that determine the functions rewards of each participants, and in individual expression of consumer choice through a free market place. In its political manifestation, capitalism may fall in a range between extreme individualisms and anorchism (no government) and the acceptance of some state sanctions.

The capitalist system is also known as free enterprise economy and market economy.

Two types of capitalism may be distinguished, viz.,

(i) The old, laissez-fair capitalism, where government intervention in the economy is absent or negligible; and

(ii) The modern, regulated or mixed capitalism, where there is a substantial amount of government intervention.

(b) Socialism :Under socialism, the tools of production are to be organized, managed and owned by the government, with the benefits occurring to the public. A strong public sector, agrarian reforms, control over private wealth and investment and national self reliance are the other planks of socialism. Socialism does not involve an equal division of existing wealth among the people, but advocates the egalitarian principle. It believes in providing employment to all and emphasizes suitable rewards to the efforts put in by every worker. Also called fabian socialism, this philosophy is followed in our country and other social democratic countries in the world.

(c) Communism

Communism goes further to abolish all private property and property rights to income. The state would own and direct all instruments of production. Sharing in the distributive process would have no relationship to private property since this right would not exist. Alternatively called maxism, communism was followed in Russia, China and East European Countries.

In the mid-1960s, India had a better industrial base and possessed more pre- requisites for industrial growth than south Korea, Malaysia, Taiwan, Thailand and Indonesia. Since then, the country has succeeded in creating a virtually autarkic economy where all outputs and factors were subject to rigid price and quantity controls; where investment was strictly rationed; where there were multiple barriers to entry, investment, foreign trade, and competition, and where the objective of the financial system was to supply subsidized development funds irrespective of returns. Consequently, all the countries, mentioned above, have overtaken India and are far ahead in industrial growth.

The various administrative controls are industrial licensing, industrial policies, MRTP and asset classification of monopolies, product reservation for small scale sector, Foreign Exchange Regulation Act (FERA), tariffs and quotas, the miniplant fetish, labour market rigidities, development finance and indege now availability and essentiality. These are explained as follows :-

Industrial policy is an important document which lays a wide canvas and sets a tone for implementing promotional and regulatory roles of the government. The term “industrial policy”

refers to government's policy towards industries - their establishment, functioning, growth and management.

The policy indicate the respective areas of the large, medium and small scale sector. It also spell out government's policy towards foreign capital, labour, tariff and other related aspect. Naturally, the industrial development of a country is shaped, guided, fostered, regulated and controlled by its industrial policy.

Industrial policy is probably the most important document which indicates the relationship between the government and the business. But, it has no legal sanction and as such its violation can not be challenged in a court as is possible in the case of Fundamental Rights guaranteed by the constitution.

This policy was announced on 30th April 1956. The resolution of 1956 made the industrial policy. More socialist oriented and widened the scope of public sector. The resolution classified industries into three categories, having regard to the role which the state would play in each of them.

(i) The first category contained industries "the future development of which will be the exclusive responsibility of the state". Industries in this category were listed in Schedule A of the resolution. Schedule A contained 17 industries. These included railways and air transport, arms and ammunitions, and atomic energy, which were to be developed as Central Government monopolies. In the remaining industries in Schedule A, the expansion of the existing privately-owned units, or the possibility of the state securing the cooperation of private enterprise in the establishment of new units where the national interest so required, was not produced. However, it was made clear that "whenever co-operation with private enterprise is necessary, the state will ensure, either through majority participation in the capital or otherwise, that it has the requisite powers to guide the policy and control the operations of the undertaking.

(ii) In the second category were included industries "which will be progressively state-owned and in which the state will, therefore, generally take the initiative in establishing new undertakings, but in which private enterprise will also be expected to supplement the efforts of state. The industries included in the second category were listed in schedule B of the resolution. It contained 12 industries like machine tools, firo-alloys and tools, drugs, fertilizers, synthetic rubber, canbonisation of coal, chemical pulp, road transport and sea transport and aluminium and other non-ferrous metals not included in Schedule A.

(iii) the rest of industries were thrown open to the private sector but this did not prevent the state from starting any new undertaking. State will facilitate and encourage the development of these industries in the private sector by providing infrastructural facilities.

Ours is one of the few countries in the world where an entrepreneur is required to obtain an industrial license from the government before venturing into a new business.

A “Licence” is a written permission issued by Central Government to an industrial undertaking stating such details as the location, the article to be manufactured, production capacity and other relevant particulars. It is also subjected to a validity period within which the licensed capacity should be implemented. The general Objectives of Licensing are:

- a) To limit industrial capacity within the targets set by the plans.
- b) To direct investment in industries according to plan priorities.
- c) To regulate the location of industrial units so as to secure a balanced regional development.
- d) To prevent monopoly and prevention of concentration of wealth.
- e) To protect small scale industries against under competition from large scale industries.
- f) To foster technology and economic improvements in industries by ensuring units of economic size and adopting modern processes.
- g) To encourage new entrepreneurs to start industrial units, broadening the entrepreneurial base.

The monopolies and Restrictive Trade Practices Act, 1969, brought into force from 1st June 1970, was a very controversial piece of legislation.

The principal objectives of the MRTP Act which extends to the whole of India except to the state of Jammu and Kashmir, viz. :

- i) Prevention of concentration of economic power to the common detriment.
- ii) Control of monopolistic, restrictive and unfair trade practices which are prejudicial to public interest.

The MRTP Act was significantly amended in 1982, 1984, 1985 and 1991. After the amendments the first objective has become irrelevant as the relevant provisions to achieve the objective have been deleted. The objectives now are :

- i) Controlling monopolistic trade practices.
- ii) Regulating restrictive and unfair trade practices.

A monopolistic trade practice is essentially a trade practice which represents the abuse of the market power in the production or marketing of goods, or in the provision of services, by

charging unreasonably high prices, preventing or reducing competition, limiting technical development, deteriorating product quality, or by adopting unfair or deceptive practices.

Two tests will determine whether a trade practice is an MTP or not :

i) abuse of market power, and (ii) unreasonableness in any practice.

Thus, the following are MTPs :

1. Maintaining the prices of goods or charges for any services at an unreasonable level.
2. Limiting technical development or capital investment to the common detriment.
3. Unreasonably preventing or lessening competition.
4. Allowing quality of goods produced, supplied or distributed or any service rendered to deteriorate.
5. Increasing unreasonably the cost of production of any goods or charges for provision or maintenance of services.

Social Environment

The social environment of business includes social factors like customs, traditions, values, beliefs, poverty, literacy, life expectancy rate etc. The social structure and the values that a society cherishes have a considerable influence on the functioning of business firms. For example, during festive seasons there is an increase in the demand for new clothes, sweets, fruits, flower, etc. Due to increase in literacy rate the consumers are becoming more conscious of the quality of the products. Due to change in family composition, more nuclear families with single child concepts have come up. This increases the demand for the different types of household goods. It may be noted that the consumption patterns, the dressing and living styles of people belonging to different social structures and culture vary significantly.

Technological Environment

Among all the segments of environment, technological environment exerts considerable influence on business. Thus this section requires more devotion. J.K. Galbraith defines technology as a systematic application of scientific or other organized knowledge to practical tasks. During the last 150 years, technology has developed beyond anybody's comprehensions. Year 1983 was particularly considered by scientists as the year of scientific success. In this year scientists put a billion dollars technology into space, produced the world's first test-tube triplets and obtained evidence of another solar system. A major break through was achieved in the field of genetic engg. to cure dwarfism. Technology, thus, is the most dramatic force shaping the destiny of people and business all over the world.

Status of Technology in India

India, like any other third world country, attained political independence after prolonged colonial rule and exploitation. The country entered the modern world in a state of economic backwardness and poverty of a large section of people. It is obvious that technology must attend to the basic problems of food, clothing, health and housing of people. At the same time rapid industrial development through latest technology is necessary to catch up with advanced countries. With these objectives in mind, Government of India set-up series of R & D establishments, space research centre, Medical research centres, agricultural research establishments, oil explorations centres, power development projects and the council of scientific and industrial research. Besides, several universities and institutes have been set-up to provide higher education in science, technology and management. As on today there are 4700 intermediate/junior colleges, 144 universities, and 44 deemed universities in the country. Also there are more than 500 science and technological institutions, and 1080 in house research and development laboratories. There is also the Department of Science and Technology, an administrative wing of Government, to coordinate the activities of all research and technical activities in the country.

Private sector in India:

India being mixed economy, has assigned a great importance on the private sector of the country for attaining rapid economic development. The Government fixed specific role to the private sector in the field of Industries. Trade and service sector. Private sector is playing a positive role in the development and expansion of industries in India.

The role of Private sector:

1. **The dominant sector:** Despite of rapid progress of PSU during planning era private sector has occupied dominant position in Indian economy. Number of private sector companies in 2010 was 2, 21, 214 out of total companies 2, 40,171. In terms of employment private sector has employed 61.5 percent of workers against 34.1 percent employed in private sector.
2. **Extensive modern industrial sector :** large number of modern industries have been set up in private sector .consumer goods industries were set up in pre independence period itself. But as regard engineering industries are concerned a start was made by TATA of iron and steel at Jamshedpur during post Independence period.

Problems : Private sector operates presently is not free from its short coming following are the some of the problems of private sector

- Profit generation in sole motive

- Attention on consumer durable sector
- Decline share of net value added in total out put
- Infrastructural bottlenecks
- Monopoly and concentration of wealth

Small Scale Industry: The significant feature of Indian economy since independence is the rapid growth of the small industry sector. The Small scale industrial sector which plays a pivotal role in Indian economy in terms Employment and growth. The concept of small Scale industry has been defined by different countries in different ways. The two main criteria used to define small business are as follows

Size refers to the scale of operations. Size may be measured in the following ways :-

- a. Total capital investment
- b. The value of total assets or fixed assets.
- c. Total investment in plant and machinery
- d. The total number of persons employed
- e. Volume/value of production
- f. Volume/value of sales turnover
- g. A combination of above

2006 The small scale industry defined undertaking investment with capital investment less than Rs 25 lakhs Micro industry; for small scale industry with investment between Rs 25 lakhs and RS 5 crore and medium enterprises with investment of Rs 5 crore to 10 crores.

Role of Small Business In Indian Economy

1. Employment
2. Optimisation Of Capital
3. Balanced Regional Development
4. Mobilisation Of Local Resources
5. Export Promotion
6. Consumer Surplus
7. Feeder To Large Scale Industries
8. Social Advantage
9. Share In Industrial Productio
10. Development Of Entrepreneurship

Fiscal Policy

Fiscal policy is an important tool of macro economic policy, which has a power to influence key variables of an economy. Fiscal policy is the part of government policy which is concerned with raising revenue through taxation and other means and deciding on the level and pattern of expenditure.

What is fiscal policy: It is a policy under which the government of a country uses taxation, public expenditure and public debt programmes to achieve pre-determined economic and social goals and to solve specific problem in economy. In short Fiscal policy refers to government's policy towards revenue, debt and expenditure.

Objectives of the Fiscal policy: Fiscal policy has both general and specific objectives. General objective are broadly the same as those of other macroeconomic policies specific objectives are in relation to the typical problem that arise in an economy.

The broad objective of the policy is

- Attainment of full employment of resources
- High rate of economic growth
- Optimum allocation of economic resources
- Equitable distribution of wealth and income
- Price stability
- Promoting foreign trade
- Establishing a welfare state.

Budget:

Fiscal policy operates through budget. In fact, fiscal policy is also know as budgetary policy. The Budget simply means an estimate of revenue and expenditure.

Philip V Taylor "Budget is a master financial plan of the Government. It bring estimates of anticipated revenues and proposed expenditures, employing scheduled of activities to be under taken toward the direction of national objectives"

The Union Budget: Our constitution provides that 1) no tax can be levied or collected except authority of law 2) no expenditure can be incurred from public funds except in the manner provided in the constitution 3) The executive authorities must spend public money only in the manner sanction parliament in the case of the Union and by the State legislature in the case of the state.

Economics for Decision Making (MBA103)

The policy of the government to promote economic growth and avoid undesirable effects on the economy through taxation, government expenditure and debt management is called fiscal policy.

The implementation of fiscal policy is through annual Union Budget exercise. The budget depicts a complete picture of the estimated receipts and expenditure of the government for ensuring financial year.

Union Budget is an important event which has great significance for the entire nation and is normally introduced in the last week of February every year. The budget is prepared and presented by the Finance Minister before the parliament. The financial Bill of the budget has to pass in the parliament to approve the tax proposal and appropriation bill has to be cleared to authorize expenditure.

The budget of the Government of India. For any year, gives a complete picture of the estimated receipts and expenditure of the government for that year on the basis of the budget figure of the two previous years. Every budget, for instance, gives three sets of figures

- A) Actual figure for preceding year
- b) budget revised figure for current year
- c) budget estimated for the following year

For instance the budget estimates for the year 2020-21 contain

“Actual ” or accounts for the year 2018-2019

Budget and revised figures for the year 2019-2020

Budget estimates for 2020-21

The Budget in India is divided into Two parts revenue budget or revenue account and capital budget or capital account.

Current or revenue budget: The revenue budget of the central government deals with receipts from taxation and from non- tax source and expenditure met out of these resources.

Tax Revenue comes broadly from three sources

- Tax on income and expenditure
- Tax on property and capital (or property) transaction
- Tax on commodities and services.

Non Tax Revenue consists of 1) currency, coinage 2) interest receipts and dividend 3) other tax revenue

Current expenditure or revenue expenditure is met out current revenues . The revenue expenditure is on a) Such general service as general administration including police, judiciary, defence, collection of tax, b) social and community service such as education medical and

public health, labour and employment. c) economic service, like agriculture, Industries Transportation trade, etc

Capital Budget: Capital Budget of the Government of India, is also known as capital account consist of capital receipts and capital expenditure. The receipts of the central government are composed of

A) Net recoveries of loans and advances made previously to state Government, Union territories and public sector undertakings

B) Net Market borrowing (gross borrowing from market less repayment of public debt)

C) Net small savings collections (gross collection less share of the state) and

D) Other capital receipts such as provident fund special deposit etc

Capital Expenditure of the Union Government consist of expenditure on capital item, mainly in the form of loans to state and Union territories for financing plan projects and other capital expenditure on economic development, on social and community development and capital expenditures on defense.

Revenues of Central Government: Revenue budget The estimates of receipts on revenue account have been grouped under two broad heading Tax Revenue and Non Tax Revenue

Total revenue of the central government consist of three types

- Tax on income and expenditure
- Tax on property or capital asset
- Tax on goods and service

The first two types of tax are known as direct tax and the other known as indirect tax

Direct Tax: Income Tax: Personal income Tax is levied on individuals by the central government. The income Tax does not fall on all people but only on those people who are better off. It is based on principle of ability to pay.

Corporate Tax : The corporation tax is a tax on the net income of the companies. The central government has been imposing corporation Tax on the profit of the large and small companies and for foreign companies

Center Tax on Property : The central government has imposed certain tax on wealth.

Wealth Tax: Wealth tax is the most important in this group. It has been imposed on accumulated wealth or property of every individual.

Estate Duty: Imposed on the estate of a person which was inherited by his heirs. The duty was imposed and collected by central Govt. but proceeds were passed on to the state.

Indirect Tax: Taxes on commodities and services

Revenue from commodity taxes is the most important source of revenue for the central government. Central excise duties and customs duties are two important taxes of the central Government.

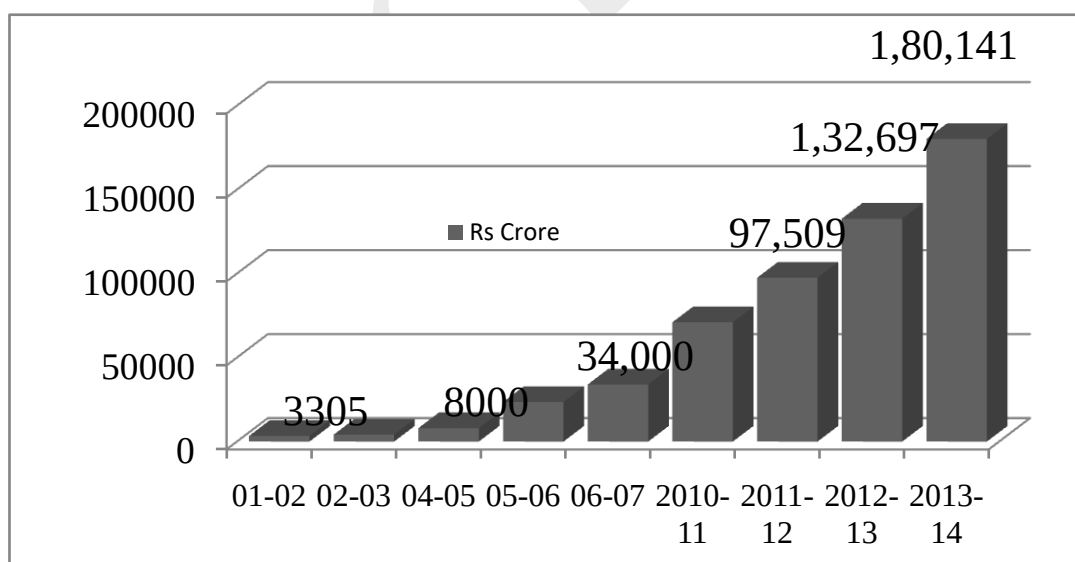
Central Excise duties: These duties are levied by the center on commodities which are produced within the country.

Customs Duties: These are duties or taxes imposed on commodities imported into India (import duties) or those goods that are exported (export duties) to India.

Service Tax: Till 1995-96 the government of India had imposed taxes only on commodities leaving out services. There was growing realization of the central finance ministry that if commodities could be taxed on their production and sales, service could be taxed. Besides in the last few decades, the service component of GDP was increasing rapidly.

Introduction of new services Taxes are. Share transfer agent, ATM operations, Recovery Agent service, Sale of Advertisement space or time, Sponsorship services, Air Transport of international travelers, Transport of goods in containers by rail, Business Support Service, Auctioneer Services, Public Relation Service, Ship management services, Internet Telephony, Transport by Cruise or ships, Credit card related services. New services would become taxable after enactment of the Finance Bill, 2006.

Service Tax



There has been a tremendous increase in the expenditure of the central Government, particularly in revenue expenditure financed through current taxation and other current non-tax revenues.

Public expenditure classified into Non Plan Expenditure, Plan Expenditure

Non Plan Expenditure: Non plan expenditure of central Government further divided into revenue expenditure and capital expenditure Revenue expenditure is financed out of revenue receipts. both tax revenue and non tax revenue.

Revenue expenditure include: A) Interest payment, defense revenue expenditure, major subsidies (food, fertilizers export promotion) and other subsidies debt relief to farmers, postal deficit, police pension, other general services like tax collection, external affairs etc

B) Social services (education, health, broadcasting etc)

C) Economic services (agriculture, industry power transport, communication science and technology etc

D) Grants to state and Union territories and grants in foreign government

Capital non plan expenditure includes such items as defense capital expenditure, loans to public enterprises. Loans to state and union territories

Plan Expenditure: The second major item of central government expenditure is plan expenditure which is composed of

A).central plans. Such as an agriculture, rural development , irrigation and flood control, energy, industry and minerals, transport, communication, science and technology and environment, social service and others. B) Central assistance for plans of the state and union territories

Expenditure of the central Government Capital expenditure of the central Government consist of

- Loans and state and union territories for financing plan project and loans to foreign government
- Capital expenditure on economic development
- Capital expenditure on social and community development

Fiscal Deficit It is the difference between the revenue receipts plus certain non-debt capital receipts and the total expenditure, including loans.

FTAs.

Monetary Policy

According to Harry G Johnson, monetary policy is a policy employing the central Bank's control of the supply of money as an instrument for achieving the objective of general economic policy. Monetary policy is generally defined, to include all measure direct and indirect which affect the supply of money , liquidity, cost, direction and availability of credit and the overall efficiency and development of the financial sector.

Economics for Decision Making (MBA103)

Monetary policy is an important tool of macro economic policy. It is designed, formulated and implemented by the central bank through a wide network of commercial banks and other financial institution. The policy is announced once or twice a year and is kept periodically tuned to take care changes in business environment and economic condition in the economy.

The monetary policy concerned changing the supply of money stock and rate of interest for the purpose of stabilizing the economy at full employment or potential out put level by influencing level of aggregate demand.

It may however be noted that in a developing country such as India , in addition to achieving equilibrium at full employment or potential out put level, monetary policy has also to promote and encourage economic growth both in the industrial and agriculture of the economy.

Monetary policy in India underwent significant changes in the 1990s as the Indian Economy became increasing open and financial sector reforms were put in place. in the 1980s, monetary policy was geared towards controlling the quantity, cost and directions of credit flow in the economy. The quantity variables dominated as the transmission Channel of monetary policy. Reforms during the 1990s enhanced the sensitivity of price Signals of price signals from the central bank, making interest rates the increasingly Dominant transmission channel of monetary policy in India. In this process, the current monetary operating framework has relied more on outright open market operations and Daily repo and reserve repo operations than on the use of direct instruments. The Monetary and Credit Policy is the policy statement, traditionally announced twice a year, through which the Reserve Bank of India seeks to ensure price stability for the economy. These factors include - money supply, interest rates and the inflation

Objectives of the monetary policy: The main objective

- Rapid Economic Growth
- Economic growth
- Full Employment
- Price stability
- Balance of payments
- Exchange Rate Stability
- Equal Income Distribution

Let us now see objectives of monetary policy in detail :-

Rapid Economic Growth : It is the most important objective of a monetary policy. The monetary policy can influence economic growth by controlling real interest rate and its resultant impact on the investment. If the RBI opts for a cheap or easy credit policy by reducing interest rates, the investment level in the economy can be encouraged. This increased investment can speed up economic growth. Faster economic growth is possible if the monetary policy succeeds in maintaining income and price stability.

Economic growth :-monetary policy can be imposed to influence the rapid economic growth. Economic growth is defined as “the process whereby the real per capita income of a country increases over a long period of time “it is measured by the increase in the amount of goods and services produced in a country. A growing economy produces more goods and services in each successive time period. Thus, growth occurs when an economy’s thus, economic growth implies raising the standard of living of the people, and reducing inequalities of income distribution.

Full Employment:- one of the objectives of monetary policy is attain full employment. It is not only because unemployment leads to wastage of potential output. But also because of the loss of social standing and self- respect. It also breeds poverty.

Price stability: - Another objective of monetary policy is to stabilize the price level. Both , rising and falling prices are bad as they bring unnecessary loss to some and undue advantage to others. They are associated with business cycles. So a policy of price stability keeps the value of money stable, eliminates cyclical fluctuations. Brings economic stability, helps in reducing inequalities of income and wealth, secures social justice and promotes economic welfare

Balance of payments:- another objective of monetary policy since the 1950s has been to maintain equilibrium in the balance of payments. It is also recognized that deficit in the balance of payments will retard the attainment of other objectives. This is because a deficit in the balance of payment leads to a sizeable outflow of gold,

Exchange Rate Stability : Exchange rate is the price of a home currency expressed in terms of any foreign currency. If this exchange rate is very volatile leading to frequent ups and downs in the exchange rate, the international community might lose confidence in our economy. The monetary policy aims at maintaining the relative stability in the exchange rate. The RBI by altering the foreign exchange reserves tries to influence the demand for foreign exchange and tries to maintain the exchange rate stability.

Equal Income Distribution : Many economists used to justify the role of the fiscal policy is maintaining economic equality. However in recent years economists have given the opinion

that the monetary policy can help and play a supplementary role in attaining an economic equality. monetary policy can make special provisions for the neglect supply such as agriculture, small-scale industries, village industries, etc. and provide them with cheaper credit for longer term. This can prove fruitful for these sectors to come up. Thus in recent period, monetary policy can help in reducing economic inequalities among different sections of society

Monetary policy instruments:

The central bank of a country employs a set of instruments to achieve the goals of monetary policy. The instrument often has different time durations during which these are applied. The various policy instruments are.

General credit control: The general controls affect the total quantity of credit and economy generally. General controls includes

- a) Bank rate
- b) Open market. operation
- c) Cash reserve requirement
- d) Statutory liquidity requirements and refinance policy

Bank Rate: Bank rate is one of the general control measures. Bank rate refers to the rate which the central bank rediscount or lends money to commercial bank However, Today The bank rate is used in broader sense and refers to the minimum rate at which the central bank provides a financial accommodation to commercial banks in discharge of its function as the lender of the last resort.

The bank rate policy seeks to affect both the cost and availability of credit. An increase in the bank rate means an increase in the rate of interest charged by the central bank on its advances to commercial banks. Hence, an increase in the bank rate compels commercial bank to raise the rate of interest they charge on their loan and advance to their customers and vice-versa.

As per the theory of bank rate, an increase in Bank rate reduces the extent of borrowings from the money market. The level of inventory holding , investment, employment and price. A reduction in the bank rate has opposite effect. The central bank may, therefore, attempt to contain an inflationary situation by raising the bank rate and fight depression or recession by lowering it.

Open Market Operation:

Open market operation refers broadly to the purchase and sale the central bank of a variety of asset, such as foreign exchange, gold, government securities and even company shares. In India, however in practice they are confined to the purchase of sales of Government securities. Under the open Market Operation, The central Bank seeks to influence the economy either by increasing the money supply or by decreasing the money supply. To increase money supply, central bank buys securities from commercial banks and public. A sale of securities by the central bank has the opposite effect.

Cash reserve Requirements (CRR): Under the RBI Act of 1935, every commercial Bank has keep a certain minimum cash reserve with the RBI. The RBI empowered to vary the CRR between 3 per cent and 15 per cent of demand and time deposit.

Statutory Liquidity Ratio (SLR) SLR mandates that every commercial Bank should Keep certain percentage of its total demand and time deposit with the RBI in the form of liquid asset like cash, gold, securities Maintenance of adequate liquid assets is a basic principles of sound Banking. The RBI empowered to change the Ratio of liquid asset.

Refinance policy: The system of refinance provided by the RBI to commercial banks affect their credit. The system is changed periodically to allow or disallow certain flow of banks. The scope of refinance as an instrument of credit control depends on the liquidity position of the bank.

Selective credit control: Selective and qualitative credit control refers to regulation of credit for specific purpose or branches of economic activity . While general credit control operates on the cost and total volume of credit. Selective control relate to the distribution or direction of available credit supplies.

Module-6

Indian Industrial Policy

Indian Industrial Policy :New industrial policy 1991, **Production Linked Incentive(PLI)** scheme for Promoting manufacturing of Telecom & Networking Products in India, New economic initiatives proposed by Indian government for economic growth Private Sector-Growth-like AtmaNirbharBharathAbhiyan.

Industrial Policies

The Industrial policy indicated the respective roles of the public, private, joint and cooperative sector; small, medium, and large scale industries and underlined the national priorities and economic development strategy. It also expressed government's policy towards foreign capital and technology, labour policy, tariff policy etc.. In short the industrial development, and thereby the economic development to a very significant extent, has been guided, regulated and fostered by the industrial policy.

Industrial Policy UP To 1991: The industrial policy of India prior to the liberalisation ushered in 1991 was characterized. By the following features.

Dominance of public sector: The policy of the government was to ensure that the public sector gained control over the commanding heights of the economy. The industrial policy resolution of 1948 established public sector monopoly in 9 industries.

Entry and growth restriction: there were a number of entry and growth restrictions on the private sector (particularly on the large firm and foreign firms) in the respect of industries where the private sector allow. A license was mandatory for establishing new units with investment above specified limit, for manufacturing new products and for substantial expansion of existing undertaking.

Restrictions on foreign capital and Technology: The scope of use of foreign capital and Technology was limited . Even in industries where foreign capital where allowed, it was normally subject to a ceiling of 40% of the total equity. Although exception were allowed in certain cases. Operation of foreign companies in India and issue of securities abroad by Indian companies were regulated under foreign exchange regulation act.(FERA) 1973.

The New Industrial Policy 1991

The Industrial policy announced on July 24 1991 which heralded the economic reforms in India, has enormously expanded the scope of private sector by opening up most of the industries for the private sector and substantially dismantling the entry and growth restriction. This new policy de-regulates the industrial economy in a substantial manner. The major objectives of the new policy to build on the gain already made, correct and distortions or weakness that might have crept in, maintained sustained growth in productivity and gainful employment and attain international competitiveness.

Objectives:

The major objectives of the new Industrial policy package are:

- To deregulate the economy in a substantial manner.
- To remove weaknesses or de distortions of the earlier policies.
- To maintain sustained growth in productivity and employment
- To encourage the growth entrepreneurship
- To upgrade technology to match the standards of international competitiveness.

In pursuit of the above objectives, the Government has decided to take a series of initiatives in respect of the policies relating to the following area.

- Industrial Licensing
- Foreign Investment.
- Foreign Technology Agreements.
- Public sector policy
- MRTP Act.

Industrial Licensing: Industrial licensing policy in India has been governed by the industries (Development & Regulation) Act 1951. over the years, keeping in view the change industrial scene the country, the policy undergone modification. Industrial licensing policy and procedures have also been liberalized from time to time. Major policy initiatives and procedural reform are also called for in order to actively encourage and assist Indian entrepreneurs to exploit and meet the emerging domestic and global opportunities and challenges. It is required to establish, to manufacture goods at a specific location before 1991.

In the new policy licensing requirement was abolished except for

1. Distillation & Brewing of alcoholic drinks
2. Manufacturing Cigarettes& Tobacco substitutes

3. All types of electronic aerospace & Defense equipments
4. Industrial Explosives
5. Certain Hazardous chemicals

Location: In the new policy projects can be established at any location except cities. In a city of 1 million population it has to be established 25kms away from city.

Foreign Investment: In order to invite foreign investment in high priority industries, requiring large investment and advanced technology, it has been decided to provide approval for direct foreign investment up to 51% foreign equity in such industries. There shall be no bottlenecks of any kind in this process. This group of industries has companies have already been known as the appendix I industries and are areas in which FERA companies have already been allowed to invest on a discretionary basis. To provide access to international markets, majority foreign equity holding up to 51% equity would be allowed for trading companies primarily engaged in export activities

- Foreign Investment allowed up to 51 % of Equity capital in FERA companies
- Foreign Investment allowed up to 26 % of Equity capital in Defence productions sector.
- Up to 100 % in certain sectors (real estate)
- Stock market opened for foreign investment
- EXIM Policy liberalised , tariffs reduced , lifted import restrictions
- Foreign exchange rate policy liberalised , Rupee was made fully convertible on current account and then recently partly on capital account.

Foreign Technology Agreements: With a view to injecting the desired level of technological dynamism in Indian industry Government provide automatic approval for technology agreement related to high priority industries within specified parameters. No permission will be necessary for hiring of foreign technicians, foreign testing of indigenously developed technologies

Public sector policy : The public enterprises have shown a very low rate of return on the capital invested. The result is that many of the public enterprises have become a burden rather than being an asset to the government. The 1991 industrial policy adopted a new approach to public enterprises. The priority areas for growth of public enterprises would be following

- Essential infrastructure goods and services.
- Exploration and exploitation of oil and mineral resources

Manufacture of products where strategic consideration predominate such as defense Equipments. There are large number of chronically sick public enterprises incurring heavy losses, operating in competitive market and serving little or no public purpose

The following measures are being adopted.

Public enterprises which are chronically sick and which are unlikely to be turned around would be referred to the Board for industrial and financial Reconstruction for formulation for rehabilitation schemes.

In order to raise resources and encourage wider public participation, a part of government shareholding in the public sector would be offered to mutual funds financial institutions, general public and workers.

Number of Industries reserved for public sector reduced to 8 and then to 2 (Atomic energy & Railway)

Growth of PSUs in the following Priority sectors

1. Essential infrastructure Goods & Services
 2. Exploration & Exploitation of Oil & Mineral resources
 3. Strategic Defence equipments
 4. Technology development & building of manufacturing capabilities in areas which are crucial for economic development
- Autonomy through MOU – Navaratnas
 - Bringing down government equity to 26 % in non-Priority sector PSUs
 - Close down PSUs which cannot be revived
 - Fully protect the interest of workers

MRTP Act: With the Growing complexity of the industrial structure and the need for achieving economies of scale for ensuring high productivity and competitive advantages in the international market, the interference of government through the MRTP Act in investment decision of large companies become deleterious in its effect on Indian industrial growth. The pre entry scrutiny of investment decision by the so called MRTP companies will no longer be required

Evaluation of the Industrial policy: The industrial policy announced by the government of India in July 1991 fulfilled a long –felt demand of the corporate sector for declaring is very clear terms that licensing was abolished for all industries except 18 industries. The new industrial policy of 1991 made a radical departure from earlier industrial policies .The various policy reforms undertaken in the subsequent years have broadly followed the spirit of the 1991 policy.

The basic thrust of the policy has been in the direction of raising productivity and efficiency and to make industrial organization more competitive. The Industrial policy is an important

instrument of industrialisation but the policy alone is not enough. Industrialisation requires the supporting force of demand, new technology, financial institutions and above all, an entrepreneurial culture

New economic initiatives proposed by Indian government for economic growth of Private Sector

Details of some of the major initiatives /schemes are as follows:

1. **Make in India initiative:** 'Make in India' is an initiative which was launched on 25th September 2014 to facilitate investment, foster innovation, build best in class infrastructure and make India a hub for manufacturing, design and innovation. It was one of the unique single, vocal for local initiative that promoted India's manufacturing domain to the world. 'Make in India' initiative is not the state/district/city/area specific initiative, rather it is being implemented all over the country.
2. **Industrial Corridor Development Programme:** In order to accelerate growth in manufacturing, Government of India (GoI) has adopted the strategy of developing Industrial Corridors in partnership with State Governments. The objective of this programme is to develop Greenfield Industrial regions/areas/nodes with sustainable infrastructure & make available Plug and Play Infrastructure at the plot level. As part of National Industrial Corridor Program, 11 Industrial Corridors are being developed in 4 phases.
3. **Ease of Doing Business:** The objective is to improve Ease of Doing Business and Ease of Living by Simplifying, Rationalizing, Digitizing and Decriminalizing Government to Business and Citizen Interfaces across Ministries/States/UTs. The key focus areas of the initiative are simplification of procedures, rationalization of legal provisions, digitization of government processes, and decriminalization of minor, technical or procedural defaults.
4. **National Single Window System:** The setting up of National Single Window System (NSWS) was announced in the Budget 2020-21 with the objective to provide "end to end" facilitation and support to investors, including pre-investment advisory, provide information related to land banks and facilitate clearances at Centre and State level. Envisioned as a one-stop shop for investor related approvals and services in the country, the National Single Window System (NSWS) was soft-launched on 22nd September, 2021 by Hon'ble Commerce & Industry Minister. Large number of States/UTs Single Window Systems have been linked with the NSWS Portal thereby providing access to approvals of these States/UTs to be applied through NSWS.
5. **PM Gati Shakti National Master Plan (NMP):** PM Gati Shakti National Master Plan (NMP), a GIS based platform with portals of various Ministries/Departments of Government, was launched in October, 2021. It is a transformative approach to facilitate data-based decisions related to integrated planning of multimodal infrastructure, thereby reducing logistics cost. Empowered Group of Secretaries (EGoS) and Network Planning Group (NPG) have been created as institutional arrangement. About 2000 data layers of various Central Ministries/Departments/State Governments have so far been uploaded on the NMP.
6. **National Logistics Policy:** National Logistics Policy (NLP) was launched on 17th September 2022, that aims to lower the cost of logistics and lead it to par with other developed countries. It is a comprehensive effort to address cost inefficiency by laying down an overarching interdisciplinary, cross-sectoral, and multi-jurisdictional framework for developing entire logistics ecosystem. This would boost economic growth, provide employment opportunities, and make Indian products more competitive in the global market.
7. **Production Linked Incentive scheme:** Keeping in view India's vision of becoming 'Atmanirbhar', Production Linked Incentive (PLI) Schemes for 14 key sectors have been

Economics for Decision Making (MBA103)

announced with an outlay of Rs. 1.97 lakh crore to enhance India's Manufacturing capabilities and Exports. These schemes have potential for creation of high production, economic growth, exports and significant employment over the next five years and more.

8. **Indian Footwear and Leather Development Programme (IFLDP):** The Central Government has approved the Central Sector Scheme 'Indian Footwear and Leather Development Programme (IFLDP)' in January, 2022 with an allocation of Rs.1700 crore till 31.03.2026 or till further review, whichever is earlier.
9. **North East Industrial and Investment Promotion Policy (NEIIPP), 2007:** North East Industrial and Investment Promotion Policy (NEIIPP), 2007 was notified for a period of 10 years from 1.4.2007 to 31.03.2017 with the purpose to boost industrialization of the region.
10. **North East Industrial Development Scheme (NEIDS), 2017:** To promote industrialization in NE States and to boost employment and income generation, a new Scheme namely North East Industrial Development Scheme (NEIDS), 2017, came into force w.e.f. 01.04.2017 for a period of five years.
11. **Aatmanirbhar Bharat:** The private sector is now entrusted as a co-partner in development. Accordingly, the government's disinvestment policy has been revived. A New Public Sector Enterprise (PSE) Policy for Aatmanirbhar Bharat has been introduced to minimise the presence of the government in the PSEs to only a few strategic sectors. Many initiatives have been introduced under the Aatmanirbhar Bharat and Make in India programmes to enhance India's manufacturing capabilities and exports across industries.

CONCLUSION

India ranked 63rd out of 190 countries in the World Bank's 'Ease of Doing Business 2023' survey, indicating a favourable business environment in the country. The recent government reforms, such as allowing private companies to operate in key industries, introducing agriculture initiatives & labour reforms and enabling commercial coal mining, serve as strong pillars to aid development of India's economy and present huge opportunities for the country to become self-reliant.

In addition, strong budgetary push for the manufacturing sector, MSMEs, along with supportive PLI reforms, will boost the domestic manufacturing companies and attract foreign companies to invest in the country; thereby, foster India's path to achieve self-reliance.

Questions

3 Marks

- 1) What is industrial Policy?
- 2) What is MRTP Act?
- 3) What is Industrial Licensing?
- 4) What are MSME.
- 5) Differentiate between fiscal policy and Monetary Policy

7 Marks

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- 1.Explain Classification of Industries in Industrial Policy Resolution, 1948
- 2.Briefly Explain Problem and prospects of SMEs
- 3) Briefly explain objective of the monetary policy

10 Marks

- 1) Critically evaluate the Industrial policy in India.
- 2) What are the major provisions of New Industrial policy 1991?
- 3) What are the New economic initiatives proposed by Indian government for economic growth of Private Sector
- 4). Critically Evaluate 1991 Industrial Policy
- 5) Briefly explain the credit control tools used by RBI to regulate credit in the Economy
- 6)Discuss the role of private sector in India and its problems

Examination Tips

1. Selection of questions
2. How much answer to write for each question
 - 3 marks maximum one paragraph
 - 7 marks minimum, One and half page answers, points with explanations
 - 10 marks minimum, Two and half page answers points with explanations
3. Write diagram, tables, if necessary properly
4. Time management -Total Time 180 minutes (3 hour)

Type of question	No of questions	Time for each question	Total question	Total minutes
3 marks	4	5	4	20
7 marks	4	15	4	60
10 marks	4	20	4	80
Case study	1	15		15
Total				175

Attend full 100 marks